Advanced Combustion via Microgravity Experiments (ACME)
Electric-Field Effects on Laminar Diffusion Flames (E-FIELD Flames)

ACME/E-FIELD Flames Final Report (2021*)

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Dunn-Rankin and Chien watch via a live video stream as their experiment runs aboard the International Space Station. [http://engineering.uci.edu/news/2018/6/uci-conducts-combustion-research-aboard-international-space-station]

**Data analysis and the publication of results is still underway, so it is expected that this report will be updated in the future. It is furthermore possible that additional tests will be conducted if the planned ACME tests are completed before the subsequent project is ready to commence operations in the Combustion Integrated Rack (CIR).*

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1. Overview

The Electric-Field Effects on Laminar Diffusion Flames (E-FIELD Flames) experiment is one of six independent experiments that make up the Advanced Combustion via Microgravity Experiments (ACME) project for which tests are conducted within the Combustion Integrated Rack (CIR) on the International Space Station (ISS).

The experiment's Principal Investigator (PI) is Prof. Derek Dunn-Rankin of the University of California, Irvine (UC Irvine). Dr. Yu-Chien (Alice) Chien, also of UC Irvine, served as the Project Scientist at the university. Meanwhile, Dennis P. Stocker of the NASA Glenn Research Center is NASA's Project Scientist (PS) for ACME including E-FIELD Flames and the rest of the ACME investigations. In preparatory drop testing, etc., Dr. Fumiaki Takahashi of the Case Western Reserve University (CWRU) served as the Deputy Project Scientist. For ACME's spaceflight operations, the NASA science team includes Drs. Uday G. Hegde and Vedha Nayagam who are also with CWRU.

The experiment had two Co-Investigators, the first of which was Prof. Felix Weinberg of London's Imperial College. Prof. Weinberg was a pioneer in researching the effects of electrical fields on flames in microgravity, but unfortunately died in 2012 (see http://www.imperial.ac.uk/news/117808/tribute-felix-weinberg/) before the hardware was built and the tests were conducted on the ISS. The second Co-Investigator was Dr. Sunny Karnani who did his doctoral research in preparation of the spaceflight experiment and afterward worked for ClearSign Combustion Corp., which markets Electrodynamic Combustion Control (https://www.clearsign.com/technology/electrodynamic-combustion-control/). The corporation provided some support to the E-FIELD Flames research effort, but Karnani left ClearSign to conduct research for the U.S. Army, effectively ending his intimate role in the project prior to the spaceflight operations. He remains a collaborator.

E-FIELD Flames operations were conducted in two rounds of testing during Increments 55-57, which were more specifically in March-May and September-November 2018. The break in testing (from May to September) was intentional and planned years in advance to allow the research team to analyze early results and use the insights obtained from the analysis to plan the second round so as to maximize its scientific output.

2. Science Overview

A. Background

With minor editing, this background is the Executive Summary from the experiment's Science Requirements Document (SRD) of March 2010.

This project utilizes electric fields in microgravity combustion science to establish a fundamental understanding of chemi-ionization and ion-driven convection in flames. It also uses this understanding to demonstrate electrical manipulation of sooting behavior and stability limits in the absence of buoyancy. This work is based on prior foundation studies that included simulating zero-g combustion of small gas jet diffusion flames in an earth-gravity laboratory, and the electric field control of flame electrodynamics. The current SRD proposes to use the extended-duration microgravity environment to further develop an understanding of electric fields in combustion processes and their potential applications.

The research centers on the fact that electrical forces, acting on charge carriers created through chemi-ionization, can provide for measurement and manipulation of flames in microgravity environments. For example, using the drag exercised on flame ions, electric fields have demonstrated observable influences on flame shape, sooting behavior, burning velocity, extinction, and stability. The source of these influences almost certainly resides in ion wind effects. In microgravity, where there are no density-driven buoyant effects to contend with, ion winds represent an important (and perhaps the only) body force that can be applied to manipulate combustion systems. This project includes both academic study of electric field effects and aspects for practical utilization of electrical control of combustion processes.

The prior laboratory experiments have shown that electrical properties of flames can be used to produce enhanced combustion performance by: (a) manipulating flame ions in order to change flame chemistry – this is generally considered a minor influence because flame ion concentration is typically small, but it may be critical at near-limit conditions; (b) creating a neutral wind (ion driven convection) locally at the flame front – this process can provide a local actuator for flame convection control; (c) using electric fields to affect soot formation and transport – ions and soot are closely linked because there are shared chemical pathways and because soot particles are charged by the flame ions; and (d) using electrical properties to characterize flames – e.g., there is a relationship between flame temperature, the carbon content of a fuel, and the ion current it can produce at saturation.

In 1-g, the above-mentioned processes are confounded by complex buoyancy interactions that cannot be unraveled. Hence, to evaluate them in detail, we propose the following experimental objectives for an extended microgravity environment: (A) determine the relationship between electric field voltage and chemi-ion current as functions of total fuel flow rate and the amount of inert (nitrogen) in the fuel; (B) compare the time response of the flame's chemiluminescence and ion current to rapid changes in electric fields; (C) determine sooting behavior in relation to the electric field influences; and (D) manipulate the limit behavior of lifted flames using electric fields.

The experimental approach includes a simple gas jet diffusion flame and a co-flow burner. The basic variations in the tests are in the burner type, the fuel type (nominally nonsooting and sooting), and in the level of inert dilution of the fuel to vary the flame strength. The co-flow or environment gas will have a nominal molar composition of 21% O2 and 79% N2. The experimental concept is to ground the burner to the test chamber and to install an electrically active planar mesh electrode downstream of the burner mouth. The potential of the mesh electrode can be either positive or negative, with a steady voltage or with a time varying potential (including a rapid step change) to examine the flame's steady state response and time response. The data to be recorded include the applied voltage, the ion current, the flame broadband luminosity, the CH* luminosity, flame images of OH* and visible light showing the flame shape as a function of electric potential and flame shape changes during transients in the electric field. In addition, a desired measurement of temperature in and near the flame will monitor the thermal field as a function of the electric field conditions.

The data from these measurements will allow a chemical and physical understanding of flame ion production and information showing the potential to use these ions to electrically control the flame both via modification of the local chemistry near extinction limits and through changes in the local convection using the ion driven wind. For example, a lifted diffusion flame near its blowoff condition is sensitive to mixing at the flame anchor point. Therefore, electric fields acting

on near-limit flames can exert a large influence because they can affect the local mixing behavior. The role of electric fields on soot formation will be dramatic under conditions where the ion driven wind has a large influence on flow residence time (or on ion residence time, if chemical effects are found important) in sooting flames.

Generally, the convective flows driven by electric fields acting on flame ions and those driven by buoyancy have been found to be of the same order of magnitude in earth gravity. In addition, raising the temperature often increases the ion production rate and the response to an electric field, but increasing temperature also increases the buoyancy effect, making it difficult, on earth, to separate ion wind effects from those due to buoyancy. In the zero-g environment, it will be possible to eliminate the buoyancy confusion and develop sufficient understanding to predict the response of flames to electric fields under all conditions. Extended run times are crucial for all of the studies because the timescales of flame property changes (which includes chemi-ionization) are dominated by convection (length scale/velocity) and past studies have shown that soot studies require high quality zero gravity timescales longer than is available in aircraft and drop towers. In addition, exploring instabilities and extinction behavior in near-limit flames requires experiments lasting tens of seconds.

Overall, the goal of this zero-gravity study is to understand chemi-ionization behavior and the resulting ion driven winds sufficiently well so that electrical properties of flames can be used to characterize (by monitoring ion current) and control them (via direct chemical or local convective influences). Electric field effects on flames have been documented for decades (even centuries), but flames in 1-g all include buoyancy influences that affect ion generation behavior. For flames near transitions and limits, buoyant influences always confound any understanding of the role of ion driven winds in using electric fields to characterize and manipulate flames. The proposed study removes this confounding influence and allows definitive measurement and analysis of the electric field effects on jet diffusion and co-flow flames.

B. Rationale for Microgravity

This is the 'Justification for Extended-Duration Microgravity' section of the experiment's SRD from March 2010. See the SRD itself for the references, etc. as they are not included in this report.

As described in the Background and Overview (Section 1.1), electric field effects and buoyancy effects have been found to be of the same order of magnitude. In addition, in some cases, raising the flame temperature will increase the O atom concentration, thereby increasing the ion production rate and increasing the response to an electric field. At the same time, increasing temperature increases the buoyancy effect, making it difficult to separate ion wind effects from buoyancy. In the zero-g environment, it will be possible to eliminate the buoyancy confusion and develop sufficient understanding to predict the response of flames to electric fields under all conditions. For electrically driven flames, microgravity is also necessary to create conditions that produce symmetric flames suitable for Abel inversion and steady-state analysis. While in general natural convection can enhance symmetry relative to non-buoyant flames, a flame at 1-g in an electric wind experiences buoyancy as a disruptive influence on symmetry. Even when buoyancy effects can be minimized, we have found that there are periodic disturbances that disrupt the ideal symmetry. Highly symmetric conditions of extended microgravity are also useful for numerically simulating these various effects by our electrokinetic model or more comprehensively by others.

Finally, in order to fully evaluate the potential for control, it is important to examine the effects separately. An example of the importance of microgravity for the electric field effects studies is shown in Figure 2.6. This figure compares the capillary flame used for 1-g experiments (which has some similarities to the jet-flame companion study) when it is oriented vertically to when it is oriented horizontally relative to gravity. These flames behave differently in shape, size, and ion production demonstrating the substantial influence played by buoyancy even in the relatively small capillary flames. Gravity effects in the co-flow flames have been described in the SRD for the CLD Flame experiment, and the same influences will be important when we apply electric fields. Only in extended microgravity will it be possible to separate the ion driven wind effects from buoyant wind effects.

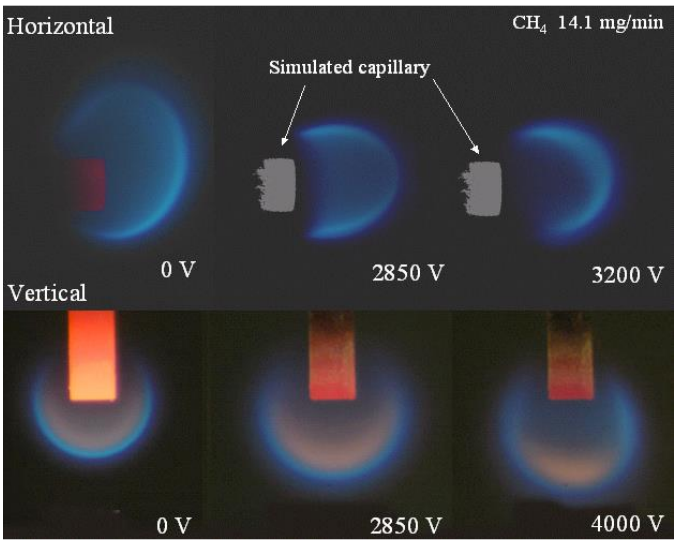


Figure 2.6. Effect of orientation on combustion of 1.8 mm jet diffusion flames in 1-g.

Extended run times are crucial for all of the studies because the timescales involved extend beyond drop tower capabilities and the microgravity quality required for symmetry and stability exceeds aircraft microgravity. The electric-field flame interaction tests carried out in the NASA Glenn 2.2 Second Drop Tower tests by Yuan and Hegde [55] showed that the flame did not appear to reach steady state during the 2.2 sec. drop. Furthermore, their results showed substantial differences in flame shape between cases where the electric field was activated half-way through the drop and at the beginning of the drop, again suggesting that steady state was not achieved. Although, we varied the range of tests performed at the Drop Tower, our research group encountered similar obstacles. Drop times were insufficient to achieve a steady flame after voltage changes in one direction. Additionally, we were unable to capture the removal of soot from the ignition event and the subsequent response of the flame to the electric field. Furthermore, testing has shown path dependence in the response time of the flame when subjected to step voltages of varying polarities. Properly mapping the steady-state ion current as a function of applied voltage requires that the flame stabilize at each applied voltage. Generally speaking, the timescale of flame property changes (which includes chemi-ionization) is dominated by convection (length scale/velocity), which is on the order of 0.1 second. In order to ensure quasi-steady conditions of the VI characteristics for a single flame configuration with some fidelity (e.g., 50 steps of 200 V each, with 2 seconds between steps for settling after the change), requires more than 100 seconds of continuous high quality microgravity. Each condition and voltage cannot be

run individually (i.e., with separate drop tests) because the 2 second settling time requires that the flame begin at a settled initial condition before taking another relatively small voltage step. If we tried to step to the initial condition, allow settling, take the base measurement, then step the voltage and again wait the required settling time, we would exceed 5 seconds for each test in nearly all cases. To run a return voltage sweep to examine hysteresis might take a further 100 seconds. Furthermore, these times are based on 1-g results which under low ion wind conditions are likely to be faster than occurs in microgravity since their the convective timescales can be substantially smaller (under low electric field forcing conditions, for example).

Understanding the electric field effects on soot is another area that requires high quality zero gravity timescales longer than is available in drop towers. A variety of zero gravity diffusion flame experience has shown that quasi-steady soot formation requires many seconds [22, 56, 57]. The long duration is necessary since we are exploring the relative importance of residence time modification of these processes in these flames. In addition, as mentioned above, instabilities and extinction behavior in near-limit flames generally have a stochastic character so that relatively long observation times are needed to quantify electric field effects. For example, lifted flames in electric fields can exhibit an oscillatory behavior where they alternately approach and recede from the burner exit until they finally reattach. Tens of seconds for this process is not unusual. Extended microgravity conditions will clarify the role of electric fields (via ion winds) on stability limits of flames.

C. Science Objectives

The proposed objectives text presented here is the 'Experimental Objectives' section of the experiment's SRD from March 2010. Although the objectives were not formally changed, the slow response time of the high-voltage power supplies (recommended in the ISRD) limited the ability to accomplish objective B. To make up in part for this shortfall, the investigation was permitted to test with two different burner types (specifically gas-jet and coflow burners), where the use of multiple burner types was disallowed for ACME experiments to limit their scope. Note that the hardware shortcoming was identified during the hardware development (and thus prior to spaceflight operations), but the project management decided to accept the shortcoming and proceed with the existing design to avoid the cost and schedule ramifications of a redesign.

The ultimate objective of this experiment is to develop an understanding of the mechanisms of chemi-ionization in flames and to describe the interplay between convective flows driven by those flame ions and their generation rate. With this information it will then be possible to detect and control flame behavior electrically. In 1-g, laminar diffusion flames are often controlled to a large extent by natural convection, and particularly by the rate at which natural convection brings fresh oxidizer to the reaction zone. Hence, the vigor with which combustion proceeds depends on local convection. As described in Section 1, an electric field acting on flame ions produces a body force and convection (analogous to buoyancy) which affects the combustion process. The change in the combustion process can in turn affect the ion production rate, which is responsible for the body force. This coupled interaction is very difficult to measure, study, and understand in 1-g because buoyant forces (of approximately equal magnitude) interfere. In microgravity, there is no such complication and the body force in the flame is associated entirely with the ion-driven wind. 1-g experiments have demonstrated clearly that electrically driven convection can be used to modify the local flow field in a manner that changes combustion behavior. For example, it can stabilize lifted flames or force them to reattach. Electric fields have been seen to change soot formation,

though whether solely by residence time effects or their combination with chemical alterations has not been confirmed. In addition, ions can attach to soot particles providing them with enhanced mobility that can be manipulated. Furthermore, the electrical properties of flames provide a measure of their character. For example, non-premixed flames exhibit different relationships between ion production and ion wind forces than do premixed flames. The hydrocarbon fuel, and in particular, the carbon/hydrogen ratio, also affects these interactions. The goal of this project is to clarify and quantify this range of electrical effects and to then demonstrate the electrical manipulation of flame stability and soot in flames using the information obtained.

To accomplish this scientific goal, there are four fundamental experimental objectives, in priority order, using a jet diffusion flame burner and a co-flow diffusion flame burner. The jet flame is the more critical configuration because it has a large 1-g data foundation and it can be more easily modeled as an ion and thermal source. In microgravity, however, it will be primarily driven by ion wind convection (no buoyant contribution and at its low fuel flow rates, the injection momentum dies rapidly with distance from the injector) which may limit its operating range. The co-flow flame allows a forced convection component to provide a wider range of flow conditions (fuel dilution and exit velocity) and a reproducible lifted flame for stability and soot behavior studies:

A) Determine the relationship between electric field voltage and chemi-ion current as functions of total fuel (methane and ethylene) flow rate and the amount of inert (nitrogen) in the fuel.
For the above determination, a low-sooting and highly sooting fuel are included, and the inert weakens the flame sufficiently to permit near-limit conditions at moderate flow velocities, where the effects of electric fields will be most acute. As shown in Figure 1.10, the resulting voltage-current (VI) curve (also known as the voltage current characteristic - VCC) from these experiments will identify the voltage and current at which the ion current first saturates, the voltage at which the current begins to increase beyond saturation, indicating a change in combustion behavior, and the voltage at which the flame extinguishes. These data can be analyzed further to provide the ion production per carbon atom of fuel flux which describes the flame character (i.e., diffusion to partially premixed), and the ion wind strength for each condition. In addition to their direct interpretative value for calculating ion-driven winds and body forces, these data can also be used in electrokinetic simulations to compute ion mobility values consistent with measured ion currents and to help validate the numerical simulations of other researchers that include ion species and electrical effects in their flame calculations.

B) Obtain the time response of the flame chemiluminescence, luminosity, and the ion current to large step changes in voltage, as functions of total fuel (methane and ethylene) flow rate and amount of inert (nitrogen) added to the fuel stream.
These data provide the dynamic system behavior of the flame. This information comprises the time response of the flame (i.e., its surface area and luminosity) as compared to the time response of the ion current. These timescales can temporally separate the direct chemical effects of electric fields from ion diffusion and ion wind convective forcing. In addition, the system response is required to eventually build flame models suitable for active electrical control, assuming linear behavior over small ranges.

C) Evaluate and manipulate sooting behavior (i.e., soot volume fraction and location of luminous soot) in relation to the electric field voltage and chemi-ion current as functions of total fuel (methane and ethylene) flow rate and amount of inert (nitrogen) added to the fuel stream.

The interaction between soot and electric fields is very complex because it combines the direct influence of electric fields on the chemistry, the ion wind effects on residence time and mixing, and the electric forces on the charged soot particles themselves. The zero-gravity environment, using the combination of the jet diffusion flame and the co-flow flame provides a wide range of conditions where these different influences can be allowed to dominate individually. For example, the forced convection environment of the co-flow burner will minimize the influence of the axial ion-driven wind, which helps to distinguish the direct electrical effects on ion chemistry in soot formation. In addition to demonstrating the direct control of soot behavior by electric fields, separating the influences can help distinguish between neutral molecule addition mechanisms of soot formation and those that presume the importance of ionic species. For example, changing the strength and polarity of the electric field can change the availability of ion precursors, and electric convection can manipulate reactants' time/temperature history. In addition, charges will accumulate onto soot, and the electric field can then act directly on the soot particles to change their residence time and path through the flame.

D) Use the variation of the chemi-ion current as a function of applied voltage to characterize the combustion behavior of near-limit lifted flames and use the associated ion-driven wind to stabilize or reattach these flames.

Because the chemi-ion current per unit flame area can be used as an indicator of flame strength, its measure (both average and fluctuating properties) with low electric field voltage can be used to detect flames approaching their limit condition. By adjusting the electric field, 1-g studies indicate that an ion driven wind can be used to reduce the liftoff height and stabilize the flame. This is an example of open loop electrical flame sensing, analysis, and control.

D. Science Success Criteria

The success criteria is from the E-FIELD Flames' portion of the 'Success Criteria' section of the ACME Integrated Science Requirements Document (ISRD, ACME-SRD-001.B) from Sept. 2013, which is revision B of the version prepared for the Requirements Definition Review (RDR).

6.2.1. Minimum Success

- Obtain V-I curve in quasi-steady conditions for undiluted methane fuel on either diameter of the gas jet burner or the co-flow burner
- Simultaneously capture color images of flame responding to electric field during V-I sweep

6.2.2. High Success

All of the above plus:

- Obtain V-I curve in quasi-steady conditions for both fuels (methane and ethylene) on one gas jet burner and the co-flow burner with a range of inert dilution of fuel
- Simultaneously capture flame images during voltage sweeps to allow measurement of liftoff height has a function of applied voltage
- Measure soot luminosity (radiometer measurement) as a function of applied voltage during VI sweeps and step changes
- Capture color images and soot luminosity (radiometer measurement) as a function of coflow velocity during the Electric Field Effects on Soot experiment.

6.2.3. Complete Success

All of the above plus:

- Demonstrate open loop control of flame near sooting and stability limits using ion current and luminosity as sensors by determining the decrease (or increase) in soot luminosity and by evaluating the extent to which stability limits can be extended using an electric field
- (Desired) Obtain thermal field information for the gas jet or coflow flame to visualize ion driven wind

3. Hardware Overview

A. General Description

This text is from the 'Methods / Experimental' section of the following conference paper:

> *Y.-C. Chien, J. Tinajero, D. Stocker, U. Hegde, and D. Dunn-Rankin, "Electric Field Effects on Flames in Microgravity on the International Space Station," 2018 Spring Technical Meeting of the Central States Section of The Combustion Institute, Minneapolis, MN, 20-22 May 2018.*

See the paper itself for the references, etc. as they are not included in this report. As a minor note, the sizes of some figures were adjusted in this report, but none were cropped.

The E-FIELD Flames experiment is being conducted in 2018 within the ISS' Combustion Integrated Rack (CIR) [15], which was launched in 2008 and used from 2009 to 2017 for a series of droplet combustion experiments [16]. The CIR facility is similar in size to a vending machine as can be seen in Figure 1. It includes a 100-liter combustion chamber capable of operation at pressures of up to 300 kPa. The chamber features a ring of 8 sapphire windows and is equipped with a set of reconfigurable imaging systems. The CIR was designed with gas delivery, venting, thermal control, electrical power, data handling, image processing and storage, and vibration isolation to enable combustion research with gaseous, liquid, and solid fuels.

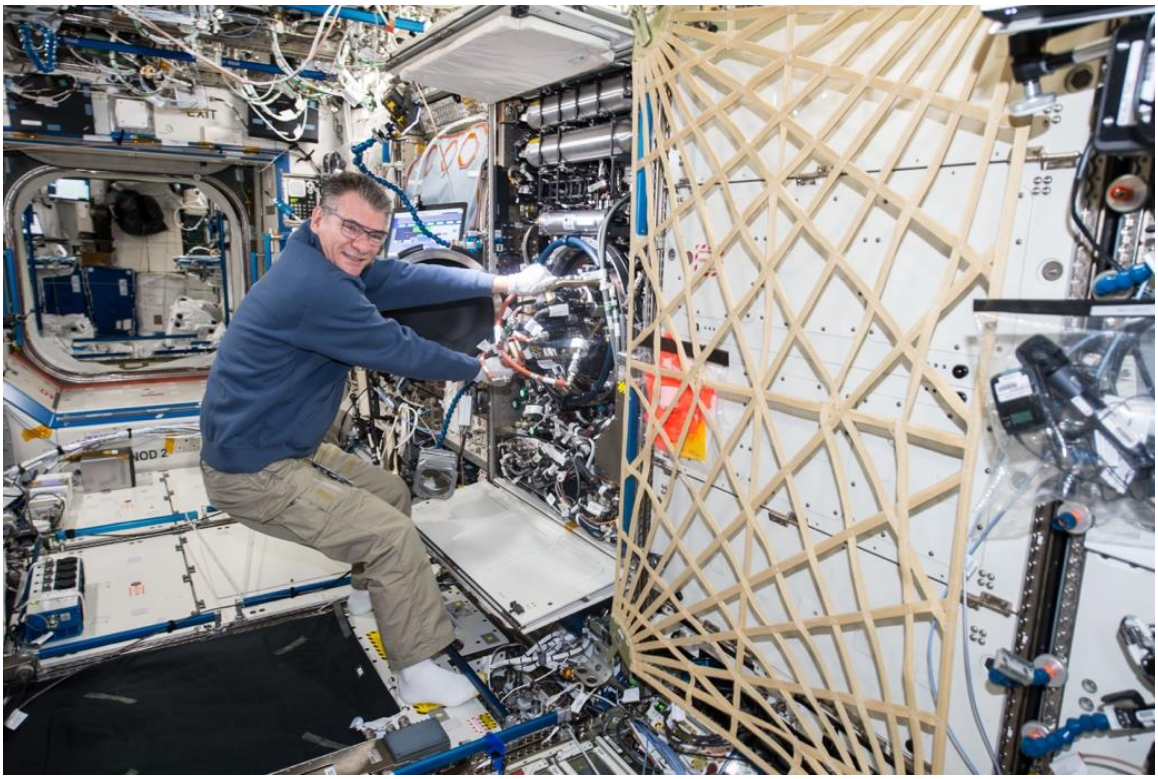


Figure 1: European Space Agency (ESA) astronaut Paolo Nespoli installing the chamber insert for the Advanced Combustion via Microgravity Experiments (ACME) in the Combustion Integrated Rack (CIR). This photograph [ISS053e098207] was taken on the International Space Station (ISS) on 12 Oct. 2017.

E-FIELD Flames is one of the six unique experiments featuring non-premixed laminar flames of gaseous fuels which currently make up the ACME project [17]. A modular set of hardware, including the critical chamber insert depicted in Figures 2 and 3, was developed to carry out the ACME investigations in the CIR facility. The 0.4-m diameter ACME chamber insert provides common mounting for a variety of burners on the chamber axis (but only one at a time) and a retracting resistively-heated igniter. The igniter has a crew-exchangeable tip and arm (to accommodate the various burners) and stepper-motor drive for radial positioning. The chamber insert includes three mass flow controllers (visible on the far left of Figures 2-3 behind an orange hose and black cables) for delivering fuel, oxidizer (or an inert), and nitrogen to the burner or chamber. The crew can exchange the mass flow controllers to provide appropriate accuracy for the desired flow ranges. ACME's gas delivery system enables on-orbit dilution of the gaseous fuel or oxidizer with nitrogen (available from the space station), but not both simultaneously. However, bottles of premixed fuel and oxidizer are also used, where the CIR's standard bottle sizes are 1.00, 2.25, and 3.80 liters. Fuels are stored in 1-liter bottles, while the 21/79 oxygen/nitrogen mixture (on a volume basis) used by E-FIELD Flames comes in 3.80-liter bottles. The ACME experiments are constant volume studies, where the chamber's free volume, with the insert installed, is about 90 liters. Burner flows are generally constrained to limit the pressure rise and vitiation of the atmosphere, where typical ACME tests are only 0.5-2 minutes in duration.

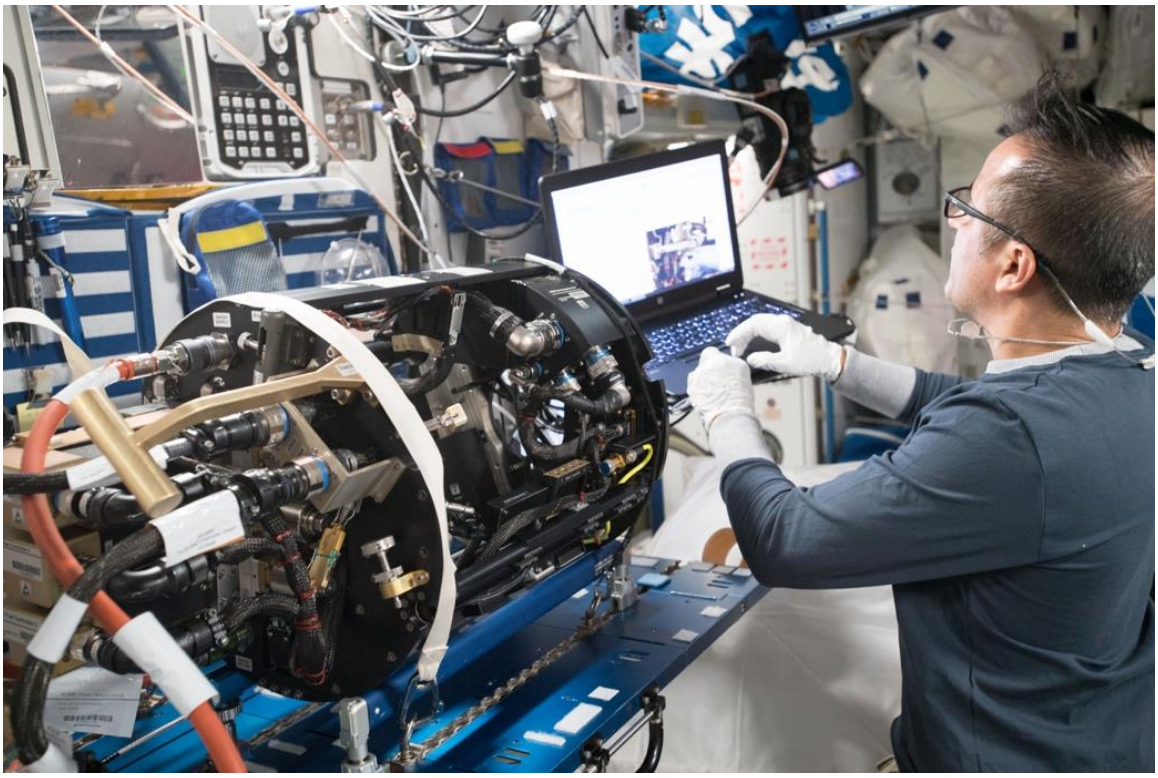


Figure 2: NASA astronaut Joe Acaba reviewing procedures to align the hot-wire igniter on the ACME chamber insert to his left, in a photograph [ISS054e014660] taken by NASA astronaut Mark Vande Hei on 5 Jan. 2018.

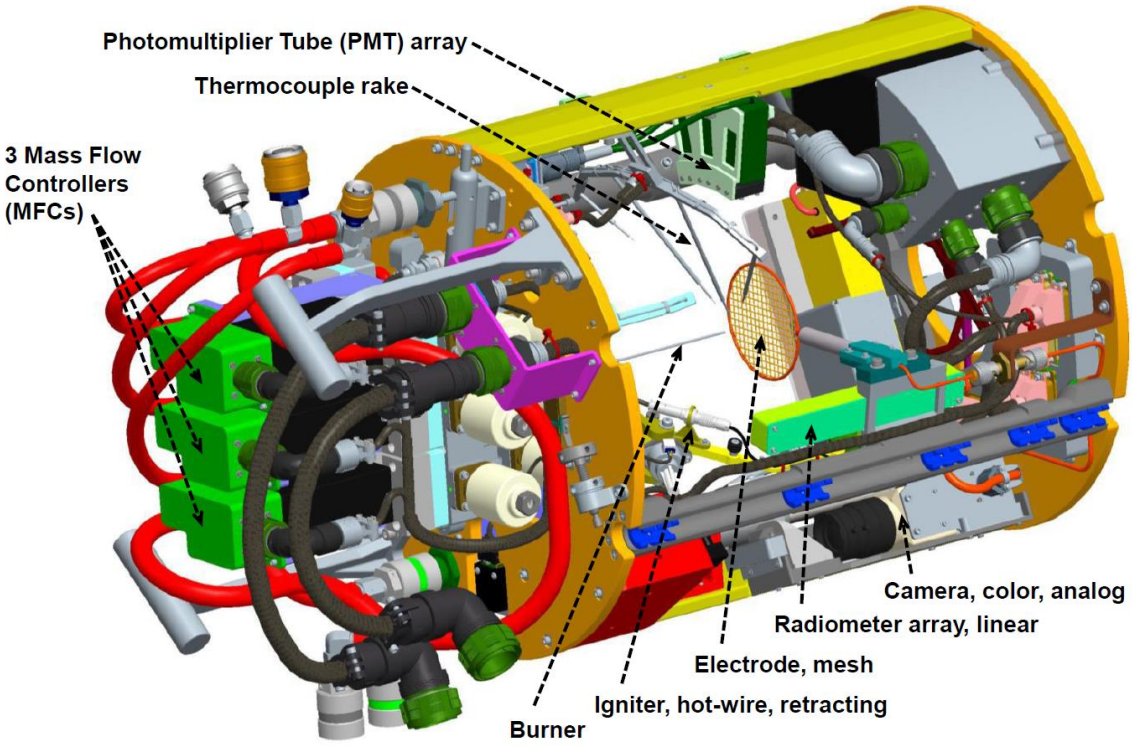


Figure 3: Solid model representation of the ACME chamber insert.

The E-FIELD Flames experiment is being conducted with two different burners: a simple gas-jet burner and a coflow burner, which can be seen in Figure 4(a). The results discussed in this paper are only from the first of the experiment's tests conducted with the coflow burner. The stainless steel fuel tube in the burner has inner and outer diameter of 2.2 and 2.4 mm, respectively. The coflow burner's steel outer tube has an inner diameter of 25 mm, where the annulus outlet is a 6.4-mm thick Hastelloy X honeycomb with 0.05-mm foil thickness and 0.81-mm cell size. The coflow burner's inner tube protrudes 3.1 mm beyond the surface of the honeycomb outlet. While the inner tube is being used exclusively for fuel flow in E-FIELD Flames, valves in the ACME chamber insert allow the fuel, oxidizer, and nitrogen to be directed to either tube, making this burner capable of inverse flame testing.

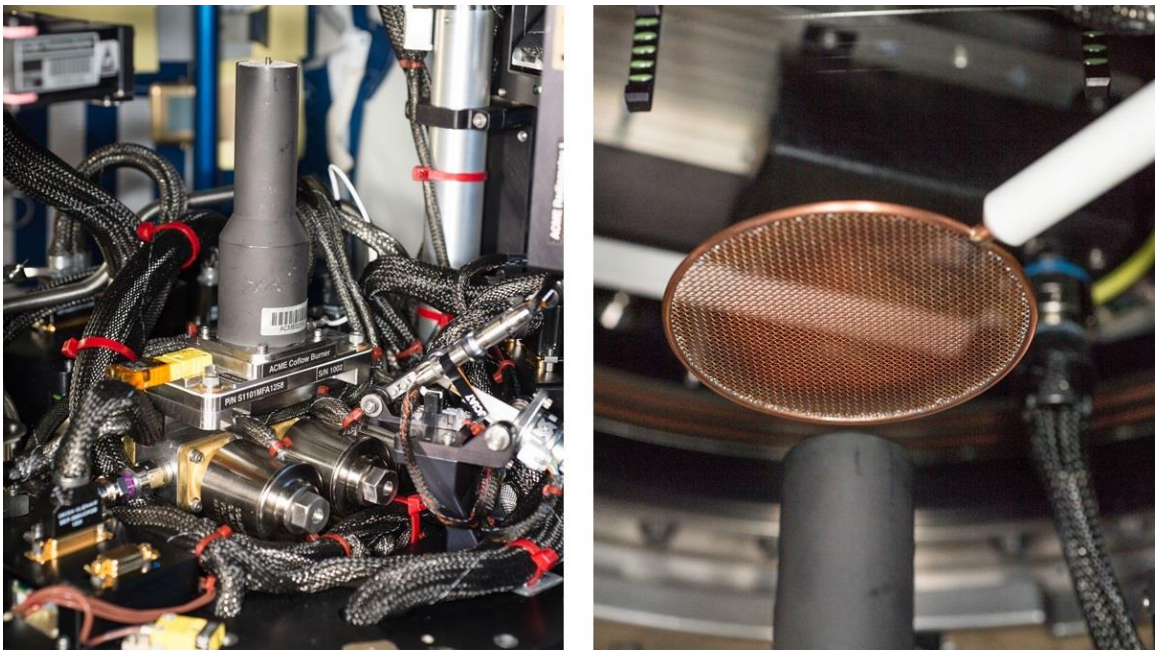


Figure 4: (a) Detail of the ACME chamber insert featuring the coflow burner (in the image center) used by the E-FIELD Flames experiment. This is from a photograph [ISS053e096074] taken by ESA astronaut Paolo Nespoli on 12 Oct. 2017 after he installed the burner on the chamber insert. (b) Detail of the ACME chamber insert where the copper electrode mesh, used by the E-FIELD Flames experiment, can be seen above the coflow burner. This is from a photograph [ISS055e001252] taken by NASA astronaut Scott Tingle on 8 March 2018 following his installation of the electrode mesh.

In this study, a high-voltage electric field is generated between the burner and a copper electrode mesh located above (i.e., downstream of) the burner outlet, as can be seen in Figure 4(b). The circular mesh is energized at up to 10,000 volts with either polarity, where the burner serves as the electrical return and the flow of flame-produced ions completes the circuit. The ion current between the burner and electrode mesh is measured from the voltage drop across a known resistor and application of Ohm's Law. With and without the copper ring, the electrode mesh is 8.9 and 8.4 cm in diameter, respectively. The square copper weave has 7.1 wires/cm, a wire diameter of 0.43 mm, and a 48% open area. The mesh permits gas to pass through while also acting as an electrical element. The electrode mesh is mounted 37 mm above the coflow burner tip and is centered on and orthogonal to the chamber axis. The ACME chamber insert can also

accommodate an alternate electrode position of 52 mm above the coflow burner tip, but the closer position is being used to produce the highest field strengths.

ACME's non-imaging measurements are recorded at 125 Hz. The measured test conditions include the chamber pressure and temperature, the three gas flow rates, and the electrical potential of the electrode mesh. Acceleration measurements are provided by the Space Acceleration Measurement System (SAMS) [18], which is the primary instrument for such measurements on ISS and previously served that role for the space shuttle.

The flame behavior is observed optically with a variety of instruments. For most ACME experiments, including E-FIELD Flames, the radiant emission is measured with 6 thermopile detectors with a spectral range of nominally 0.2-11 microns. Five radiometers are in a crew-exchangeable linear array that is mounted parallel to the chamber axis. The other detector is downstream and off axis from the burner. Chemiluminescent emission is measured with a set of three Photomultiplier Tube (PMT) modules. Their spectral range is 230-700 nm, i.e. in the visible and near ultraviolet, but two are equipped with narrow bandpass filters for measurement of OH* and CH* emissions, respectively. The third PMT is equipped with a neutral density filter to prevent saturation, although the gain values for all three PMTs are independently and readily adjustable. Like many of the components, the array of PMT modules can be removed from the chamber insert by the crew. In addition to their scientific value, PMT measurements are used operationally to detect ignition and automatically deactivate and retract the igniter. The radiometer or PMT measurements can similarly be used to detect extinction and automatically end the gas flow and test.

Imaging measurements of the flames complement the optical sensors described above. Color imaging is provided by an analog color camera within the chamber and a high-definition (1360 x 1024), 12-bit color camera mounted outside the chamber on CIR's optical bench. The latter is equipped with a motorized lens with independent control of the zoom, iris, and focus. It has a filter barrel with a fused silica window (for color imaging), a blue-green BG7 Schott glass filter, and bandpass filters for imaging CH* and HCHO (formaldehyde) chemiluminescence. The digital color camera was characterized prior to launch for color-ratio pyrometry and measurement of the soot volume fraction, via the techniques of [19, 20]. The camera makes soot pyrometry possible as well as Thin Filament Pyrometry (TFP), where the chamber insert is equipped with a translating array of silicon-carbide (SiC) fibers. A 14-micron fiber array can be seen at the top of Figure 4(b). However, the fibers interfere with the electric field preventing the use of Thin Filament Pyrometry in the E-FIELD Flames experiment.

Two intensified monochrome cameras mounted outside the combustion chamber are available for ACME use, although only one is being used by E-FIELD Flames. It is a high-definition (1390x1040), 12-bit camera equipped with a filter barrel similar to that used by the digital color camera discussed in the preceding paragraph, although it doesn't have a blue-green filter. The CH* filter has been used in the intensified photography of the E-FIELD Flames tests. While unused by E-FIELD Flames, the CIR is also equipped with a high-definition (1024x1024) camera with an intensifier extending into the ultraviolet which can be equipped with a bandpass filter and corresponding optics for OH* imaging.

Finally, a high-definition (1024x1024), 12-bit monochrome camera is mounted outside the chamber opposite a collimated light source with two fiber-coupled laser diodes emitting red light

of 650-655 nm. This camera has a bandpass filter corresponding to the illumination source, allowing for determination of soot volume fraction through an extinction technique previously employed in microgravity research [21-23]. Unfortunately, because of trade-offs demanded by limited chamber volume and large number of instruments and components, the collimated light is obstructed substantially by the electrode mesh mount, limiting the utility of this camera for the E-FIELD Flames experiment.

B. Additional Details

The preceding section described the hardware was used in the first round of E-FIELD Flames testing, where that was conducted with the coflow burner. The second round was conducted with the 2.1-mm gas-jet burner, which was built with the same tubing as the coflow burner's inner tube. However, it is important to note that the nominal inner diameter (ID) in both cases was 2.16 mm, i.e., approximately 2.2 mm rather than 2.1 mm. While potentially confusing, this met the tolerance requirement for the 2.1-mm gas-jet burner. Meanwhile, the nominal outer diameter (OD) was 2.41 mm, again for the 2.1-mm gas-jet burner and the inner tube of the coflow burner. It is also important to understand that the gas-jet burner was slightly taller than the coflow burner, where the electrode mesh was nominally 35 and 37 mm downstream of the gas-jet and coflow burners, respectively (based on the outlet of the inner tube in the case of the coflow burner).

It is expected that a separate supporting document will be created in the future to describe the ACME hardware in greater detail. Meanwhile, it can be helpful to consult the ISRD, relevant drawings, and various reports. Some important reports include the characterization reports for the data camera (see ACME-RPT-010), radiometers (ACME-RPT-011), and Photomultiplier Tubes (PMTs, ACME-RPT-013). Key hardware dimensions and more are confirmed in the inspection report (ACME-VER-001 Rev. A), while many science requirements are addressed in the ACME ERD Verification Report (ACME-VER-002). In this regard, ERD stands for Engineering Requirements Document and many of those requirements were directly derived from the science requirements. Contact ACME's Project Scientist Dennis Stocker (216-433-2166, dennis.p.stocker@nasa.gov) with questions or for more information.

C. Hardware Shortcomings

As previously discussed, the slow response time of the high-voltage power supplies (recommended in the ISRD) did not meet the requirements necessary to achieve the requested time response performance, and as such inhibited determination of the flame's response time to rapid changes in the electric field strength.

The negative and positive high-voltage power supplies were to have been coupled, as described for example in Appendix C of the ISRD. However, switching between the two high-voltage power supplies led to excessive Electromagnetic Interference (EMI) so the power supplies were completely decoupled, where only one was installed at a time. As such, testing was limited to a single polarity at a time.

The imaging for soot volume fraction via light extinction was largely blocked by the bracket holding the electrode mesh, where this was a design trade-off necessitated primarily by ACME's large number of cameras and other optical instruments. However, Prof. Marshall Long (Yale

University), who is also an ACME investigator, has an alternate technique for determining the soot volume fraction from ACME's data camera (DataCam) imaging.

The DataCam is unfortunately not as sensitive as had been hoped to the dim microgravity flames. Therefore, the imaging with that camera has typically been at a long exposure time, e.g., 0.65 s, even though it is a video camera capable of 30 frames per second (fps). The principal exception has been when making soot pyrometry or Thin Filament Pyrometry (TFP) measurements.

Between its thickness and the anti-reflection coating, the DataCam's blue-green filter excessively attenuates the red light making its use undesirable. The filter's purpose intended was to balance the intensity of the three colors in flame imaging to facilitate pyrometry. The accuracy of the pyrometric temperature measurements was diminished as result of this shortcoming.

It is expected that lesser shortcomings will be addressed in the envisioned document describing the ACME hardware in greater detail than presented here.

4. Preliminary Results

A. Changed Test Plans

To limit their scope, ACME investigations were limited to a single burner type and the E-FIELD Flames tests identified as required in the ISRD were with a gas-jet burner. However, during the hardware development (and thus prior to spaceflight operations) it was determined that the slow response of the high-voltage power supplies (suggested in the ISRD) inhibited determination of the response time of the flame to the electric field, i.e., objective B. To make up for that science loss, the experiment was permitted to conduct a portion of its tests with ACME's coflow burner, where such tests had been identified as desired in the ISRD. The gas-jet burner used for the E-FIELD Flames investigation matched the inner tube of the coflow burner. Therefore, the tests conducted allowed assessment of both the coflow and jet entrainment, where tests with the coflow burner were carried out with and without a coflow.

It had also been conceived that some tests would be conducted in which the polarity of the electric field was varied. However, the switching between the two high-voltage power supplies led to excessive Electromagnetic Interference (EMI). Therefore, the electrical design was altered such that the power supplies were completely decoupled, where only one was installed at a time. This avoided the excessive EMI but prevented changes of polarity within a test, as had been planned for the step response procedure described in Table 4.2.2 of the ISRD. It also led to the need for crew to swap the power supplies to enable testing at each polarity.

To be clear, these significant changes to the experiment were made prior to the spaceflight operations where the decisions were made by the project management to manage the budget and schedule.

In addition to the above hardware-driven changes to the test matrix, a series of other smaller changes ensued as choices for more consistent scientific outcomes were sought. All adjustments were based on finding more effective ways to utilize the space time experiments. The change history included the following.

Fall 2016: Removed the 1.3-mm jet burner from the test plan - The original concept was to compare results from two jet burner sizes, but a single jet burner with the co-flow burner was clearly the better combination once the use of two burner types was allowed.

September 2017: Changed tests with 30% CH4 to 40% CH4 - This change occurred because ground testing with the coflow burner indicated that the more dilute methane flames were not stable at similar conditions. Therefore, to ensure stable starting conditions, the slightly higher fuel concentration was chosen.

January 2018: Changed tests with 80% CH4 to 70% CH4 - This modification was made in order to better span the dilution range; we decided that changes of 30% dilution in two steps (i.e., 40%, 70%, 100%) was more consistent.

December 2017: Added parameters to the draft script (i.e., software) - Parameters were added to make it convenient to adjust the Photomultiplier Tube (PMT) gains and thus maintain useful PMT signal strengths over the wide range of flame conditions. Parameters were also added to provide more flexibility in the step response script, where independent specification of on and off step durations was implemented so that testing wasn't restricted to square waves.

Before operation in 2018: Added script parameters for the initialize voltage and duration - The parameters were helpful in clearly defining the starting point for the electric field strength.

During operation in 2018: Took actions to improve gas flow performance - A nitrogen (N2) priming time was added to the script to allow the flow to stabilize. Mass Flow Controller (MFC) performance was improved by remote adjustment of the cracking and Proportional, Integral, and Derivative (PID) settings.

During operation in 2018: Implemented improved control of the ACME Data Camera (DataCam) settings - More detailed camera settings were added to the script. This importantly included specification for camera bracketing to allow photography with a set of exposure times. All of these changes gave more control over the camera and improved the data quality; the changes to camera settings were designed to help get images over a wide dynamic range (where part of the flame was bright with soot and other parts were dim). The bracketing was only partly successful as the hardware was unable to maintain reliable shutter times when the request was shorter than the maximum framing rate of the camera, but the brackets were effective when long exposures were used.

During operation in 2018: Did not use open loop and step response script - During testing we discovered that the standard sweep and ramp scripts could be used to provide the kind of information that would have been possible with the open loop and step scripts because of the hardware limitations on the power supplies described earlier. That is, the time response and single polarity features of the power supplies meant that a creative sweep design could mimic what would be obtained for a single target voltage with fixed step duration.

(a) Setting the electric field to a zero voltage in the step response procedure deactivated the High-Voltage Power Supply (HVPS) between steps. The recovery time of the HVPS between two steps during shut offs is more than two seconds, while there is limited fuel flow time (999 seconds without resetting it by evacuating the chamber). Hence, testing the flame response

time to various sudden electric field changes in voltage would require more than 20 seconds of precious fuel time for 10 voltages (for example). If there are 10 test points planned for the test day, the recovery time alone will consume more than 200 seconds of the fuel time. These numbers neglect the time required for the fuel to fill the lines from the bottle, which is another 20-30 seconds depending on the test points. These facts would lead to the actual experiment duration being cut by half. Therefore, after evaluating the trade off, we made the decision to not use step response procedure to save the valuable fuel time and we instead expanded the voltage sweep tests. As a future remedy, we would conduct step response tests by using a low non-zero voltage instead of turning off the HVPS.

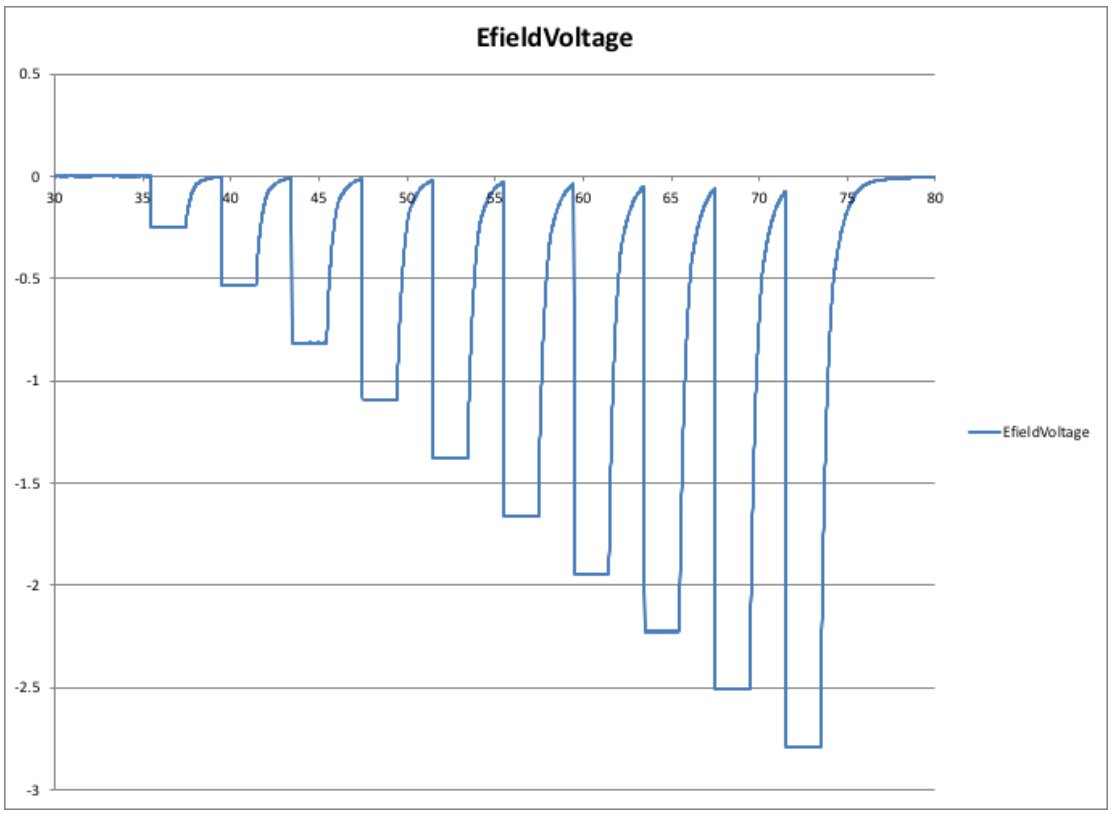


Figure 6: High voltage power supply test results using the step response script. The x axis is time in seconds while the y axis is the measured electric potential in kilovolts.

(b) The open loop outcomes can be accomplished with an appropriate choice of parameters in the voltage sweep script. Therefore, the open loop script was not needed during testing. We chose to only use the voltage sweep script for stepped and ramped changes in the field strength

B. Overview of Experiment Test Operations

The narrative description of the operations is from the 'Methods / Experimental' section of the following conference paper:

Y.-C. Chien, J. Tinajero, D. Stocker, U. Hegde, and D. Dunn-Rankin, "Electric Field Effects on Flames in Microgravity on the International Space Station," 2018 Spring Technical Meeting of the Central States Section of The Combustion Institute, Minneapolis, MN, 20-22 May 2018.

See the paper itself for the references, etc. as they are not included in this report. As a minor note, the size of some figures was adjusted in this report, but none were cropped.

The CIR experiments are configured, but not conducted, by the ISS crew. Thus far, ACME's activity has included the majority of the crew members, regardless of their nationalities. Most often the astronauts and cosmonauts exchange gas bottles, mass flow controllers, and igniter tips. In the latter case, the igniter tip exchange can occur after failure or proactively before failure when the crew already has changes to make changes in the interior of the insert. While these activities may seem mundane, the crew play a critical role in setting up the hardware for each experiment. As an extreme example, there were crew activities on 8 separate days in Sept.-Oct. 2017 to configure the CIR and ACME hardware for the first ACME experiment, namely the Coflow Laminar Diffusion Flame (CLD Flame) experiment. In contrast, only two crew days were required to reconfigure the hardware for the first E-FIELD Flames tests because of the initial use of the same coflow burner. Planning is critical because crew activity is scheduled three weeks in advance, for which detailed procedures must be prepared and uplinked to the ISS.

All CIR tests are remotely commanded from the ground from NASA's Glenn ISS Payload Operations Center (GIPOC) in Cleveland, Ohio. Typically, four to six people staff four consoles during the testing. The CIR hardware is controlled from the Rack Officer (RO) and Data Management Officer (DMO) consoles, and the ACME hardware is controlled from the Payload Developer (PD) console. Two shifts of personnel are scheduled to staff these three consoles on test days because they typically last 14 hours with the many pre- and post-testing activities. Given that the tests are remotely commanded, the operations are carried out in the day and evening in Cleveland, e.g., from 7 AM to 9 PM. A day's tests, i.e., with flames, are normally conducted in a period of 10 hours of less. Tests are normally conducted on three weekdays per two weeks. On intervening weekdays, the image data is downlinked and the crew often swap one or both gas bottles. Transferring and downlinking data from a day's tests also requires a 2-shift day for the RO and DMO consoles, where it is always necessary to verify that data was downlinked without errors and that data is not missing. Those two positions are also staffed to guide and oversee the crew activity. As a result, the CIR consoles are normally actively staffed on four weekdays per week. A team of four carry out the detailed planning for the CIR and ACME console staff, and the daily operational plans typically exceed 50 pages on days including testing.

The science console allows for review of the analog video and non-imaging telemetry (at 1 Hz) in near real time but doesn't allow for commanding. In the case of E-FIELD Flames, arrangements were made so that the analog video is streamed to a secure computer at UC Irvine to allow the research team to participate in test operations and to recommend adjustments to the day's planned parameters. That team does not see the telemetry during the testing, but is in communication via phone and e-mail with their NASA research partners at the GRC ISS Payload Operations Center (GIPOC). While there are many preparatory steps, the tests are ultimately conducted using an uploaded software script and parameter files. In advance of the test day, the

UC Irvine science team plans the nearly 60 parameters in the voltage sweep script and specifies other parameters that are configured outside of the script, such as camera settings. It is also possible to effectively conduct tests in a 'manual' mode, where parameters are changed in response to the analog video downlink or telemetry. For example, the electric field potential can be adjusted to approach a stability or soot limit.

Typically, ten E-FIELD Flames tests are conducted per test day. Despite the short test durations, the low rate results from several unavoidable factors. After a test is conducted, the 125-Hz non-imaging data is downlinked for immediate review by the science team. A small sample of images are often downlinked to ensure that the camera exposures are appropriate, where the times for those images are typically selected based on non-imaging data to provide extreme examples. The review helps to verify parameters or plan for adjustments in subsequent tests, e.g., to select the voltage range after determining where the ion current saturates for the given flame conditions. After a test is conducted, a fan circulates the chamber atmosphere after which the chamber is vented (to space) back to atmospheric pressure, if needed. Midway through a day's E-FIELD Flames tests, the atmosphere is circulated for 15 minutes through a filter cartridge filled with molecular sieves to capture any soot and adsorb at least a portion of the combustion products, e.g., carbon dioxide, water vapor, and incomplete combustion products. The tests are only conducted when both Ku and S band communication is available through NASA's Tracking and Data Relay Satellite (TDRS) System. Depending on the various orbits, there are periodic communication outages with a loss of signal (LOS) that can be a short handover (from one satellite to another) or can last as long as 30 minutes.

A variety of constraints limit the number of tests that can be performed, both overall and on a daily basis. The next combustion experiments in the CIR will feature solid fuels and focus on flame spread and material flammability. That follow-on plan limits the duration of ACME operations, where each ACME experiment is allotted four months for testing. Although it requires additional crew time, these allotments are split into two rounds so that the investigators have a break from testing for careful review of the early results so that they can maximize the scientific outcome of their second round. This paper reports on early results from the first round of E-FIELD Flames.

While nitrogen is available from the space station, the combustion testing is limited by the available gas bottles that were and are launched to support the tests. For E-FIELD Flames and most of ACME's experiments, the test matrices used to plan those gas resources were established in 2010. At that time, NASA limited the scope of the ACME tests matrices with the intent of focused experiments with limited objectives rather than comprehensive experiments. As such, the general target was 50 tests per experiment. For most ACME experiments, the limiting gas constraint is the oxygen/nitrogen bottles used to fill the chamber. This resource constraint is exacerbated by the CIR because it is equipped with a pair of redundant safety timers to limit the fuel that can flow into the combustion chamber (where the timers can be set for the oxidizer flow in the case of inverse flame testing). The timers can only be reset by venting the combustion chamber. Therefore, the timer limit combined with other constraints effectively corresponds to the day's maximum flow time for the fuel (or more generally, the limiting reactant). For E-FIELD Flames, the safety timers are set to the maximum value of 999 seconds, which corresponds to the 10 tests typically completed per day.

Another constraint is the available time in a day, where the console personnel and ISS resources, such as the video downlink, are planned for a limited number of hours. When necessary for the

research, NASA does have continuous 24-hour/day operations for some non-CIR experiments. But as discussed, the ACME tests are typically two minutes or less in duration, so the testing is almost always limited to the planned durations.

Data limits are yet another constraint, where the on-orbit hardware can store a limited amount of data. More importantly, 127,000 MB is the most that can be reasonably managed in a two-shift transfer and downlink day. Therefore, tests days are limited to this amount of data, where image data from select cameras is often archived at reduced rates to mitigate this challenge. Nonetheless, the limit sometimes forces decisions to abandon recorded images.

This detailed description of operations and hardware provides the context under which the ACME experiments are conducted. Their complexity makes the flame conditions seem trivial in comparison. It is a simple coflow flame, with just a small jet of fuel surrounded by a weak coflow of 'air' which also fills the chamber, but where tests are also conducted with no coflow. A test includes starting the gas flows to prime the system, energizing the igniter to initiate a flame, retracting the igniter, and allowing the flame to steady slightly before energizing the electric field to its initial value. The flame is again allowed to steady before beginning a voltage sweep over a specified range with steps or smooth ramps of specified voltage changes and durations. The sweep can be in one direction or can include cycles of sweeps. After each test, a duplicate field sweep is made to ensure that there are no current leaks, i.e., other than the ion current through the flame.

The first phase tests in E-FIELD Flames employ the coflow burner geometry shown in Figure 4, with the fuel tube extending slightly above the burner surface. The results shown in this paper are for tests conducted with 100% methane in a chamber atmosphere and coflow (when used) of synthetic 'air' (i.e., 21/79 oxygen/nitrogen by volume) at a nominal pressure of 100 kPa. The mesh is held at a negative potential relative to the burner in these tests. The sweeps shown in this paper extend to at least -7 kV, where the voltage is increased in small steps (e.g., of 100 V) lasting 1-2 seconds. The details of the voltage changes can be seen in the resulting plots.

Meanwhile, Table 1 is a summary log of the flames ignited, including a limited overview of the test conditions, where it is an excerpt from Appendix C. 30-Day Operational Accomplishments Report for ISS Increments 57 and 58. Note that it is comprehensive and includes the earlier testing conducted in the preceding increment pair.

Table 1.

E-FIELD Flames log of flames ignited in Increments 55-57
(*where the investigation was completed prior to Increment 58*)

Test Day[1]	**Year**	**Month**	**Date**	**Mesh Polarity[2]**	**Fuel bottle**	**Fuel conc.[3]**	**Flames ignited[4]**
First round of tests, with the coflow burner, in March to May 2018 (Increment 55)							
18073	2018	3	14	negative	100% CH4	100%	10
18078			19			70%, 100%	10
18080			21		40% CH4	40%	8
18087			28		100% C2H4	80%, 100%	10
18092		4	2		30% C2H4	30%	11
18095			5	positive		30%	10
18102			12		40% CH4	40%	8
18106			16			40%	13
18113			23		100% CH4	100%	15
18130		5	10			70%	13
18137			17		100% C2H4	80%, 100%	12
Second round of tests, with the gas-jet burner, in Sept.-Nov. 2018 (Increments 56-57)							
18269	2018	9	26	negative	40% CH4	40%	11
18274		10	1		30% C2H4	30%	11
18282			9		100% CH4	70%, 100%	13
18285			12		100% C2H4	100%	10
18290			17			80%	11
18298			25	positive	40% CH4	40%	9
18302			29		30% C2H4	30%	10
18312		11	8		100% CH4	70%, 100%	13
18319			15		100% C2H4	25%, 50%, 80%, 100%	11

[1]**Test Day** includes the year as the first two digits followed by the 3-digit day of the year.
[2]**Mesh Polarity** refers to the polarity of the high-voltage power supply and the copper electrode mesh relative to the burner.
[3]**Fuel conc.** (as diluted with nitrogen on a volume basis) was varied between tests, but typically matched the bottle. The list only accounts for the flames which ignited.
[4]**Flames ignited** is the number ignited on that day (neglecting the day change on ISS).

C. Summary of Preliminary Results

The following text is the abstract from a 2019 conference paper on early results from the first round of testing:

> *Y.-C., Chien, J. Tinajero, D. Stocker, D., U. Hegde, U. and D. Dunn-Rankin, "Ion Current and Flame Changes with Electric Fields in Microgravity," 11th U.S. National Combustion Meeting 2019, Pasadena, CA, 24-27 March 2019.*

This work presents changes in reaction zone shape and location, as well as ion currents obtained after applying an electric field onto a diffusion flame with fuel dilution in the International Space Station (ISS). As one of the projects of Advanced Combustion via Microgravity Experiments (ACME), the E-FIELD Flames experiments were initiated in March and completed in November of 2018 with a series of flame studies on the coflow burner and the jet burner within the Combustion Integrated Rack (CIR). The coflow burner, presented in this paper, consists of a small central fuel jet and a concentric outer ring flowing with synthetic air. The research examines fundamental hydrocarbon flame behaviors driven by the chemi-ion response during the combustion process. The experiment is conducted by applying high voltage onto a copper mesh situated 37 mm above the burner tip, where measurements include images and ion currents over a range of electric field strengths and time histories depending on the region of interest (e.g., near the saturation transition). This paper specially investigates the flame behavior while flowing different methane/nitrogen mixture compositions. Among other effects, the results show that the electric field can stabilize the microgravity diffusion flame at very high dilution conditions, especially when there is no coflowing air.

The preliminary results are discussed in greater detail in sections based on the burner configuration (coflow burner and jet burner), fuel (methane and ethylene), and dilution levels tested. First presented are the results from the methane flames with the coflow burner and then with the jet burner. Then, the ethylene flame results are presented. Before providing the results, it is important to clarify the terminology of polarity and field direction adopted for this system.

Field direction and polarities

The electric field is established between the two electrodes, burner and mesh, which are connected separately with positive and negative polarity, as the test requires. Because this research has no gravitationally relevant direction, any orientation can serve as the spatial reference coordinate. The vast majority of research in electric field related combustion study is conducted at earth gravity, where the buoyancy is one of the key properties considered and repeatedly discussed. In microgravity, a facing upright burner is equivalent to one facing upside down (or even sideways) scientifically because the buoyancy/gravitation force is zero. To maintain the concept where the nominal field strength coincides with the direction of an ion driven wind that results from positive ions flowing from high to low potential, the positive field direction aligns with the fluid flow direction (where the jet flows) from burner to mesh, as positive ions repel from the burner ($E_{nominal} = -\Delta V/d$); a negative electric field is the opposite direction from mesh to burner, and the positive ions move toward the burner, as indicated in Figure 7.

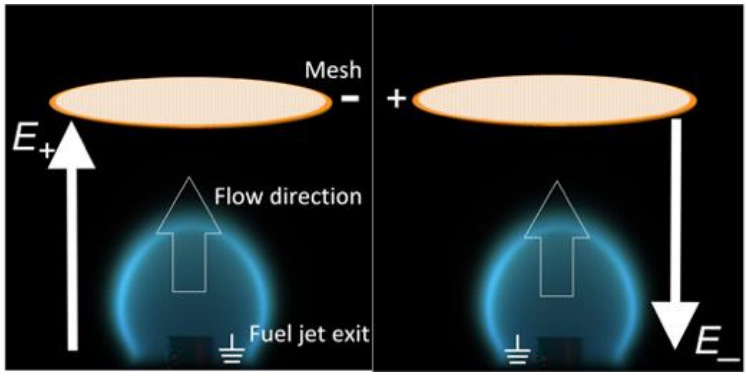


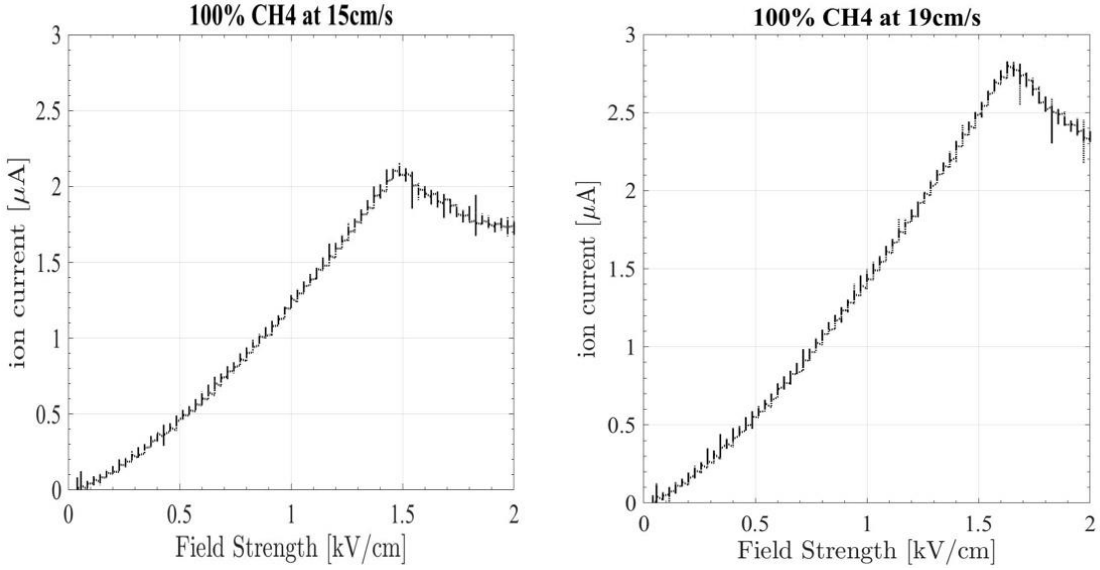
Figure 7: Illustrations of positive electric field and negative electric field directions.

100% CH_4 at 15 and 19 cm/s using the coflow burner with a positive field

As shown in Table 2, the cases presented in this section include two methane flow rates and two coflow conditions, specifically a weak coflow and no coflow where the flame relied on entrained oxygen from the chamber's atmosphere.

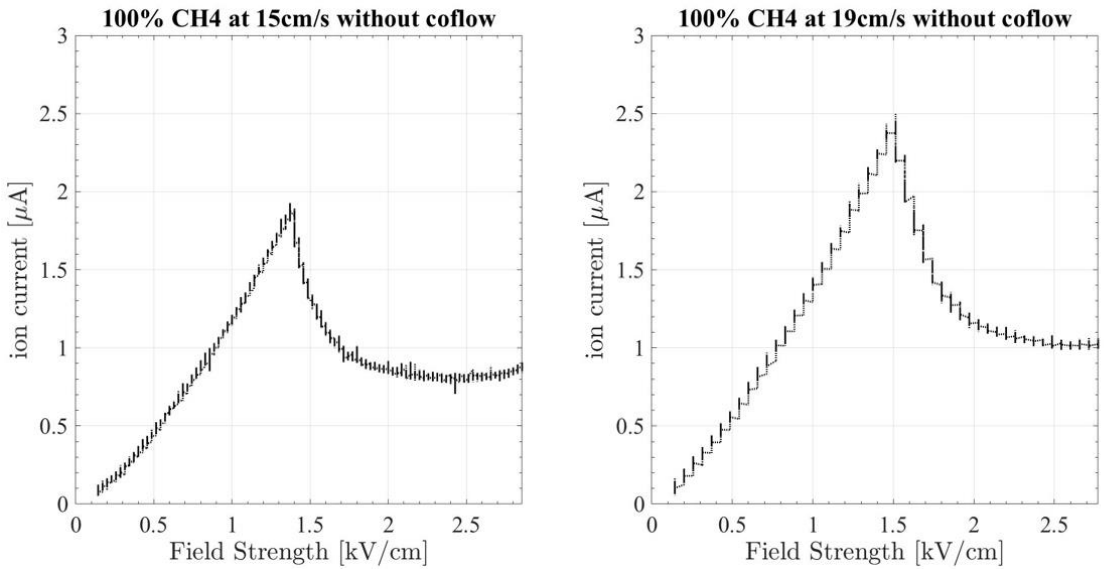
Table 2: Experimental conditions and the figure numbers with the coflow burner, negatively-charged electrode mesh, and 100% methane burning in a 21/79 oxygen/nitrogen mixture (by volume).

Coflow Velocity / Fuel Velocity	3.1 cm/s	0.0 cm/s
15 cm/s	8(a), 11(a)	9(a), 12(a)
19 cm/s	8(b), 11(b)	9(b), 12(b)



(a) (b)

Figure 8: Ion current as a function of the field strength and fuel velocity with a 3.1 cm/s coflow. (a) 15 cm/s methane. (b) 19 cm/s methane.



(a) (b)

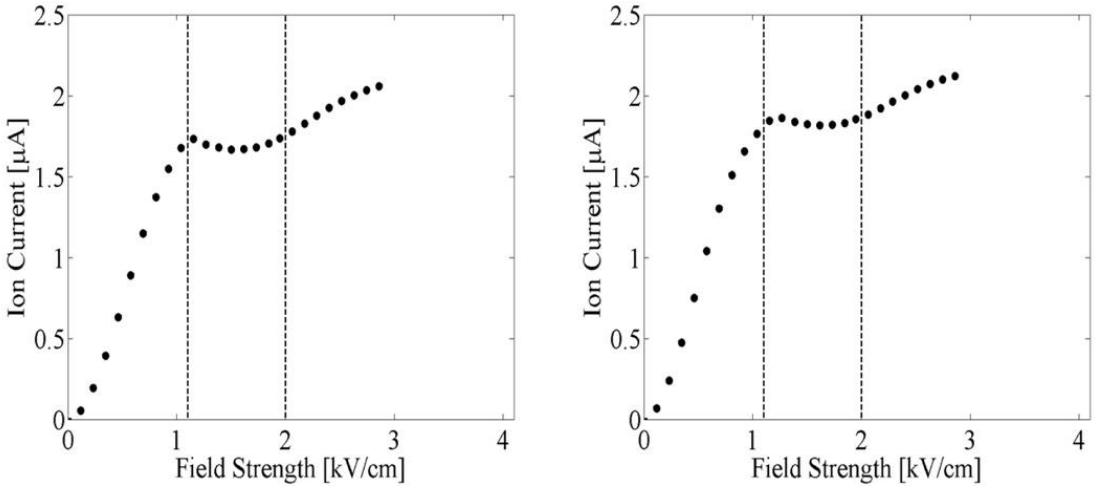
Figure 9: Ion current as a function of field strength and fuel velocity with no coflow.
(a) 15 cm/s methane. (b) 19 cm/s methane.

The first set of results show the voltage-current (V-I) curve for flames in microgravity. The curves are qualitatively similar, with a parabolic increase in ion current with increasing nominal field strength, up to a peak. Following the peak, increasing the voltage creates a distinct decrease in ion current, towards a saturation plateau, where the ion current remains constant with increasing field strength. Although the plateau is not apparent in Figure 8, it is clearly seen in all cases when a sufficient field strength is applied (e.g., Fig. 9), where the sufficient field strength depends on the fuel flow rate and coflow. In all the curves, the jagged features are spikes following a voltage step change that are mostly associated with the transient capacitance of the system when rapid steps are requested. These overshoots notwithstanding, the V-I curves follow smooth and reproducible trajectories.

For comparison, the equivalent V-I curves for 1-g flames with the same burner configuration with no coflow are shown Figure 10 [J. Tinajero, dissertation, 2015]. The results are too recent to allow a detailed analysis and discussion, but some significant observations from these results and the comparison follow.

1. The ion current peaks at approximately the same field strength for both velocities in 1-g while in microgravity the ion current peak for a lower fuel velocity occurs at a lower field strength, where the field strength at the peak current seems to scale linearly with the fuel velocity.

2. In microgravity, the ion current does not smoothly increase to saturation nor does it show a gentle peak with a mild decrease as is seen in the normal gravity results. Instead, there is a distinct peak in the ion current with a severe decrease to a saturation condition. That is, the microgravity saturation is not the maximum ion current observed. This difference is more dramatic when there is no coflow, which suggests that entrainment plays an important role whether caused by buoyancy or a forced coflow.



(a) (b)

Figure 10: Ion current as a function of field strength and fuel velocity with no coflow in normal gravity. (a) 16 cm/s methane. (b) 19 cm/s methane. [J. Tinajero, 2015]

3. The peak ion current is higher for microgravity than comparison 1-g flames. The reason is not obvious at this point but may result from a more focused upward flow from the electric body force and ion-driven wind as compared to the combined effects of the ion wind and buoyancy.

4. The peak ion current is approximately proportional to fuel flow velocity, which, as described in prior studies, reflects the relationship between ion production and the carbon flux into the system [4]. The peak ion current also increases with the coflow which suggests that healthier oxidizer transport helps increase the ion production.

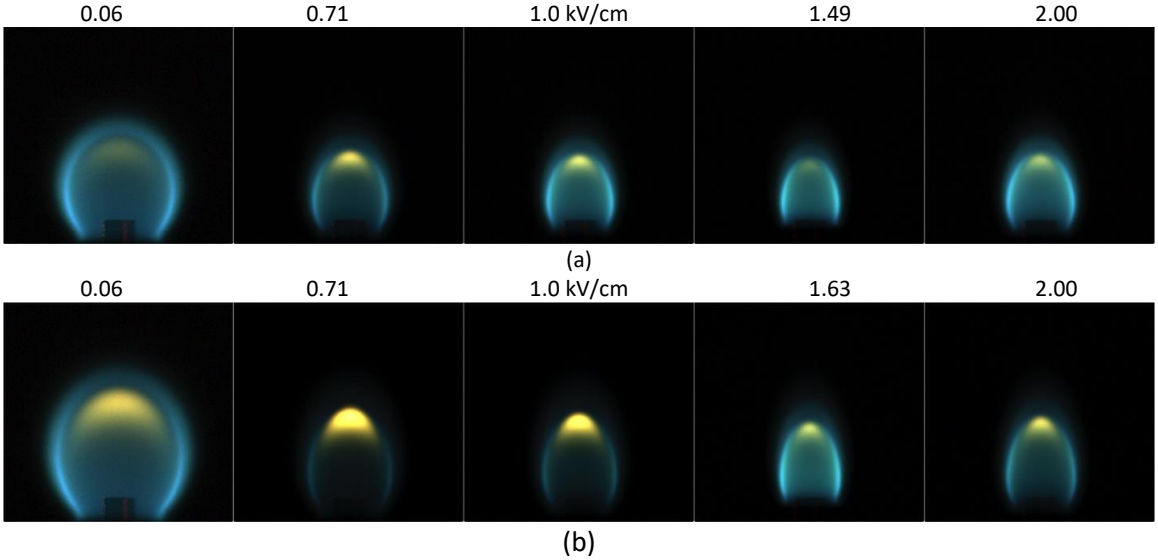


Figure 11: Flame appearance as a function of the field strength and fuel velocity with a 3.1 cm/s coflow. (a) 15 cm/s methane. (b) 19 cm/s methane.

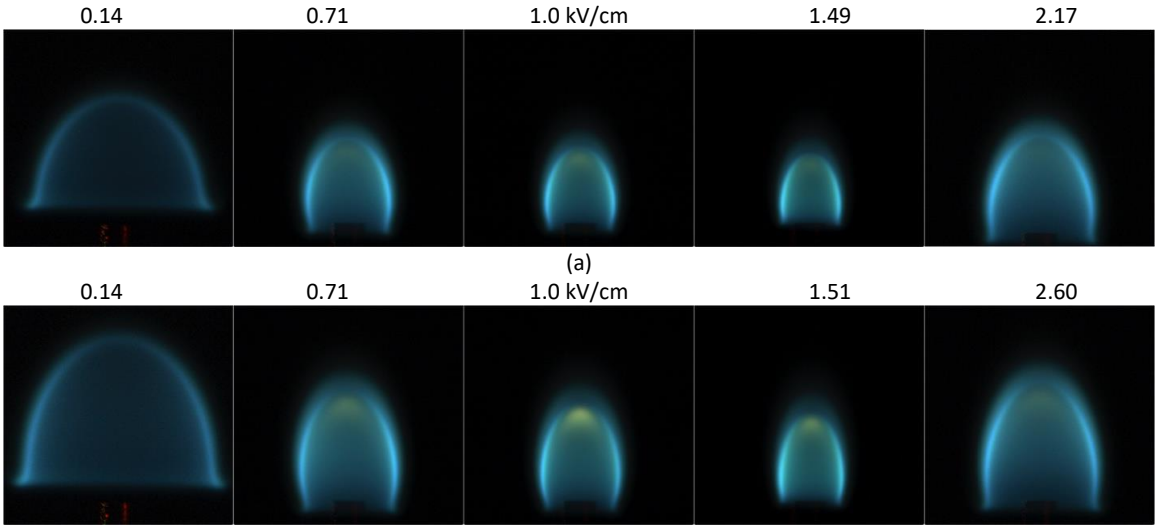


Figure 12: Flame appearance as a function of field strength and fuel velocity with no coflow. (a) 15 cm/s methane. (b) 19 cm/s methane.

To help clarify some of the ion current behaviors, Figures 11-12 show representative flame images from the color digital camera at different field strengths for the four cases presented. The flame geometry changes very little in the presence of a weak electric field. With a weak coflow, these low-momentum microgravity flames form nominally spherical shapes with a weak (or no) electric field, as seen in the leftmost images. But in the absence of the coflow, the flame broadens to reach more oxygen and lifts from the burner.

From the middle three images, we can see that as the electric field increases, the ion driven wind pulls upwards on the flame and entrains oxidizer into it. This increases the intensity of the reaction producing more ions per unit surface area and increasing the overall ion current. At field strengths greater than that resulting in the peak ion current, voltage increases cause an expansion of the flame and a decrease in reaction intensity, which drops the overall ion current. This does not happen in normal gravity because the buoyancy dominates the flame shape. The cause for the expansion in the flame size at higher voltage is not yet clearly understood, though it may be that there is a stronger radial ion flux at higher voltages. Regardless, the behavior of a peak ion current is different than that seen at normal gravity.

70% and 40% CH_4 at 19 cm/s using the coflow burner with positive and negative fields

The experimental flow concentrations in this section with different polarities and the figures showing the reflected results are tabulated as follows.

Table 3: Experimental conditions and the figure numbers with the coflow burner with no coflow, and different methane dilutions burning in a 21/79 oxygen/nitrogen mixture (by volume) environment.

Polarity / Fuel Concentration	Positive Field	Negative Field
0.7 methane, 19 cm/s	13(a), 15(a)	14(a), 16(a)
0.4 methane, 19 cm/s	13(b), 15(b)	14(b), 16(b)

The flame images of methane diluted with nitrogen at positive and negative field strengths are shown in Fig. 13 and 14, respectively. The images are taken with different shutter speeds based on the flame luminosity and where the average of several background images is subtracted. There is no coflow in the outer concentric region to enhance the air/fuel mixing, and hence the flame is relatively less stable and less bright as compared to the equivalent coflow flames, in general. At the start, a weak electric field is applied to help maintain the dim jet flame, demonstrated as the very first image of each condition (e.g., at 0.05 kV/cm). The flame quickly responds to this weak electric field force and reduces the time, and thus fuel, needed for the flame to reach a relatively steady initial condition. We call this the initial voltage, which is applied before the voltage sweep. During the initial voltage period, the electric field is held for at least two seconds and typically four seconds where the flame is nominally hemispherical (as shown in the figure) as it gradually lifts away from the burner tip. As shown in Figures 13 and 14, for both positive and negative field strengths, the ion current peaks when the flame is the most compact; specifically at field strengths of 1.27 (Fig. 13(a)), 0.86 (Fig. 13(b)), 0.38 (Fig. 14(a)) and 0.32 kV/cm (Fig. 14(b)).

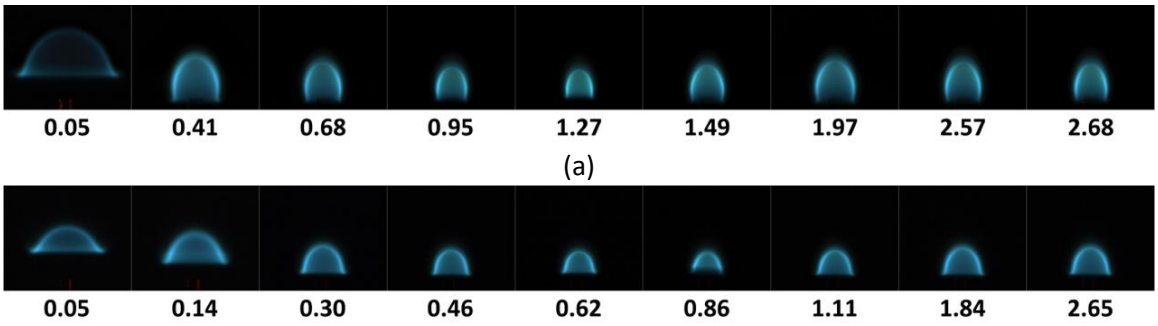


Figure 13: Flame appearance as a function of the positive field strength and fuel concentration with no coflow, (a) 70% methane. (b) 40% methane. (Unit: kV/cm)

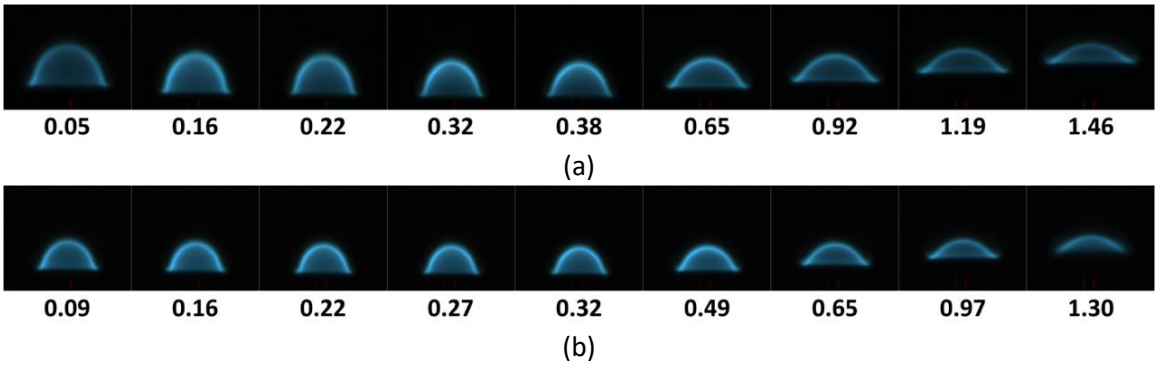
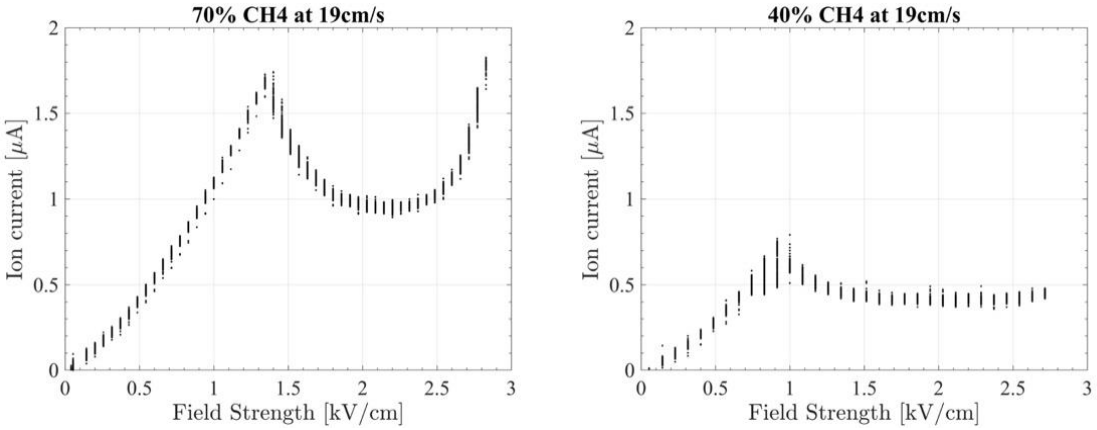


Figure 14: Flame appearance as a function of the negative field strength and fuel concentration with no coflow, (a) 70% methane. (b) 40% methane. (Unit: kV/cm)

For the positive field, the voltage-current (V-I) curve in microgravity shows a very similar trend as is observed from the very first sets of results presented in the 2018 Spring meeting of the Central States Section with the 100% methane fuel condition but with additional results from over a larger range of voltage. The ion current increases parabolically with the nominal field strength and then reaches a peak. With a further increase in voltage, the current undergoes another parabolic-like decrease and slowly increases back up again. This is shown for 70% methane in Fig. 15(a). Similar increasing parabolic behavior is seen for 40% methane (Fig. 15(b)) but with a less steep development with the field strength. The second increase of the ion current is not complete due to the low fuel concentration, power supply limits, and the fixed distance of the mesh electrode.

The flame images also relate to the ion current to field strength curve, showing that the flame reduces in size as the ion current increases and, as the current drops, the flame size increases (Fig. 13(a), 1.97 kV/cm). The flame size reduces again (Fig. 13(a), 2.57 and 2.68 kV/cm) as the ion current climbs back up close to the end of voltage-current curve, where this phenomenon is not observed in Fig. 13(b).

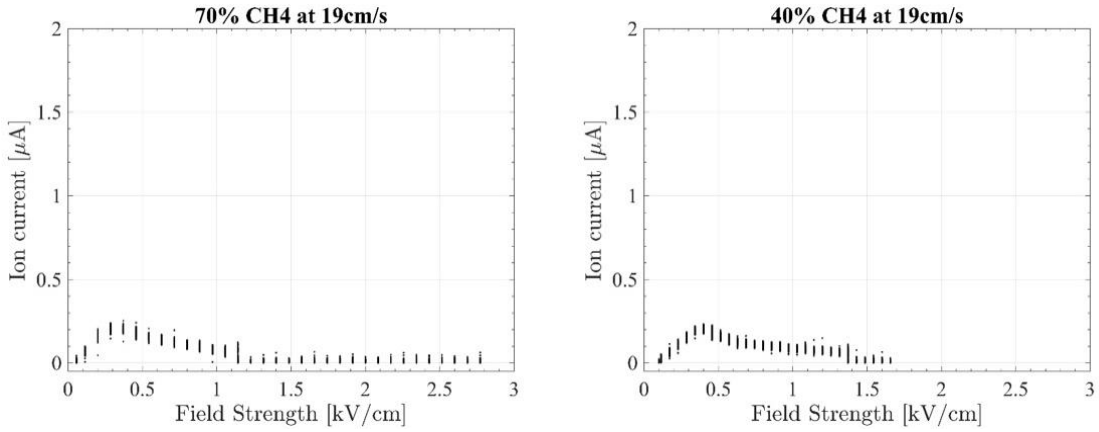


(a) (b)

Figure 15: Ion current as a function of the positive field strength and fuel velocity with no coflow. (a) 70% methane. (b) 40% methane.

With the negative field, the overall ion current remains low (below 0.3 microamps) for both the 70% and 40% methane conditions in comparison to the positive field condition. In contrast to

tests with the positive electric field, the ion current also reaches its peak sooner, at around 0.4 kV/cm, and then decays immediately with the increasing of electric field strength until the flame extinguishes. The ion current decay also corresponds to the flame appearance, shown in Fig. 14(a) and Fig. 14(b). For both fuel dilutions, the flame extends wider and the flame base is gradually lifted after the saturation peak. The flammability limit (i.e., just before flame blowout with the field) in the perspective of nominal electric field strength is 1.46 kV/cm and 1.30 kV/cm for 70% and 40% methane, respectively.



(a) (b)

Figure 16: Ion current as a function of the negative field strength and fuel velocity with no coflow. (a) 70% methane. (b) 40% methane.

This section shows the initial flame appearance and voltage-current (V-I) plots for the unique condition of a nonpremixed flame under the influence of an electric field in microgravity conditions using coflow burner (without turning on the coflow). The flame images in microgravity with no coflow are distinct in the change of size and reaction zone location while there are relatively small differences observed in normal-gravity laboratory experiments. The ion current results at low gravity do not exhibit the typical sub-saturation, saturation and supersaturation trend as recorded in terrestrial tests, suggesting that buoyancy driven flows play an important role in this characteristic ion current behavior.

100% CH_4 at 24 and 19 cm/s using a gas-jet burner with positive and negative fields

The results in this paragraph are for flow rates of 24 and 19 cm/s methane jet. The image data and ion current plots presented are preliminary results without any further polishing or detailed processing. The experimental test points with NASA data format and figure numbers are integrated in Table 4.

Table 4: Experimental conditions and the figure numbers with the jet burner, burning in a 21/79 oxygen/nitrogen mixture (by volume) environment.

Polarity / Fuel Velocity (cm/s)	Positive field	Negative field
24	18282B1; Fig. 18a	18312B1; Fig. 20a
19	18282D1; Fig. 17, 18b	18312D1; Fig. 19, 20b

Images of a methane jet flame at 19 cm/s for positive and negative field strengths, are shown in Fig. 17 and Fig. 19, respectively. The very first image of each condition, e.g., 0.05 kV/cm, demonstrates the initial voltage stage before the voltage sweep experiment. The jet flames are relatively more stable initially and less stable under high voltage than the previously reported jet flames using a coflow burner with no coflow. As the positive field is applied to the jet, the flame becomes compact in width and height. The immediate change observed is the bright soot aggregating around the flame tip. With an increase of electric field strength, the flame shrinks and the soot is less distributed at the tip. The voltage is swept at 350 volts per step and held at each voltage for two seconds. The ion current is sampled at 125 Hz (like the rest of the non-imaging data) and the monitored current fluctuation at the same voltage step is shown in the graph. The ion current to electric field strength (V-I) graph is plotted for 24 and 19 cm/s flow velocity in Fig. 18, and the current is initially increasing parabolically with the voltage applied. The ion current corresponds to the flame appearance in that the current increases and reaches a peak at the most compact flame, e.g. 1.28 kV/cm for 19 cm/s (Fig. 18b) where the soot is almost gone (Fig. 17). The flame base starts rising slightly around the current peak 1.28 kV/cm and lifts higher, with one side of the reaction wing attached to the tube, as the electric field increases further. The flame is not always lifted at the same side/edge of the burner tube, therefore, the attached reaction wing sometimes rotates along the tube edge with approximately the same lift angle under the same electric field strength but even the lifted edge flame remains relatively stable. With further electric field increase, the flame starts fluctuating and a distinct dancing phenomenon can be seen from the I-V curve where the ion current oscillates widely at the same field strength; the fluctuation is crossing more than 3.5 μA (between ~ 1.2 to 5 μA) and reaching 5.0 μA for 24 cm/s. Similarly, this oscillation at the same voltage is observed for 19 cm/s with the ion current spanning around 2.5 μA (~ 1.4 to 4 μA) and reaching 4.0 μA; the flame images are shown as 1.76 and 2.04 kV/cm in Fig. 17. These very high peak values may represent an incipient corona or discharge event. The combustion extinguishes with the oscillation at 1.84 kV/cm for 24 cm/s and 2.04 kV/cm for 19 cm/s.

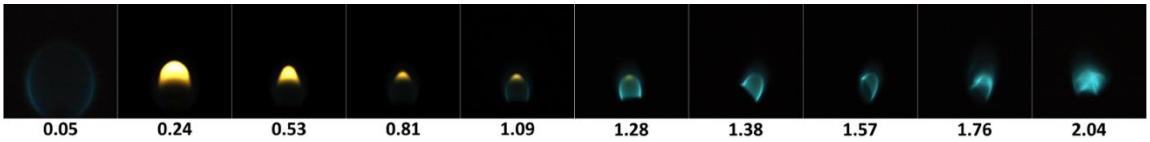
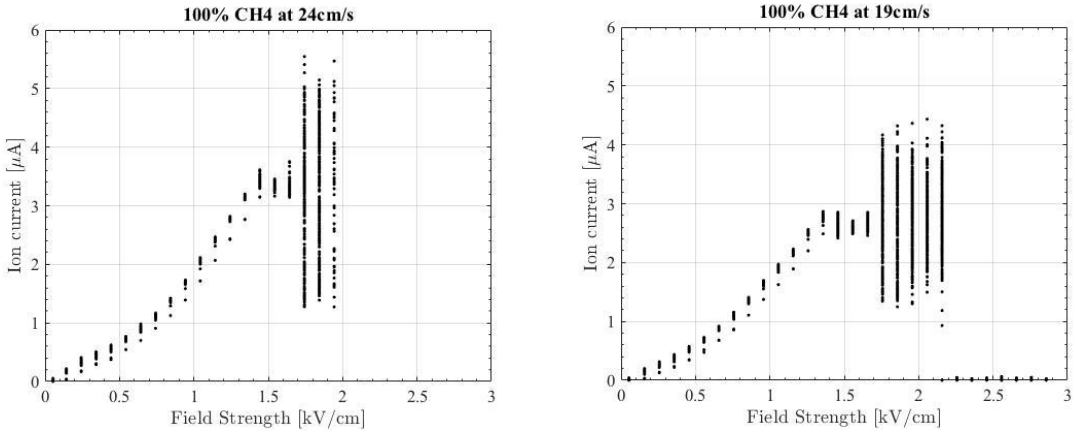


Figure 17: Flame images observed from the data camera at 19 cm/s methane jet with a positive field applied. (Unit: kV/cm)



(a) (b)

Figure 18: I-V curve at (a) 24 and (b) 19 cm/s methane jet with a positive field applied.

Under the negative field, the methane jet flame encounters the downward ion wind (around the flame) and is not sustained after 1.76 kV/cm for 24 cm/s, being blown out by the electric field. The 19 cm/s flame lasts until 1.93 kV/cm. When the flame experiences a negative field, its height decreases and the full flame is moved onto the tube, similar to a downward pointing jet flame in 1g [Papac et al., Ann. N. Y. Acad. Sci., 2006, 1077, 585-601]. The change of the flame width is not as clear or obvious as with a positive field but it still changes, shown in Fig. 19 for 0.05, 0.14, and 0.23 kV/cm. The ion current (V-I curve, Fig. 20) also increases with the field strength but differently than with the positive field without a clear curved slope. Instead, it decreases parabolically after reaching the current peak. The current peak corresponds to the most compact flame at 0.32 kV/cm for 19 cm/s and 0.41 kV/cm for 24 cm/s. The peak ion current is 0.53 and 0.55 μA, as shown in Fig. 20(b) and 20(a) respectively. Soot also appears at the flame tip but with a wider distribution along the reaction sheet. The soot reduces with increasing negative field strength, and it disappears at 0.80 kV/cm (Fig. 19) along the parabolic decay line after passing the ion current peak. The flame width and height also expand after the most compact flame (at the peak current). This trend is demonstrated in the images in Fig. 19 of 0.51 kV/cm and higher nominal field strength. The flame is pushed down around the jet burner so that it glows red. Notice also that there are several sparkler-like soot particles at 1.18 and 1.93 kV/cm. Further evaluation of this soot sparking is underway.

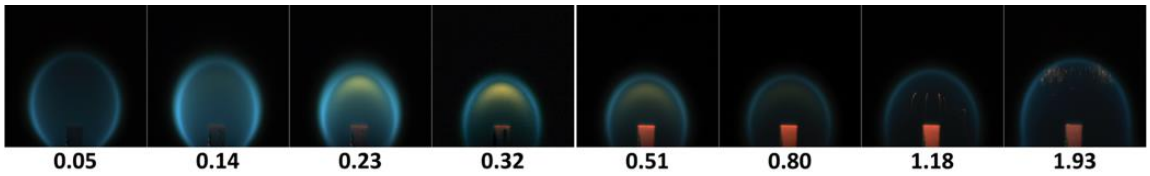
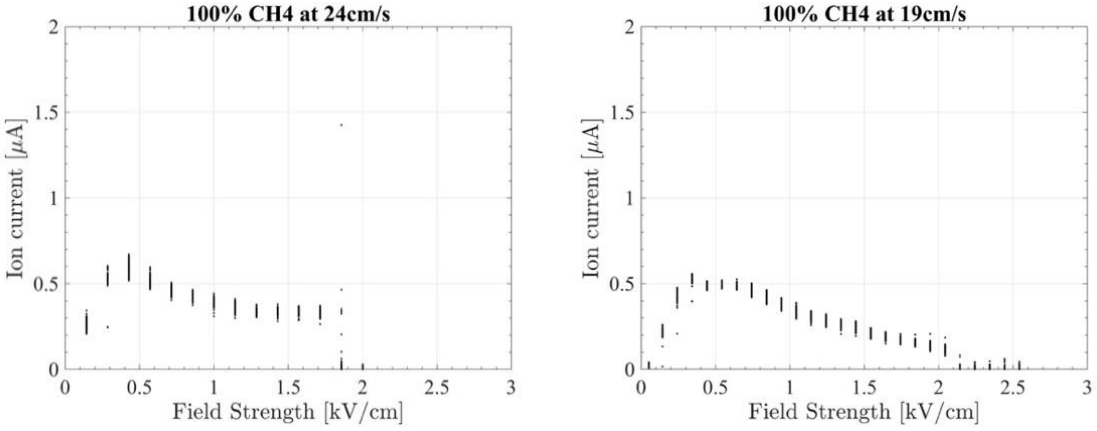


Figure 19: Flame images observed from the data camera at 19 cm/s methane jet with negative field applied. (Unit: kV/cm)



(a) (b)

Figure 20: I-V curve at (a) 24 and (b) 19 cm/s methane jet with a negative field applied.

Overall comparison to existing 1-g electric field flame results

This section includes 1-g flame images that were acquired in our 1-g laboratory in preparation for the 0-g experiment to provide an overall preliminary comparison. Figs. 21 and 22 are the coflow burner results from the microgravity electric field flames, with and without coflow, side-by-side with 1-g results [Tinajero, J. 2017 and Chien, Y.-C. 2015 dissertations]. Fig. 21 demonstrates what happens under positive field influence: the microgravity flame size, including height and width, changes with electric field while the flame height remains fairly constant in 1-g under the buoyant force influence. Fig. 22 presents flames in negative fields, and in this case the flame width increases with increasing field strength, while the width of the flame base alters in 0-g. The detailed comparison for understanding the relative effects of electric body force and buoyant force requires further evaluation and analysis.

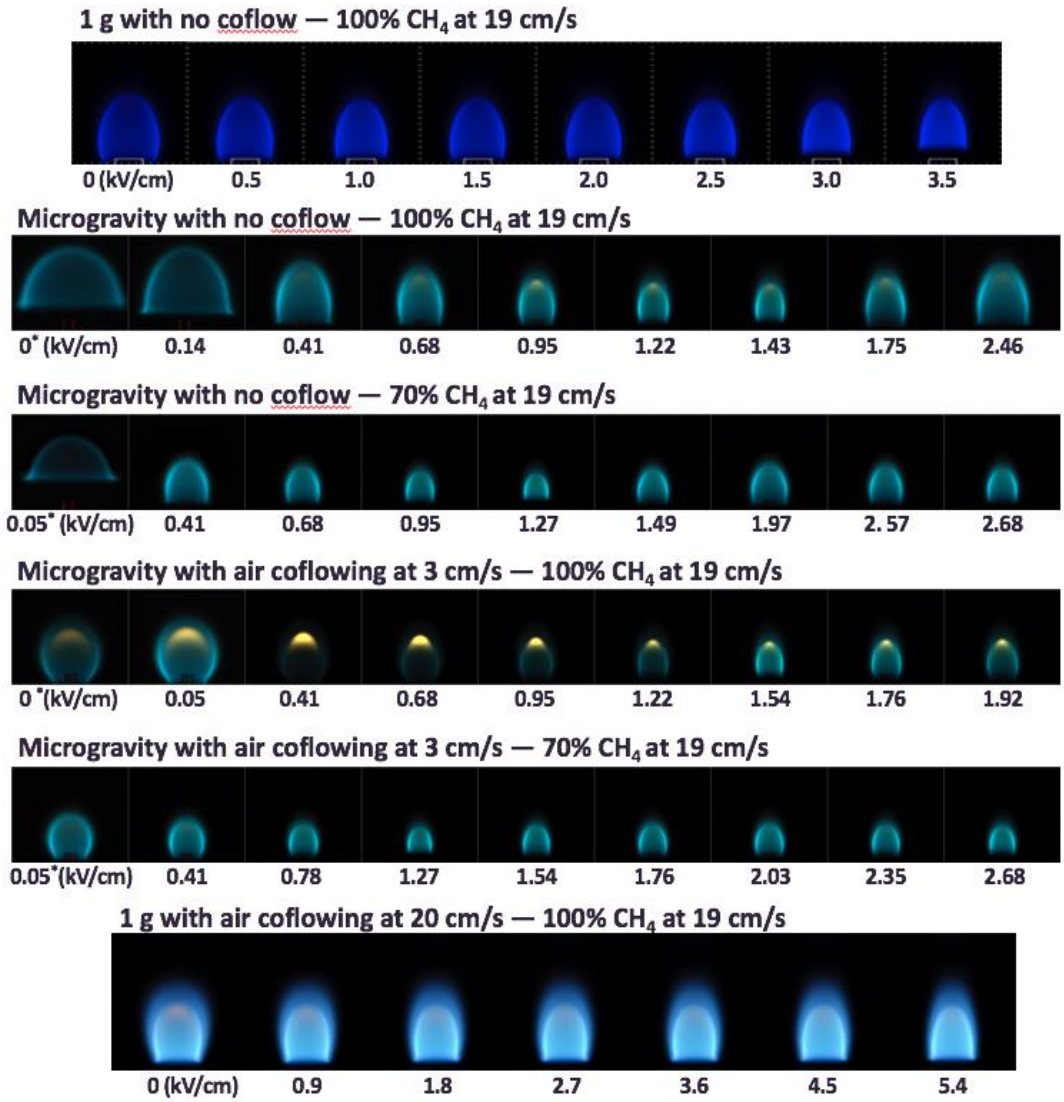


Figure 21: The overall initial comparison of 0-g flames to 1-g coflow methane flames with positive field strength. (Unit: kV/cm)

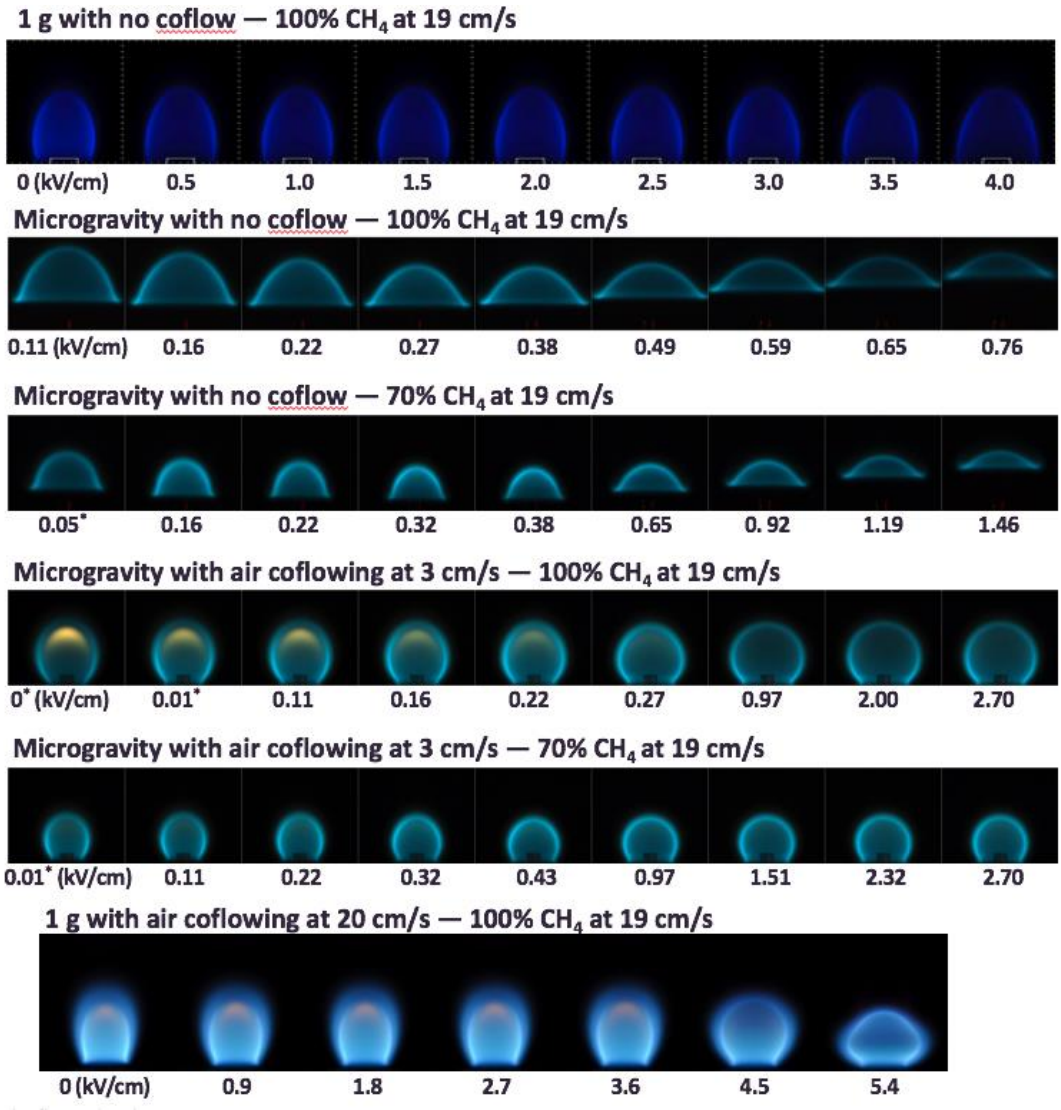


Figure 22: The overall initial comparison of 0-g flames to 1-g coflow methane flames with negative field strength. (Unit: kV/cm)

30% C_2H_4 at 15 and 19 cm/s using coflow burner applying positive and negative field

The experiment results reported above involve two types of burner and configurations using methane as fuel at different dilution percentage, but this section of the research is focused on the science of how the chemi-ions react during combustion of multi-carbon content fuel, ethylene (C2H4), under the influence of an electric field in a microgravity environment. Only the more highly diluted ethylene results are included here because the higher concentrations produce so much soot that the electric field effects on the flame are secondary to those effects on the soot directly. In some cases, we found soot buildup on the burner and the mesh. The lower ethylene concentration cases did not suffer from these complications. Positive and negative electric fields are applied onto the planar mesh oriented orthogonally and around 37 mm above the fuel jet exit.

The preliminary results of flame shape, current-voltage characteristics, and soot responses of ethylene/air diffusion flames to varying electric fields are presented in Fig. 23 and the overall comparison is shown in Fig. 24. Soot is observed reducing and disappearing when applying positive electric fields onto the mesh. Further detailed analysis is underway.

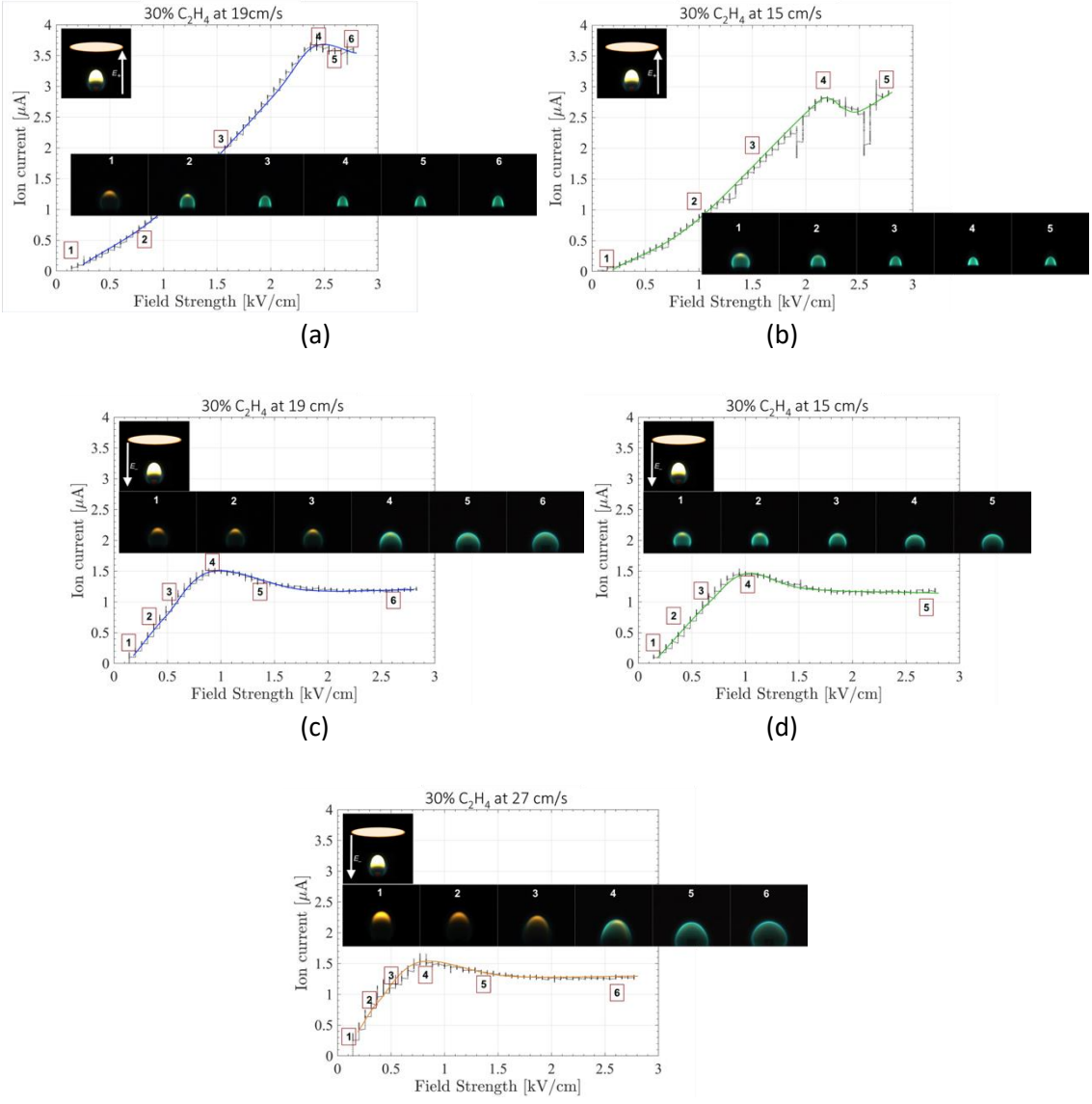
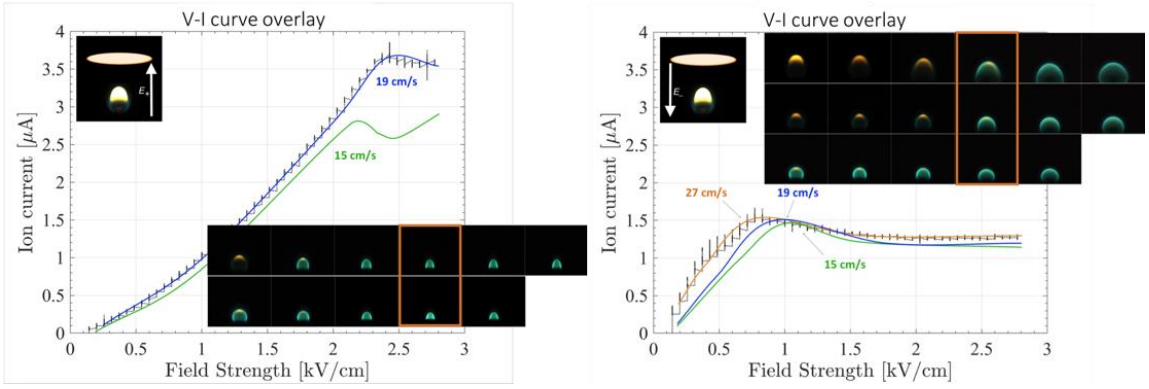


Figure 23: V-I curve with corresponded flame image of 30% ethylene (diluted with nitrogen) and flow velocities with a positive field at (a) 19 and (b) 15 cm/s; and a negative field at (c) 19, (d) 15, and (e) 27 cm/s.



(a) (b)

Figure 24: The overlay comparison of the V-I curve from Fig.23 with (a) positive field and (b) negative field. The orange box represents the flame image at peak current.

D. Surprises and New Discoveries

We expect many more discoveries to arise as we sort through the detailed data sets, but the initial surprises and discoveries include:

(1) The behavior of the V-I curve is distinctly different in microgravity as compared to 1-g flames, where in microgravity there is not a distinctive saturation ion current plateau but rather the flames exhibit a distinct peak ion current that corresponds to the most compact flame. These results show that buoyancy driven flows are an important influence on 1-g flames even when an electric field is applied.

(2) When soot is present in a flame, electric fields create a wide range of unpredictable behaviors that include elimination of the soot through what appears to be enhanced mixing for combustion, showers of luminous particles that are likely to be incandescent soot agglomerates, and unstable dancing flames. Sooting is far more significant in flames in microgravity than is seen in similar conditions with 1-g flames.

(3) There are clear correlations between flame luminosity, combustion intensity, and ion current. The results suggest that the most robust burning also produces the highest ion current, even though the flame is most compact.

(4) The very weak low fuel concentration flames were stabilized at ignition with very small electric fields, where without the fields the flames would quickly quench. This stabilizing feature of electric fields for weak zero gravity flames was a surprise, and it suggests that electric fields may be a good tool for approximating buoyancy body forces at partial gravity.

(5) Jet flames under the influence of the electric field were unstable at high field strengths, and these flames experienced corona and then arc discharges due to the large field concentration near the jet burner tip. The flame behavior was dramatic but difficult to characterize, suggesting a spatially- and time-varying field/flame interaction.

E. Accomplishment of Approved Objectives

All major science objectives were accomplished. As identified below, objectives A, B, C, and D, were completed with differing degrees of success.

A) Determine the relationship between electric field voltage and chemi-ion current as functions of total fuel (methane and ethylene) flow rate and the amount of inert (nitrogen) in the fuel.

This was accomplished through the collection of V-I curves at all experiment conditions. This objective was completed with great success.

B) Obtain the time response of the flame chemiluminescence, luminosity, and the ion current to large step changes in voltage, as functions of total fuel (methane and ethylene) flow rate and amount of inert (nitrogen) added to the fuel stream.

Although the power supplies were not capable of rapid downward voltage changes, the upward steps were sufficiently fast to permit the time response evaluation. In microgravity, the preliminary results suggest that the time response is faster than is seen in 1-g suggesting that the changes in the buoyancy driven flows play a role in the time response of these flames on earth. The scientific understanding associated with this objective was sufficient.

C) Evaluate and manipulate sooting behavior (i.e., soot volume fraction and location of luminous soot) in relation to the electric field voltage and chemi-ion current as functions of total fuel (methane and ethylene) flow rate and amount of inert (nitrogen) added to the fuel stream.

This objective was only partially completed because the key soot volume fraction instrument was unavailable to the experiment because of the obscuration of the High-Bitdepth Multispectral (HiBMs) camera by the bracket holding the electrode mesh. The location of the luminous soot was identifiable, however, so the qualitative results were completed. There is, however, the potential to use additional calibration of the ACME color camera and the associated pyrometry and total luminosity to provide the temperature and soot volume fraction from the images that were collected. This objective was met sufficiently.

D) Use the variation of the chemi-ion current as a function of applied voltage to characterize the combustion behavior of near-limit lifted flames and use the associated ion-driven wind to stabilize or reattach these flames.

Both aspects of flames mentioned in the objective were determined. As mentioned above, the electric field was definitely successful in stabilizing flames that would otherwise quench after ignition or not burn stably. There was not an opportunity to produce true near-limit lifted flames, but the effect of the ion-driven wind for generating more intense combustion and compact flames was clearly demonstrated. This objective was met sufficiently.

The formal success criteria were presented in Section 2.D. As can be seen from the science objectives completed, the project definitely met the high success criteria and partially met the complete success criteria. Complete success was only missed because, as described under the objectives, only some time response data was obtained, and there were specific cases where the ion-driven wind helped stabilize flames that would otherwise be difficult to achieve a sustained ignition. The electric field was also able to blow out flames under some conditions. The response time of the power supplies, and the inability to switch polarity during a single experiment

(different power supplies were needed) made it impossible to fully test the open loop control of the flame.

We also had a desired objective of imaging the thermal field around the flame. This information would be very valuable for providing direct imaging of the ion-driven hot gas plume. Schlieren or shadowgraph imaging was not available to the ISS experiment. One alternative would be to run a limited number of long-duration drop tests using a schlieren rig and an equivalent electric field burner, but the path to this outcome is not yet formally defined.

F. Additional Objectives Achieved

As we became more familiar with the conditions that provided the most impactful data, we reconstructed some test cases to provide more comprehensive matching of fundamental conditions. For example, it is known that flames of similar character (i.e., temperature and intensity) have saturation ion currents that scale with the input carbon to the flame. This is because the dominant chemical pathway to the key ion species includes HCO+. Hence, to examine this feature of flames under electric field influence, we ran additional conditions with matching input carbon content.

Matching carbon content with higher or lower flow rates: In addition to the core objectives achieved, experiments were run to match the carbon content of flames as a test for the role of fuel carbon in the ion-current peak production. Classical theory shows that ion current scales with carbon content for premixed flames, and some 1-g data shows similar trends for diffusion flames, but no one had determined if the concept held for zero-g flames.

Flame response hysteresis: We also obtained data for flames where the electric field was ramped up and down to determine any effects of hysteresis in the ion current or flame response. This was an additional objective achieved.

G. Impacts, Benefits, and Products

The impact of this work is, as the original objectives identified, the delivery of a unique data set of chemi-ion behavior in flames under sufficiently stable and symmetric conditions that models and theories can be evaluated against the results. As an example, the very distinctive shape of the V-I curve in microgravity, when matched in models, will give a very good indication that the feedback between ion driven wind and changes in flame shape and burning behavior that then further change the ion current is captured accurately.

These results have already garnered the interest of other groups working on electric field effects on flames.

The use of electric field effects on flames can be as sensor (variations of ion current as associated with combustion behavior) or actuator (ion-driven wind changes the flame shape and characteristics). This work, as was originally planned, provides quantitative evaluation of the sensing and response potential of electric fields in combustion. The applications can include the control of flames.

Follow-On Experiments

We hope to have a third round of testing so that we can concentrate on flames in the coflow burner configuration that we found to be the more useful and symmetric than those with the jet burner. It is likely that the more uniform electric field is responsible for this observation.

As noted above, useful results regarding the flame response time were obtained despite the slow response time of the power supply time through tests with fast upward voltage steps in some configurations. The step response is not as comprehensive as we would like, however, to confirm the links between the ion driven wind as actuator and sensor.

For objective D, we obtained very interesting information about flame stabilization and destabilization using electric fields but not in the manner we expected. Rather the destabilizing influence of the field producing fluctuations in the flame was more pronounced a phenomenon than we had seen when buoyancy was also part of the body force involved. We did not get to limit flames except that we showed clearly how in microgravity a flame can be stabilized with a small field rather than simply expanding and extinguishing. So that is some information again at the limit. We can discover more by concentrating on very low field strengths.

The outcomes of the first two rounds justify further testing because the results were so successful. The success showed how perfectly the E-FIELD Flames experiment fits the capabilities and limitations of the CIR. It is a configuration that can use the imaging capabilities at their current performance effectively. The experiments manipulate flames between quasi-steady behaviors in microgravity rather than passively observing non-steady flames evolving. The first rounds of testing spanned the experimental configuration space in terms of gases, flow rates, field ranges, and burner types. These spanning experiments mapped the phenomena but did not allow a deep dive into the conditions that showed the most dramatic changes of flame character with changes in electric field strength. In addition, excessive amounts of soot made unraveling some of the gas phase ion chemistry more complicated in the pure fuel cases, so we can now concentrate on moderate sooting cases. In general, the bonus tests let us go back to the known "best case" situations and resolve the processes (e.g., flame shape, luminosity, soot formation and disappearance, ion current). That is the matrix we proposed.

It is not within the scope of the current work, but we continue to advocate for a Schlieren imaging capability in zero-g combustion systems. The ability to image hot gas flows would benefit a range of experiments and would be particularly useful for the flames in electric field situations because alternative methods for determining flows would not be possible in the high field environments. It may be that the 5-second drop facility is appropriate for many of these companion tests.

Appendix A. Abbreviations and Acronyms

#####	ACME days are identified by year (2 digits) and the day of the year (3 digits)
#######	ACME tests are identified by day (5 digits; see above), a letter, and a 1-digit number
0-g	microgravity
1-g	normal gravity (on the surface of the Earth)
ACME	Advanced Combustion via Microgravity Experiments
ASGSR	American Society for Gravitational and Space Research
BRE	Burning Rate Emulator (*an ACME experiment*)
C2H4	ethylene
CH4	methane
CFI	Cool Flames Investigation (*where ACME uses the CFI intensified camera*)
CIR	Combustion Integrated Rack (*ISS facility used for ACME experiments*)
CLD Flame	Coflow Laminar Diffusion Flame (*an ACME experiment*)
cm	centimeter
Co-I	Co-Investigator
CWRU	Case Western Reserve University
CRS	Commercial Resupply Service (*a designation of ISS cargo missions)*
DataCam	Data Camera (*ACME's primary camera which was characterized for pyrometry)*
E-FIELD Flames	Electric-Field Effects on Laminar Diffusion Flames (*subject ACME experiment*)
EMI	Electromagnetic Interference
ERD	Engineering Requirements Document
ESA	European Space Agency
Flame Design	(*an ACME experiment*)
GC	Gas Chromatograph (*where CIR is equipped with one for chamber gas analysis*)
GigE	Gigabit Ethernet (*an interface standard for high performance cameras*)
GIPOC	Glenn ISS Payload Operations Center
GMT	Greenwich Mean Time (*where most references in this report are the day*)
HiBMs	High-Bitdepth Multispectral (*a CIR camera*)
hobj	Halcon Iconic Object (*an image format*)
HVPS	High-Voltage Power Supply
ICDERS	International Colloquium on the Dynamics of Explosions and Reactive Systems
ID	inner diameter (*e.g., of burner tube*)
IPSU	Image Processing and Storage Unit (*CIR device for recording images on ISS)*
IPSU-G	IPSU for GigE cameras (*including the DataCam and CFI intensified camera*)
ISS	International Space Station
JAXA	Japan Aerospace Exploration Agency
kPa	kilopascal
kV	kilovolt
MFC	Mass Flow Controller
mm	millimeter
N2	nitrogen
O2	oxygen
ops	operations
PARIS	Passive Rack Isolation System (*with which the CIR is equipped*)
PI	Principal Investigator
PMT	Photomultiplier Tube
PS	Project Scientist

PSI	Physical Science Informatics System
RSA	Russian Space Agency, i.e., Roscosmos
s	second
s-Flame	Structure and Response of Spherical Diffusion Flames (*an ACME experiment*)
SiC	Silicon Carbide (*material of the TFP fibers*)
tif	Tagged Image File (*an image format*)
ts	Transport Stream (*a video format*)
TDRS	Tracking and Data Relay Satellite (System)
TFP	Thin Filament Pyrometry (*diagnostic for determining flame temperature*)
UC Irvine	University of California, Irvine
VCC	voltage current characteristic (*synonymous with voltage-current*)
V-I	voltage-current (*e.g., plot of current as a function of the voltage or field strength*)
zero-g	zero gravity, i.e., microgravity
ZIN	ZIN Technologies, Inc. (*the SLICE experiment developer*)
µA	microamp

While the list of abbreviations and acronyms is long, it is not comprehensive. Contact ACME Project Scientist Dennis Stocker (216-433-2166, dennis.p.stocker@nasa.gov) for additional clarification.

Appendix B. 30-Day Operational Accomplishments Report for ISS Increments 55 and 56

15 Nov. 2018

30-day Operational Accomplishments Report

… for the International Space Station (ISS) Increments 55 and 56

ACME Advanced Combustion via Microgravity Experiments
E-FIELD Flames Electric-Field Effects on Laminar Diffusion Flames

E-FIELD Flames is one of several independent experiments that are part of the ACME project, which uses a single modular set of hardware to carry out studies of flames burning gaseous fuels in the ISS' Combustion Integrated Rack (CIR).

Prepared by the PI team and Dennis Stocker (dennis.p.stocker@nasa.gov), the ACME Project Scientist, who is the primary contact for E-FIELD Flames and ACME information.

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I. Introduction

The Electric-Field Effects on Laminar Diffusion Flames (E-FIELD Flames) experiment is conducted in the Combustion Integrated Rack (CIR) on the International Space Station, as part of the Advanced Combustion via Microgravity Experiments (ACME) project. In this experiment, an electric field with voltages as high as 10,000 volts is established between the burner and a flat, circular mesh electrode suspended above (i.e., downstream of) the burner. The motion of the charged ions, which are naturally produced within the flame, are strongly affected by a strong electric field. The resulting ion-driven wind can dramatically influence the stability and sooting behavior of the flame. Measurements are made of electric-field strength, the ion current passing through the flame, and flame characteristics such as the size, structure, temperature, soot, and stability. Conducting the tests in microgravity allows for great simplifications in the analysis of ion-driven convection and flame interaction, enabling new understanding and the development of less polluting and more efficient combustion technology for use on Earth.

An electric field can strongly influence flames because of its effect on the ions produced as a result of the combustion reactions. The induced ion wind can modify the flame shape, alter the soot and stability limits, and direct the heat transfer. The purpose of the E-FIELD Flames experiment is to gain an improved understanding of flame ion production and learn how the electric field can be used to manipulate the ions and control non-premixed flames for practical benefit. For example, electric fields can enable stable burner operation at fuel-lean conditions with reduced pollutant emission, i.e., where the flame is not reliably sustained without the electric field. As another example, the role of electric fields on soot formation can be dramatic. It appears that the ion driven wind alters the local convection and strongly influences the residence time of the soot precursors within the hot flame. The experiment contributes to our critical understanding of combustion processes in the presence of electric fields.

More specifically, E-FIELD Flames has the following experimental objectives:

A. Determine the relationship between electric field voltage and chemi-ion current as functions of total fuel (methane and ethylene) flow rate and the amount of inert (nitrogen) in the fuel.
B. Obtain the time response of the flame chemiluminescence, luminosity, and the ion current to large step changes in voltage, as functions of total fuel (methane and ethylene) flow rate and amount of inert (nitrogen) added to the fuel stream.
C. Evaluate and manipulate sooting behavior (i.e., soot volume fraction and location of luminous soot) in relation to the electric field voltage and chemi-ion current as functions of total fuel (methane and ethylene) flow rate and amount of inert (nitrogen) added to the fuel stream.
D. Use the variation of the chemi-ion current as a function of applied voltage to characterize the combustion behavior of near-limit lifted flames and use the associated ion-driven wind to stabilize or reattach these flames.

The effect of the electric field on the flame is studied through tests conducted with a range of flame and electric field conditions. The fuel, fuel jet velocity, air flow, electric-field voltage and polarity are the primary parameters of the E-FIELD Flames experiment. Tests are conducted with methane and ethylene, including nitrogen dilutions of each, which provide a broad range of soot-producing conditions, i.e., from soot-free to highly sooting. They are conducted with both a simple gas-jet burner and the coflow burner that was also used for the Coflow Laminar Diffusion Flame (CLD Flame) experiment. Tests with the coflow burner is conducted with and without a coflow of air, where the results of the latter are compared with those from the gas-jet burner to assess the effect of air entrainment on the flame. The coflow burner also provides a different electrode configuration (close to a plane-to-plane) than the jet burner (close to a point-to-plane). In a typical test, the mesh electrode is energized at increasing potentials (relative to the burner), where the flame's response (e.g., ion current, luminosity, shape) to the resulting field strength is measured. The field strength is changed in repeated steps or is instead smoothly ramped.

Microgravity studies of flames allow for a range of simplifications and provide various advantages over normal-gravity studies. For example, the flicker of flames, which is a buoyancy-driven (i.e.,

gravity-driven) instability, is eliminated yielding quasi-steady flames in microgravity. Microgravity flame flicker is then the result of electric body forces alone. Length scales are increased in microgravity flames facilitating analysis of the flame structure. Momentum-dominated flames, which are important for most practical combustion, can be studied at low velocities to simplify analysis. Microgravity flames tend to have a much stronger sensitivity to their atmosphere and exhibit a much broader range of characteristics than normal-gravity flames because of the near absence of buoyant entrainment. The long residence times in microgravity flames can lead to strong soot production, but many microgravity flames are soot free. The absence of buoyancy makes microgravity flames advantageous for studies of limit and stability behavior where chemical kinetics are important. With these advantages, microgravity testing can lead to new understanding of the complex nature of combustion – regardless of the gravitational environment of the combustion application. Chemi-ion species, electronically excited chemiluminescent species are very low concentration in typical flames, making the reduced buoyancy environment valuable for evaluating the minor species chemistry.

II. Methods/Research Operations

A. Discussion of Method/Protocol

E-FIELD Flames and other ACME experiments are configured, but not conducted, by the ISS crew. The crew's role is to set up the hardware, gas bottles, etc. for each set of tests. The most common changes include the gas bottles on the Combustion Integrated Rack (CIR) and the mass flow controllers on the modular ACME chamber insert.

All testing is commanded from the ground, specifically from the Glenn ISS Payload Operations Center (GIPOC). The science portion of the testing is typically conducted in a nominally automated mode using pre-programmed scripts. However, it may alternately be conducted in a 'manual' mode, where changes are made to a control parameter in response to the analog video downlink and the numerical data available in the telemetry stream. For example, the mesh voltage can be adjusted to approach a flame's stability or soot limit. This was not done during Increments 55-56, where the console commanding during a test was typically limited to halting the test when it was observed in the analog ops video that the flame had extinguished.

Testing is not conducted when thrusters fire or spacecraft dock or undock from ISS to ensure an acceptable acceleration environment.

Data is downlinked between each set of tests, i.e., day of testing, to allow for preliminary analysis and requisite planning for subsequent testing.

III. Results

A. Baselined Activities Completed – Pre-flight, In-flight, Post-Flight

A total of 142 flames were ignited for the E-FIELD Flames experiment during Increments 55 and 56. The first round of testing occurred from March to May 2018, which was followed by a second round, which began in September 2018 and was still underway at the beginning of Increment 57. The first round of tests resulted in 120 ignited flames conducted with the same coflow burner used by the Coflow Laminar Diffusion Flame (CLD Flame) experiment, where CLD Flame's second round occurred between the two rounds of the E-FIELD Flames experiment. The 'interruption' in the E-FIELD Flames testing was intentionally planned years in advance to enable the PI team to review the initial results in detail and with that knowledge plan the second round of tests to maximize its scientific outcome.

E-FIELD Flames does not have living subjects like a life science experiment, but the tests are independently conducted with two different gaseous fuels, methane and ethylene. Both are simple hydrocarbon gases for which the chemistry is well understood. Methane (CH_4) is the primary component of natural gas and tends to produce low levels of soot given the absence of carbon-carbon bonds. Meanwhile, ethylene (C_2H_4), is highly sooting because of its carbon double bond. For both fuels, tests were conducted with pure fuel and fuel mixed with nitrogen over a broad range of dilutions. For limited cases, the fuel and nitrogen was premixed on the ground prior to launch. ACME bottles were launched with methane concentrations of 30%, 40%, 45%, and 100% with a balance of nitrogen (on a volume basis). Similarly, bottles were launched with ethylene concentrations of 30%, 50%, and 100%, again with a balance of nitrogen (on a volume basis). But the ACME hardware also allows for simultaneous flow of fuel and nitrogen to the burner, allowing for on-orbit fuel dilution and testing at a continuum of fuel concentrations. In all cases, the coflow was 21% oxygen with a balance of nitrogen (on a volume basis).

There are no samples (e.g., of soot) collected for E-FIELD Flames, where all of the ISS results are electronically transmitted to the ground. That is furthermore the case for all currently planned ACME experiments.

A summary log of the E-FIELD Flames ignited for E-FIELD Flames during Increments 55 and 56, including a limited overview of the test conditions, is found on the next page in Table 1.

Table 1.

E-FIELD Flames log of flames ignited in Increments 55 and 56							
Test Day[1]	**Year**	**Month**	**Date**	**Mesh Polarity[2]**	**Fuel bottle**	**Fuel conc.[3]**	**Flames ignited[4]**
First round of tests, with the coflow burner, in March to May 2018 (Increment 55)							
18073	2018	3	14	negative	100% CH4	100%	10
18078			19			70%, 100%	10
18080			21		40% CH4	40%	8
18087			28		100% C2H4	80%, 100%	10
18092		4	2		30% C2H4	30%	11
18095			5	positive		30%	10
18102			12		40% CH4	40%	8
18106			16			40%	13
18113			23		100% CH4	100%	15
18130		5	10			70%	13
18137			17		100% C2H4	80%, 100%	12
Start of the second round of tests, with the gas-jet burner, in Sept.-Oct. (Increment 56)							
18269	2018	9	26	negative	40% CH4	40%	11
18274		10	1		30% C2H4	30%	11

[1]**Test Day** includes the year as the first two digits followed by the 3-digit day of the year.
[2]**Mesh Polarity** refers to the polarity of the high-voltage power supply and the copper electrode mesh relative to the burner.
[3]**Fuel conc.** (as diluted with nitrogen on a volume basis) was varied between tests, but typically matched the bottle. The list only accounts for the flames which ignited.
[4]**Flames ignited** is the number ignited on that day (neglecting the day change on ISS).

Pre-flight and post-flight activities are generally not addressed in this report.

B. List of Preflight, In-flight, Postflight Anomalies

1. Mission anomalies

The primary hard drive for the CIR's IPSU-G Image Processing and Storage Unit - for Gigabit Ethernet (GigE) cameras – failed to boot on April 27, 2018 (18117). For ACME experiments, the IPSU-G is used for two key cameras, the ACME color digital 'data' camera (i.e., DataCam) and the Cool Flames Investigation (CFI) intensified camera, which is used for CLD Flame with a narrow bandpass filter to image CH* chemiluminesence. Use of the IPSU-G was recovered by astronaut Scott Tingle on May 7 (18127) by the removal of that primary drive. The removed drive was reinstalled on Aug. 14 (18226) by astronaut Ricky Arnold during the second round of CLD Flame testing. Although the hard drive was installed in the secondary position, the IPSU-G surprisingly again became unusable. Therefore, it was removed on Aug. 22 (18234) by astronaut Alex Gerst, returning it to use, albeit without the intended storage capacity.

The CIR's IPSU-1 Image Processing and Storage Unit, which is used with the High Bit-Depth Multispectral (HiBMs) camera and the collimated illumination package, was only intermittently operable in mid-Sept. 2018 and was not operable at the start of E-FIELD Flames' second round. Those capabilities were not seen as critical for E-FIELD Flames, where the illumination is significantly blocked by the mounting bracket for the mesh electrode. Therefore, the tests on Sept. 26 (18269) and Oct. 1 (18274) were conducted without HiBMs imaging or the CIR illumination. It was later determined to be an issue with a cable connection, which was corrected by astronaut Alex Gerst on Oct. 22 (18295) when he carried out other ACME tasks.

2. Any other difficulties in obtaining scientific data

E-FIELD Flames tests sometimes automatically abort (due to the controlling software script), where that is typically prior to gas flow and the activation of the igniter. As a result, there is little lost other than the time in the failed test attempt. The aborts normally occur because of communication issues with IPSU-1. These are infrequent, but there are some days when tests abort repeatedly.

Tests are of course also unsuccessful when the flame fails to ignite. The first test of the day has sometimes failed to ignite, presumably because the fuel line has not been sufficiently primed. Ignition is sometimes unsuccessful because the settings for the automatic flame detection are such that detection threshold is reached because of the igniter glow rather than a flame. As a result of the detection, the hot-wire igniter is deactivated and retracted, which is of course an issue if the flame has yet to ignite. This is typically avoided by detecting ignition with measurement of the flame's OH* chemiluminescence with a Photomultiplier Tube (PMT) module. The OH* chemiluminescence, in the ultraviolet at 310 nm, is a good indicator of an ignited flame rather than the glowing igniter, which emits primarily in the infrared and visible spectrum. However, the combination of igniter settings and PMT gains can lead to false ignition detection.

ACME's Mass Flow Controllers (MFCs) used to control the flow of fuel, nitrogen, and synthetic air (i.e., 21% oxygen with a balance of nitrogen, on a volume basis) have unfortunately not always performed adequately. The fuel and especially nitrogen flows sometimes oscillate. Furthermore, the MFC used for the 'air' coflow behaves as if it is sticking in a few places, where the flow rate won't linearly ramp when commanded (as required for the CLD Flame experiment). It tends to stick in some places and then abruptly overshoot where that has sometimes been seen to extinguish the flame. As a result, mass flow controllers are tested - without flames - after they are first installed. Adjustments are often then made to (1) the settings that control when the valves open and (2) the Proportional-Integral-Derivative (PID) settings. This activity has involved significant testing, separate from flame tests, and has taken place during CLD Flame operations on 17333, 17334, 17347, 17354, 18008, 18022, 18038, 18046, 18143, and 18241. In this regard, there are an assortment of MFCs with different flow ranges for each of the three gases. While the MFC tuning

has been beneficial in some cases, it has not eliminated all gas flow problems. It was seen that nitrogen oscillations tend to damp with time, so a nitrogen pre-flow was implemented so that the oscillations would typically end before the flame measurements. That approach has been partially successful and was introduced during E-FIELD Flames' first round. Later, a procedure of re-zeroing the MFCs prior to each test was also implemented during E-FIELD Flames' second round. A replacement MFC for the 'air' coflow is being prepared for spaceflight, where a limited third round of CLD Flame testing is being planned to accomplish its extinction limit objective.

The final day of testing during E-FIELD Flame's first round was conducted with highly sooty ethylene flames and a positively charged mesh, which tends to push the flame down toward the burner. A significant amount of soot deposited on the burner, both on the inner tube from which the fuel issues as well as the honeycomb in the burner annulus from which the air flows. The extent of the soot wasn't appreciated until flame testing for CLD Flame's second round began on May 25 (18145). Additional evaluation tests were performed on May 31 (18151), and the decision was made to have the crew clean the soot from the burner because it was clear that the gas flow from the burner was affected by the soot buildup. A cleaning procedure was developed, approved, and carried out by astronaut Ricky Arnold on June 19 (18170). The cleaned burner was imaged with the CIR cameras on June 21 (18172), but the evaluation testing was deferred because of a cable mating issue. The cable mating was corrected by Arnold on June 25 (18176) and the evaluation testing of the cleaned burner was conducted on the following day, June 26 (18177). Regular flame testing in CLD Flame's second round finally commenced on July 5 (18186), six weeks after the soot accumulation was identified as an issue. However, useful scientific results were obtained on the June 26 evaluation day, so the burner cleaning effectively led to a 1-month delay.

C. Completeness/Quality of Data

The number of flames ignited significantly exceeds the planned tests which were (1) specified as required in the ACME Integrated Science Requirements Document (ISRD), using the gas-jet burner, and (2) later expanded to include tests with the coflow burner to partially make up for science loss resulting from the slow response time of the high-voltage power supply. Following the revision, a total of 74 E-FIELD Flames tests were identified as required. In contrast, 120 flames were ignited in the experiment's first round, i.e., first half. In that round, roughly 200% more flames were ignited than had been planned.

It should be understood that a complete set of image data is not available for all tests. As discussed in section III.B, the IPSU-1 was not working at the beginning of the second round of testing, preventing use of both the HiBMs camera and the CIR's collimated illumination package. Those are not critical for E-FIELD Flames, so tests continued without their use.

Furthermore, recorded image data is sometimes deleted without downlinking. This is done because of operational constraints which limit the amount of data that can be managed per test day. Given the constraint, the decision is sometimes made to sacrifice some image data (e.g., from the HiBMs camera) to allow additional tests to be conducted. This is intentionally done in order to maximize the scientific return of each day of testing within the project constraints.

Data quality is of course dependent on the experimental hardware. But neglecting that limitation on data quality, the cameras are sensitive to radiation where pixels gradually become 'hot' over time. While these bright spots are addressed in image post-processing, they are a degradation of the image quality.

It is possible that the tests are subtly affected by the acceleration environment, but there is not obvious indication of this and it has thus far not been evaluated with available measurements from the Space Acceleration Measurement System (SAMS). In this regard, note that the Combustion Integrated Rack (CIR), in which the experiment is conducted, is equipped with a Passive Rack Isolation System (PaRIS).

Regardless of the shortcomings discussed above, a great wealth of experimental data has been secured and E-FIELD Flames is considered to be highly successful.

D. Status of Data Analysis

Data from the E-FIELD Flames experiments includes video recordings from the operations camera for general flame behavior, the ACME data camera for high resolution images at various exposures and frame rates as dictated by the flame luminosity, intensified camera images and non-imaging data at 100 Hz. The non-imaging data is extracted using Matlab scripts (modified from those developed by CLD Flame researchers) to show trends of ion current and flame luminosity as a function of the applied field, and the flame parameters (fuel, flowrate, burner type). Image data is processed using the OMA software (also developed and used by CLD Flame researchers).

The overall data volume is large with a storage limit of 127,000 MB per test day, the data has been downloaded to the hard drive with a backup system at UC Irvine. At this stage, with operations ongoing, the analysis is comparative and qualitative, just to show the major differences in behavior in microgravity and normal-gravity (1-g) flames.

E. Preliminary Research Findings

Results from the early E-FIELD Flame experiments have demonstrated clear differences between normal-gravity and microgravity flame behaviors. A few examples include the distinct maximum of ion current for co-flow burner flames in microgravity, where the 1-g flames more smoothly reach a saturation condition. The results also show that the electric body force can produce flame oscillations similar to those driven by buoyancy in normal gravity, and microgravity flame shapes with an electric field can mimic normal-

gravity flames without electric fields. Burner type plays a role in the flame/field interaction, and in particular, reversing the field polarity for the jet flame does not produce an inverted result over the entire voltage range tested. Results also demonstrate the interactions between electric fields and soot are more complex in microgravity. At the same time, the preliminary findings suggest that some of the key expectations generally hold (e.g., ion current scales approximately with flow rate for a fixed fuel and burner).

100% methane for positive field strength with and without coflowing air

This section only includes representative findings as the ISS experiments covered a wide range of conditions and outcomes. Color camera images, intensified camera images, total flame luminosity, and ion current, along with all test parameters and flow monitoring, were recorded in all cases. Figure 1 gives example flame images from the first tests conducted with methane as fuel. Two fuel flow rates are shown, with the voltage indicated as nominal field strength (kV/cm). Since the physical spacing from the burner to the mesh is 37 mm, 2 kV/cm represents 7000 V across the flame.

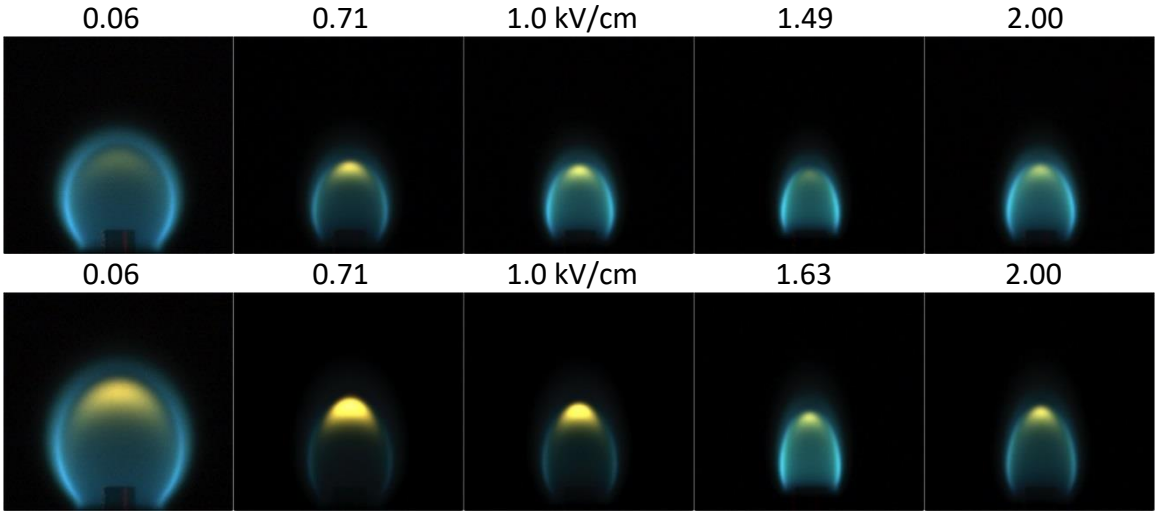


Figure 1. Typical flame appearance as a function of the field strength and fuel velocity with a 3.1 cm/s coflow. (a) 15 cm/s methane. (b) 19 cm/s methane.

The key measurement for the E-FIELD Flames experiments is the ion current as a function of applied voltage, i.e., the voltage-current or V-I curve. This curve provides information on the flames ability to produce chemi-ions based on its saturation ion current, and it provides a measure of combustion performance in that ion production is a function of carbon influx into the flame and the vigorousness of the combustion process. More compact, higher temperature flames produce more chemi-ions than do larger more diffuse reaction zones of lower overall temperature for the same fuel flow. The first effect (i.e., the relationship between carbon influx and peak ion current) can be seen in Figure 2, where the flow of 15 cm/s methane produces slightly above 2 microamps of ion current at the peak (1.5 kV/cm) and the flow of 19 cm/s methane has a peak of 2.75 microamps. While the ratio is not exact, the peak ion current is nearly proportional to the flow

velocity. Another interesting feature of these curves is the peak value. Traditional electrical aspects of flames studies look for a saturation ion current plateau, where the ion current remains fixed despite an increase in applied electric field. For these microgravity flames, there is a pronounced peak in the ion current at saturation and then a gradual decrease to a global saturation plateau. This is a strikingly different behavior than is seen in 1-g, and the behavior is even more pronounced for the case where there is zero coflow.

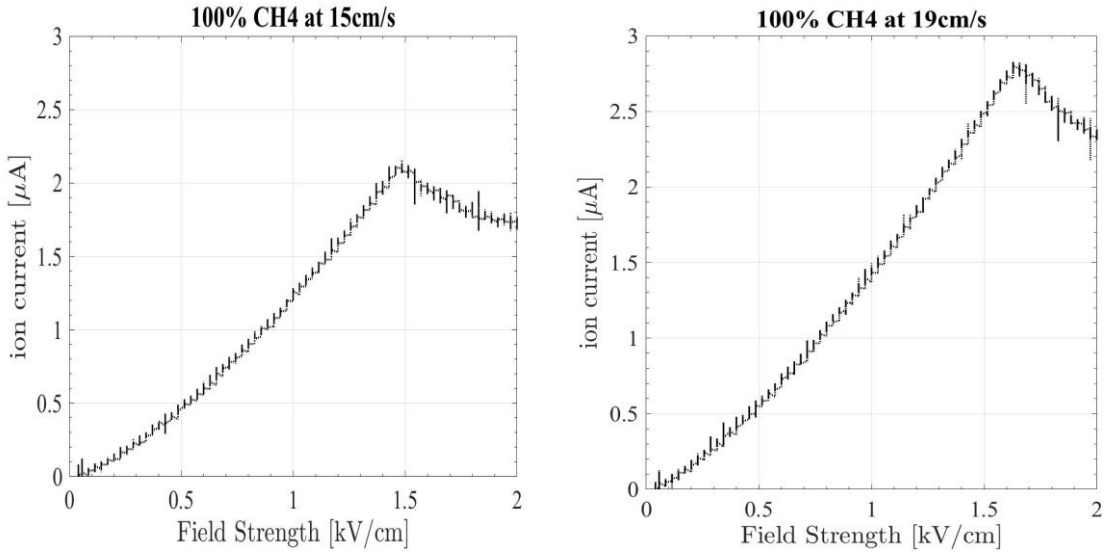


Figure 2. Ion current as a function of the field strength and fuel velocity with a 3.1 cm/s coflow. (a) 15 cm/s methane. (b) 19 cm/s methane.

As can be seen in Figure 3, which is the same V-I curve result in microgravity for the case where there is no coflowing air, and Figure 4 shows the flame images. These figures again show a reasonably linear scaling of peak ion current with fuel flow velocity, but they demonstrate a very pronounced peak ion current with a rapid drop to a global saturation ion current plateau. Since these are diffusion flames, essentially all of the fuel is being burned independent of the flame's shape and the applied electric field, but it seems clear that the electric field is affecting the chemi-ion production capability of the flame. As is apparent from the flame images, the flame is most compact at the voltage consistent with the peak ion current. With increasing voltage, the flame becomes less compact and the ion current decreases. We presume that the flame remains at saturation throughout this period (that is, the flame is producing as many ions as it can, and those ions are being removed from the reaction zone as soon as they are created) so the decrease in ion current results from a change in combustion intensity. We can explore this hypothesis further using the PMT data collected in the experiments.

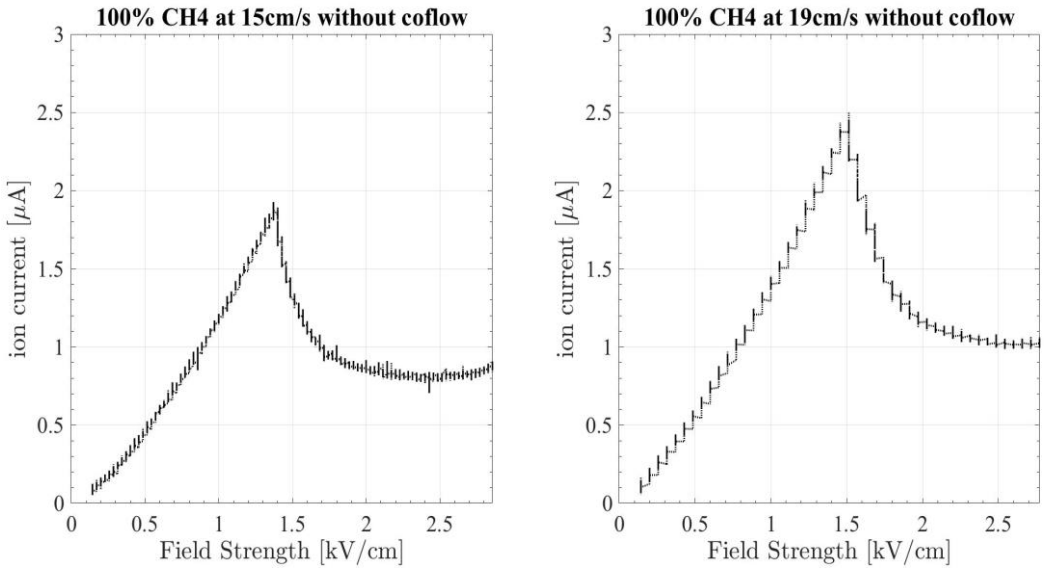


Figure 3. Ion current as a function of field strength and fuel velocity with no coflow. (a) 15 cm/s methane. (b) 19 cm/s methane.

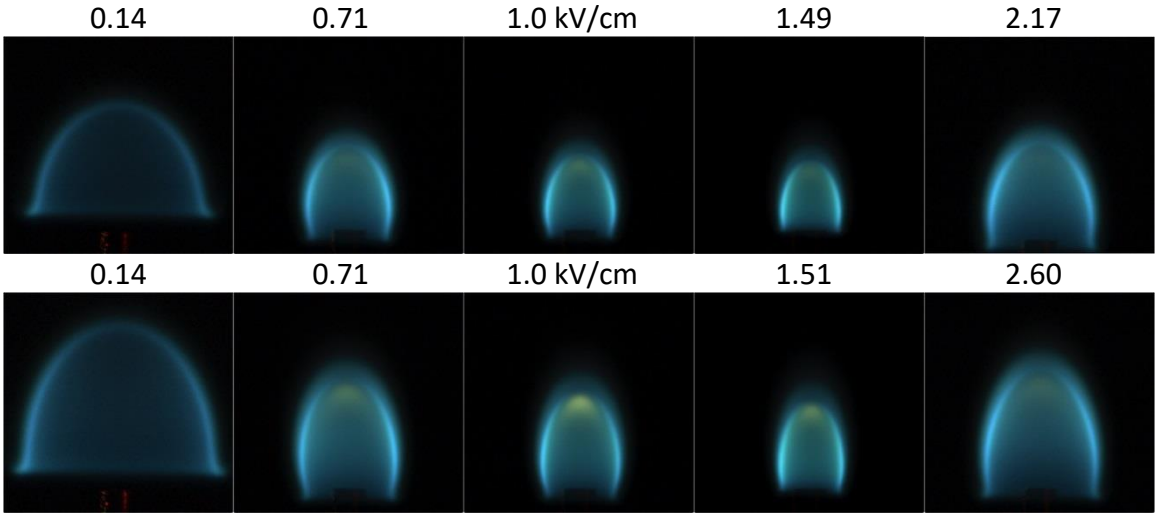


Figure 4. Flame appearance as a function of field strength and fuel velocity with no coflow. (a) 15 cm/s methane. (b) 19 cm/s methane.

In all of the curves, the jagged features are spikes following a voltage step change that are mostly associated with the transient capacitance of the system when rapid steps are requested. These overshoots notwithstanding, the V-I curves follow smooth and reproducible trajectories.

For comparison, the equivalent V-I curves for 1-g flames with the same burner configuration with no coflow are shown in Figure 5. The results are too recent to allow a detailed analysis and discussion of the differences, but some significant observations from these results and the comparison follow.

(1) The ion current reaches a peak at approximately the same field strength for both velocities in 1-g while in microgravity the ion current peak for a lower fuel velocity occurs at a lower field strength. The field strength at the peak current seems to scale linearly with the fuel velocity.

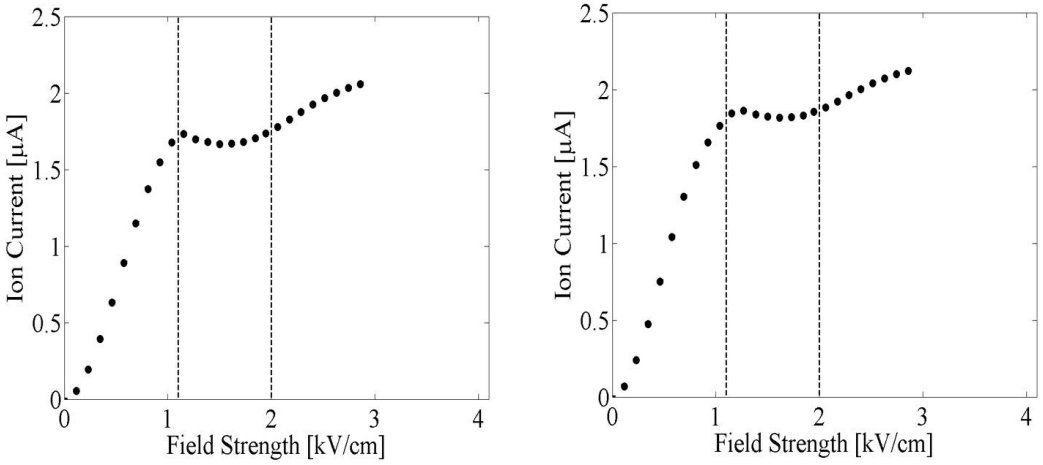


Figure 5. Ion current as a function of field strength and fuel velocity with no coflow in normal gravity. (a) 16 cm/s methane. (b) 19 cm/s methane.

(2) In microgravity, the ion current does not smoothly increase to saturation nor does it show a gentle peak with a mild decrease as is seen in the normal gravity results. Instead, in microgravity there is a distinct peak in the ion current with a severe decrease throughout the saturation condition, with a final saturation plateau observed at voltages substantially above the voltage at peak ion current. That is, the microgravity final saturation is not the maximum ion current observed. This difference is more dramatic when there is no coflow, which suggests that entrainment plays an important role whether caused by buoyancy or a forced coflow.

(3) The peak ion current is higher for microgravity than comparison 1-g flames. The reason is not obvious at this point but may result from a more focused upward flow from the electric body force and ion-driven wind as compared to the combined effects of the ion wind and buoyancy.

(4) The peak ion current is approximately proportional to fuel flow velocity, which, as described in prior studies, reflects the relationship between ion production and the carbon flux into the system. The peak ion current also increases with the coflow which suggests that healthier oxidizer transport helps increase the ion production.

30% ethylene with coflowing air at 3.1cm/s

The results from ethylene flames with coflow are shown in Figures 6 and 7, positive field and negative field respectively. At 15 cm/s, the ion current reaches a peak around 2.7 microamps which is similar to the 19 cm/s 100% methane coflow air case but with only 0.4 of the volumetric carbon content. The results for 15 cm/s also presents a rapid parabolic behavior after reaching the saturation peak. In the negative field case, the peak ion current decreases to 0.56 of the positive field case at the same flow rate 19 cm/s (or 0.4 for 15 cm/s). Another aspect is that the ion current peaks sooner for the higher flow rate/carbon content in the negative field configuration as compared to the positive field experiments. These phenomena, particularly for the positive field strength have not yet been measured in the laboratory at 1-g. The difference in behavior appears to be related to the removal of the buoyancy force so that the momentum flow coupled with the ion driven flow tends to resettle under the electric field body force. Detailed analysis is underway.

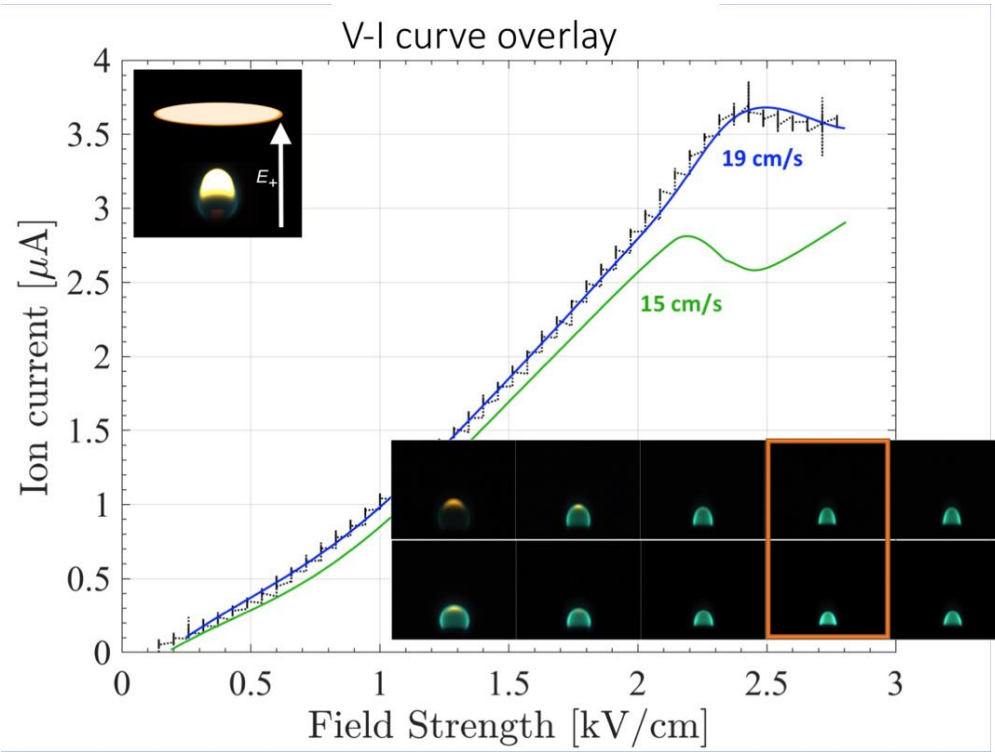


Figure 6. Ion current to electric field strength per unit distance for positive field of 100% ethylene with coflow air. The ion current peaks around 3.5 microamps at 19 cm/s and 2.7 microamps for 15 cm/s.

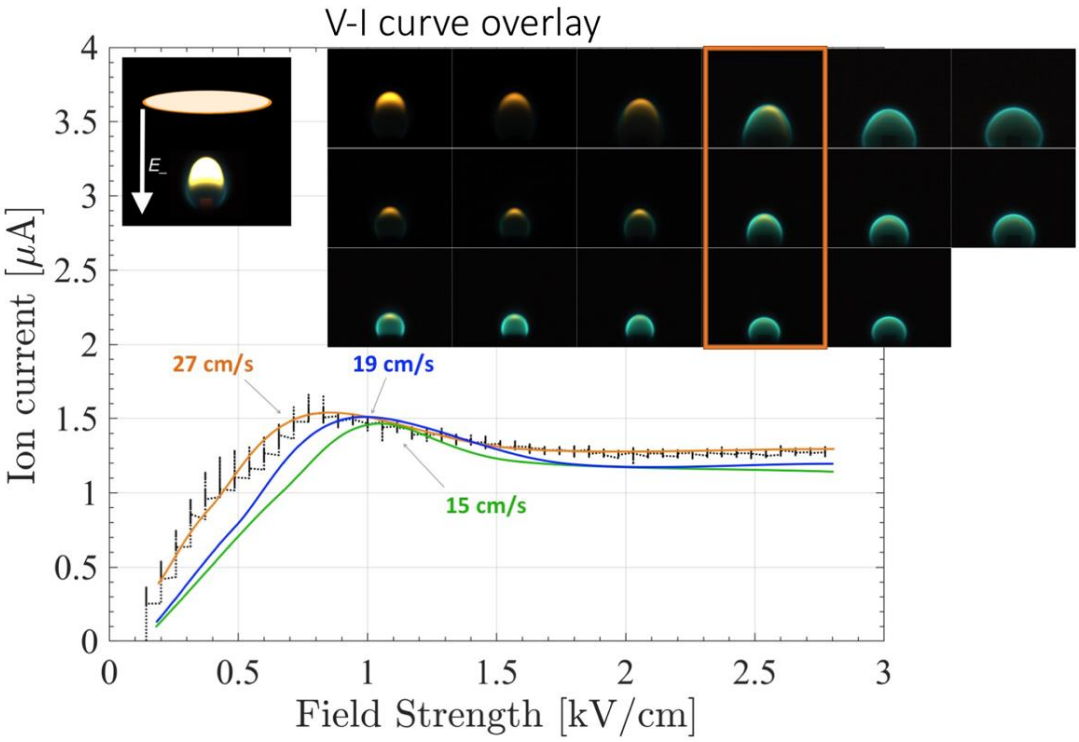


Figure 7. Ion current to electric field strength per unit distance for negative field of 100% ethylene with coflow air. The peak ion current reduces to around 1.5 microamps for all the fuel flows: 27, 19, and 15 cm/s.

Early E-FIELD Flame results from the ISS testing were presented in following oral and poster presentations:

1. Derek Dunn-Rankin and Yu-Chien (Alice) Chien, "Electrical Field Effects on Laminar Diffusion Flames," 11th Multi-Agency Coordinating Committee for Combustion Research (MACCCR) Annual Fuel and Combustion Research Review, Livermore, CA, 9-11 April 2018. (oral presentation)
2. Chien, Y.-C., Tinajero, J., Stocker, D., Hegde, U. and Dunn-Rankin, D. "Electric Field Effects on Flames in Microgravity on the International Space Station," Central States Section Combustion Institute Spring Technical Meeting, St. Paul, MN, 20-22 April 2018. (conference paper and oral presentation)
3. Chien, Y.-C., Tinajero, J., Stocker, D., Hegde, U. and Dunn-Rankin, D. "Small Diffusion Flames under the influences of Electric-Field: Microgravity and Gravity," 37th International Symposium on Combustion, Dublin, Ireland, 29 July to 3 Aug. 2018. (Work-in-Progress poster)
4. Yu-Chien (Alice) Chien, Jesse Tinajero, Dennis Stocker, Uday Hegde, Derek Dunn-Rankin, "Microgravity Experiments of Electric Field Effects on Laminar Ethylene/Air Diffusion Flames," 34th Annual Meeting of the American Society for Gravitational and Space Research (ASGSR) Meeting, Rockville, MD, 31 Oct. to 3 Nov. 2018. (oral presentation)

F. Communication Imagery

1. Still Imagery

Photos are available online at www.flickr.com/photos/space-flames. Although the Facebook page at www.facebook.com/space.flames is meant to provide updates on ACME and related combustion research, it is good source for recent photos, e.g., of flames, that may not be available on the Flickr site. Additional photographs may be requested from the Project Scientist, Dennis Stocker (dennis.p.stocker@nasa.gov).

2. Video

'The Power of Studying Combustion on the ISS,' which was developed with input from the ACME Project Scientist, is available at https://science.nasa.gov/science-news/news-articles/the-power-of-studying-combustion-on-the-iss.

Videos describing the ACME project, its hardware, and the value of combustion research can be found at:
https://spaceflightsystems.grc.nasa.gov/sopo/ihho/psrp/fcf/fcf-investigations/acme/.

Meanwhile, E-FIELD Flames videos can be found at:
https://spaceflightsystems.grc.nasa.gov/sopo/ihho/psrp/fcf/fcf-investigations/acme/e-field-flames/

Videos related to E-FIELD Flames and other microgravity combustion research can also be seen on Facebook at www.facebook.com/space.flames.

Contact Dennis Stocker (dennis.p.stocker@nasa.gov) for access to those videos or for additional videos.

IV. Additional Comments from science team

None.

Appendix C. 30-Day Operational Accomplishments Report for ISS Increments 57 and 58

1 June 2019

30-day Operational Accomplishments Report

… for the International Space Station (ISS) Increments 57 and 58

ACME Advanced Combustion via Microgravity Experiments
E-FIELD Flames Electric-Field Effects on Laminar Diffusion Flames

E-FIELD Flames is one of several independent experiments that are part of the ACME project, which uses a single modular set of hardware to carry out studies of flames burning gaseous fuels in the ISS' Combustion Integrated Rack (CIR). The investigation was carried out in two rounds of tests, where the first (with ACME's coflow burner) occurred in Increment 55 and the second (with a gas-jet burner) began in Increment 56 and ended in Increment 57. As a result, there is nothing to be reported for Increment 58.

Prepared by the PI team and Dennis Stocker (dennis.p.stocker@nasa.gov), the ACME Project Scientist, who is the primary contact for E-FIELD Flames and ACME information.

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PI project scientist:	Dr. Yu-Chien (Alice) Chien, University of California, Irvine, CA
Lead collaborator:	Prof. Sergey Minaev, Far Eastern Federal University, Vladivostok, Russia
NASA Project Scientist (PS):	Dennis P. Stocker, NASA Glenn Research Center, Cleveland, OH
PS team member:	Dr. Uday G. Hegde, Case Western Reserve University, Cleveland, OH

I. Introduction

The Electric-Field Effects on Laminar Diffusion Flames (E-FIELD Flames) experiment was conducted in the Combustion Integrated Rack (CIR) on the International Space Station, as part of the Advanced Combustion via Microgravity Experiments (ACME) project. In this experiment, an electric field with voltages as high as 10,000 volts was established between the burner and a flat, circular mesh electrode suspended above (i.e., downstream of) the burner. The motion of the charged ions, which are naturally produced within the flame, are strongly affected by a strong electric field. The resulting ion-driven wind can dramatically influence the stability and sooting behavior of the flame. Measurements were made of electric-field strength, the ion current passing through the flame, and flame characteristics such as the size, structure, temperature, soot, and stability. Conducting the tests in microgravity allowed for great simplifications in the

analysis of ion-driven convection and flame interaction, enabling new understanding and the development of less polluting and more efficient combustion technology for use on Earth.

An electric field can strongly influence flames because of its effect on the ions produced as a result of the combustion reactions. The induced ion wind can modify the flame shape, alter the soot and stability limits, and direct the heat transfer. The purpose of the E-FIELD Flames experiment is to gain an improved understanding of flame ion production and learn how the electric field can be used to manipulate the ions and control non-premixed flames for practical benefit. For example, electric fields can enable stable burner operation at fuel-lean conditions with reduced pollutant emission, i.e., where the flame is not reliably sustained without the electric field. As another example, the role of electric fields on soot formation can be dramatic. It appears that the ion driven wind alters the local convection and strongly influences the residence time of the soot precursors within the hot flame. The experiment contributes to our critical understanding of combustion processes in the presence of electric fields.

More specifically, E-FIELD Flames has the following experimental objectives:

A. Determine the relationship between electric field voltage and chemi-ion current as functions of total fuel (methane and ethylene) flow rate and the amount of inert (nitrogen) in the fuel.
B. Obtain the time response of the flame chemiluminescence, luminosity, and the ion current to large step changes in voltage, as functions of total fuel (methane and ethylene) flow rate and amount of inert (nitrogen) added to the fuel stream.
C. Evaluate and manipulate sooting behavior (i.e., soot volume fraction and location of luminous soot) in relation to the electric field voltage and chemi-ion current as functions of total fuel (methane and ethylene) flow rate and amount of inert (nitrogen) added to the fuel stream.
D. Use the variation of the chemi-ion current as a function of applied voltage to characterize the combustion behavior of near-limit lifted flames and use the associated ion-driven wind to stabilize or reattach these flames.

The effect of the electric field on the flame is studied through tests conducted with a range of flame and electric field conditions. The fuel, fuel jet velocity, air flow, electric-field voltage and polarity were the primary parameters of the E-FIELD Flames experiment. Tests were conducted with methane and ethylene, including nitrogen dilutions of each, which provide a broad range of soot-producing conditions, i.e., from soot-free to highly sooting. They were carried out with both a simple gas-jet burner and the coflow burner that was also used for the Coflow Laminar Diffusion Flame (CLD Flame) experiment. Tests with the coflow burner were conducted with and without a coflow of air, where the results of the latter are compared with those from the gas-jet burner to assess the effect of air entrainment on the flame. The coflow burner also provided a different electrode configuration (close to a plane-to-plane) than the jet burner (close to a point-to-plane). In a typical test, the mesh electrode was energized at increasing potentials (relative to the burner), where the flame's response (e.g., ion current, luminosity, shape) to the resulting field strength was measured. The field strength was changed in repeated steps or was instead smoothly ramped.

Microgravity studies of flames allow for a range of simplifications and provide various advantages over normal-gravity studies. For example, the natural flicker of flames, which is a buoyancy-driven (i.e., gravity-driven) instability, is eliminated yielding quasi-steady flames in microgravity. Therefore, the flicker in the E-FIELD Flames was a result of electric body forces alone. Length scales are increased in microgravity flames facilitating analysis of the flame structure. Momentum-dominated flames, which are important for most practical combustion, can be studied at low velocities to simplify analysis. Microgravity flames tend to have a much stronger sensitivity to their atmosphere and exhibit a much broader range of characteristics than normal-gravity flames because of the near absence of buoyant entrainment. The long residence times in microgravity flames can lead to strong soot production, but many microgravity flames are soot free. The absence of buoyancy makes microgravity flames advantageous for studies of limit and stability behavior where chemical kinetics are important. For example, chemi-ion species, electronically excited chemiluminescent species, are at very low concentration in typical flames, making the reduced buoyancy environment valuable for evaluating the minor species chemistry. With these advantages, microgravity testing can lead to new understanding of the complex nature of combustion – regardless of the gravitational environment of the combustion application.

II. Methods/Research Operations

A. Discussion of Method/Protocol

E-FIELD Flames and other ACME experiments are configured, but not conducted, by the ISS crew. The crew's role is to set up the hardware, gas bottles, etc. for each set of tests. The most common changes include the gas bottles on the Combustion Integrated Rack (CIR) and the mass flow controllers on the modular ACME chamber insert.

All testing is commanded from the ground, specifically from the Glenn ISS Payload Operations Center (GIPOC). The science portion of the testing is typically conducted in a nominally automated mode using pre-programmed scripts. However, it may alternately be conducted in a 'manual' mode, where changes are made to a control parameter in response to the analog video downlink and the numerical data available in the telemetry stream. For example, the mesh voltage can be adjusted to approach a flame's stability or soot limit. This was not done during Increments 55-56 or 57-58, where the console commanding during a test was typically limited to halting the test when it was observed in the analog ops video that the flame had extinguished.

Testing is not conducted when thrusters fire or spacecraft dock or undock from ISS to ensure an acceptable acceleration environment.

Data is downlinked between each set of tests, i.e., day of testing, to allow for preliminary analysis and requisite planning for subsequent testing.

III. Results

A. Baselined Activities Completed – Pre-flight, In-flight, Post-Flight

A total of 219 flames were ignited for the E-FIELD Flames experiment, with 142 and 77 respectively ignited during Increments 55-56 and 57-58. Note that the investigation's testing was completed during Increment 57, so no flames were ignited during Increment 58. The first and second rounds of testing respectively occurred in March-May 2018 and September-November 2018. The two rounds of tests resulted in 120 and 99 ignited flames conducted with coflow and gas-jet burners, respectively. The coflow burner was developed for the Coflow Laminar Diffusion Flame (CLD Flame) experiment, where CLD Flames' second round occurred between the two rounds of the E-FIELD Flames experiment. The 'interruption' in the E-FIELD Flames testing was intentionally planned years in advance to enable the PI team to review the initial results in detail and with that knowledge plan the second round of tests to maximize its scientific outcome.

E-FIELD Flames does not have living subjects like a life science experiment, but the tests are independently conducted with two different gaseous fuels, methane and ethylene. Both are simple hydrocarbon gases for which the chemistry is well understood. Methane (CH_4) is the primary component of natural gas and tends to produce low levels of soot given the absence of carbon-carbon bonds. Meanwhile, ethylene (C_2H_4), is highly sooting because of its carbon double bond. For both fuels, tests were conducted with pure fuel and fuel mixed with nitrogen over a broad range of dilutions. For limited cases, the fuel and nitrogen was premixed on the ground prior to launch. ACME bottles were launched with methane concentrations of 30%, 40%, 45%, and 100% with a balance of nitrogen (on a volume basis). Similarly, bottles were launched with ethylene concentrations of 30%, 50%, and 100%, again with a balance of nitrogen (on a volume basis). But the ACME hardware also allows for simultaneous flow of fuel and nitrogen to the burner, allowing for on-orbit fuel dilution and testing at a continuum of fuel concentrations. In all cases, the coflow was 21% oxygen with a balance of nitrogen (on a volume basis).

No samples (e.g., of soot) were collected for E-FIELD Flames; all of the ISS results were electronically transmitted to the ground. That is furthermore the case for all currently planned ACME experiments.

A summary log of the E-FIELD Flames ignited for E-FIELD Flames during Increments 55-57, including a limited overview of the test conditions, is found on the next page in Table 1.

Table 1.

E-FIELD Flames log of flames ignited in Increments 55-57 (*where the investigation was completed prior to Increment 58*)							
Test Day[1]	**Year**	**Month**	**Date**	**Mesh Polarity[2]**	**Fuel bottle**	**Fuel conc.[3]**	**Flames ignited[4]**
First round of tests, with the coflow burner, in March to May 2018 (Increment 55)							
18073	2018	3	14	negative	100% CH4	100%	10
18078			19			70%, 100%	10
18080			21		40% CH4	40%	8
18087			28		100% C2H4	80%, 100%	10
18092		4	2		30% C2H4	30%	11
18095			5	positive		30%	10
18102			12		40% CH4	40%	8
18106			16			40%	13
18113			23		100% CH4	100%	15
18130		5	10			70%	13
18137			17		100% C2H4	80%, 100%	12
Second round of tests, with the gas-jet burner, in Sept.-Nov. 2018 (Increments 56-57)							
18269	2018	9	26	negative	40% CH4	40%	11
18274		10	1		30% C2H4	30%	11
18282			9		100% CH4	70%, 100%	13
18285			12		100% C2H4	100%	10
18290			17			80%	11
18298			25	positive	40% CH4	40%	9
18302			29		30% C2H4	30%	10
18312		11	8		100% CH4	70%, 100%	13
18319			15		100% C2H4	25%, 50%, 80%, 100%	11

[1]**Test Day** includes the year as the first two digits followed by the 3-digit day of the year.
[2]**Mesh Polarity** refers to the polarity of the high-voltage power supply and the copper electrode mesh relative to the burner.
[3]**Fuel conc.** (as diluted with nitrogen on a volume basis) was varied between tests, but typically matched the bottle. The list only accounts for the flames which ignited.
[4]**Flames ignited** is the number ignited on that day (neglecting the day change on ISS).

Pre-flight and post-flight activities are generally not addressed in this report.

B. List of Preflight, In-flight, Postflight Anomalies

3. Mission anomalies

The CIR's IPSU-1 Image Processing and Storage Unit, which is used with the High Bit-Depth Multispectral (HiBMs) camera and the collimated illumination package, was only intermittently operable at the end of Increment 56 and was not operable at the start of Increment 57. Those capabilities were not seen as critical for E-FIELD Flames,

where the illumination is significantly blocked by the mounting bracket for the mesh electrode. Therefore, the tests on Sept. 26 (18269), Oct. 1 (18274), Oct. 9 (18282), Oct. 12 (18285), and Oct. 17 (18290) were conducted without HiBMs imaging or the CIR illumination. It was later determined to be an issue with a cable connection, which was corrected by astronaut Alex Gerst on Oct. 22 (18295) when he carried out other ACME tasks. As a result, the subsequent E-FIELD Flames tests were carried out using the HiBMs camera and illumination package.

4. Any other difficulties in obtaining scientific data

E-FIELD Flames tests sometimes automatically aborted (due to the controlling software script), where that was typically prior to gas flow and the activation of the igniter. As a result, there was little lost other than the time in the failed test attempt. The aborts normally occurred because of communication issues with IPSU-1. These are infrequent, but there are some days when tests aborted repeatedly.

Tests are of course also unsuccessful when the flame fails to ignite. The first test of the day has sometimes failed to ignite, presumably because the fuel line has not been sufficiently primed. Ignition was sometimes unsuccessful because the settings for the automatic flame detection are such that detection threshold is reached because of the igniter glow rather than a flame. As a result of the detection, the hot-wire igniter is deactivated and retracted, which is of course an issue if the flame has yet to ignite. This is typically avoided by detecting ignition with measurement of the flame's OH* chemiluminescence with a Photomultiplier Tube (PMT) module. The OH* chemiluminescence, in the ultraviolet at 310 nm, is a good indicator of an ignited flame rather than the glowing igniter, which emits primarily in the infrared and visible spectrum. However, the combination of igniter settings and PMT gains can lead to false ignition detection.

ACME's Mass Flow Controllers (MFCs) used to control the flow of fuel, nitrogen, and synthetic air (i.e., 21% oxygen with a balance of nitrogen, on a volume basis) have unfortunately not always performed adequately. The fuel and especially nitrogen flows sometimes oscillate. Furthermore, the MFC used for the 'air' coflow behaves as if it is sticking in a few places, where the flow rate won't linearly ramp when commanded (as required for the CLD Flame experiment). It tends to stick in some places and then abruptly overshoot where that has sometimes been seen to extinguish the flame. As a result, mass flow controllers are tested - without flames - after they are first installed. Adjustments are often then made to (1) the settings that control when the valves open and (2) the Proportional-Integral-Derivative (PID) settings. This ACME activity has involved significant testing, separate from flame tests, and has taken place on 17333, 17334, 17347, 17354, 18008, 18022, 18038, 18046, 18143, 18241, 19059, and 19112. In this regard, there are an assortment of MFCs with different flow ranges for each of the three gases. While the MFC adjustments have been beneficial, they have not eliminated all gas flow problems. It was seen that nitrogen oscillations tend to damp with time, so a nitrogen pre-flow

was implemented so that the oscillations would typically end before the flame measurements. That approach was partially successful and was introduced during E-FIELD Flames' first round. Later, a procedure of re-zeroing the MFCs prior to each test was also implemented during E-FIELD Flames' second round. A replacement MFC for the 'air' coflow is being prepared for spaceflight, where a limited third round of CLD Flame testing is being planned to accomplish its extinction limit objective.

C. Completeness/Quality of Data

The number of flames ignited significantly exceeds the planned tests which were (1) specified as required in the ACME Integrated Science Requirements Document (ISRD), using the gas-jet burner, and (2) later expanded to include tests with the coflow burner to partially make up for science loss resulting from the slow response time of the high-voltage power supply. Following the revision, a total of 74 E-FIELD Flames tests were identified as required. In contrast, 219 flames were ignited in the experiment, so nearly three times as many flames were ignited than required.

It should be understood that a complete set of image data is not available for all tests. As discussed in section III.B, the IPSU-1 was not working at the beginning of the second round of testing, preventing use of both the HiBMs camera and the CIR's collimated illumination package. Those are not critical for E-FIELD Flames, so tests continued without their use.

Furthermore, recorded image data is sometimes deleted without downlinking. This is done because of operational constraints which limit the amount of data that can be managed per test day. Given the constraint, the decision is sometimes made to sacrifice some image data (e.g., from the HiBMs camera) to allow additional tests to be conducted. This is intentionally done in order to maximize the scientific return of each day of testing within the project constraints.

Data quality is of course dependent on the experimental hardware. But neglecting that limitation on data quality, the cameras are sensitive to radiation where pixels gradually become 'hot' over time. While these bright spots are addressed in image post-processing, they are a degradation of the image quality.

It is possible that the tests are subtly affected by the acceleration environment, but there is not obvious indication of this and it has thus far not been evaluated with available measurements from the Space Acceleration Measurement System (SAMS). In this regard, note that the Combustion Integrated Rack (CIR), in which the experiment is conducted, is equipped with a Passive Rack Isolation System (PaRIS).

Regardless of the shortcomings discussed above, a great wealth of experimental data has been secured and E-FIELD Flames is considered to be highly successful.

D. Status of Data Analysis

Data from the tests making up the E-FIELD Flames experiment includes video recordings from the operations camera for general flame behavior, high resolution images from the ACME data camera at various exposures and frame rates as dictated by the flame luminosity, intensified camera images, and non-imaging data at 125 Hz. The non-imaging data file is processed using Matlab scripts (modified from those developed by CLD Flame researchers) to separately show trends of ion current and flame luminosity as a function of the applied field, and the flame parameters (fuel, flowrate, burner type). The PMT data is also extracted to evaluate flame luminosity oscillations. Image data is processed using the OMA software (also developed and used by CLD Flame researchers).

The overall data volume is large with a storage limit of 127,000 MB per test day. All of the data has been downloaded to the hard drive with a backup system at UC Irvine. At this stage, the data has been sorted into appropriate segments so that comparisons can be made between burner geometries (jet versus coflow from the prior round of testing), fuel type and dilution (methane versus ethylene; no dilution, modest dilution, high dilution) and fuel flow velocity. The fuel flow velocity and dilution conditions were sometimes selected to hold total carbon flux into the flame constant. At this stage, the analysis remains comparative and qualitative, to show the major differences in behavior in microgravity and normal-gravity (1-g) flames, and to show particularly how buoyancy modifies the electrical response of the flames.

E. Preliminary Research Findings

Distinctly different results were seen in E-FIELD Flames' second round tests with the gas-jet burner when compared with the first round tests with the coflow burner. For example, the ion current response to the electric field strength with an upward ion driven wind in microgravity is a clear parabolic trend (which is similar to the trend observed in 1-g flames with the coflow burner as presented in the prior report). The parabolic behavior is evident even with the strong oscillations observed during the rise. This rising trend is not observed with the coflow burner.

The behavior of the soot and its charge state are complicated since in the regions of high field strength, particularly near the fuel tube where the field lines converge, soot can emit or collect free charges. Hence, the soot transport is driven directly by the field and by the ion-driven wind (and thermophoresis) which means it can travel differentially to the bulk flow. These various transport elements can push the soot through the flame to burnout, hold it in stasis in the flame for additional growth, or drive it to the burner where it accumulates. More work is needed to sort through these elements.

Positive electric field for 100% methane fuel with jet burner

The preliminary findings for the gas-jet burner in this section highlight the representative information of electric fields applied to a small hydrocarbon flame in the 0-g condition. The ISS tests covered a wide range of conditions which allowed the flame to settle as its

image was captured with the color camera and the intensified camera. The total flame luminosity and the ion current were also recorded in all cases. Figure 1 shows example flame images representative of the jet tests conducted with methane as fuel. The voltage is indicated as the nominal field strength (kV/cm). Since the physical spacing from the burner to the mesh is 35 mm, 1.16 kV/cm represents nominal potential of around 4000 V across the flame. It is important to note that with the sharp-edged jet burner, there is an electric field concentration near the burner tip, so the field in that region will be stronger than is indicated by the nominal field strength and stronger than would be seen for the coflow burner where the coflow portion of the burner acts as a planar electrode to mitigate the field concentration at the jet burner's tip.

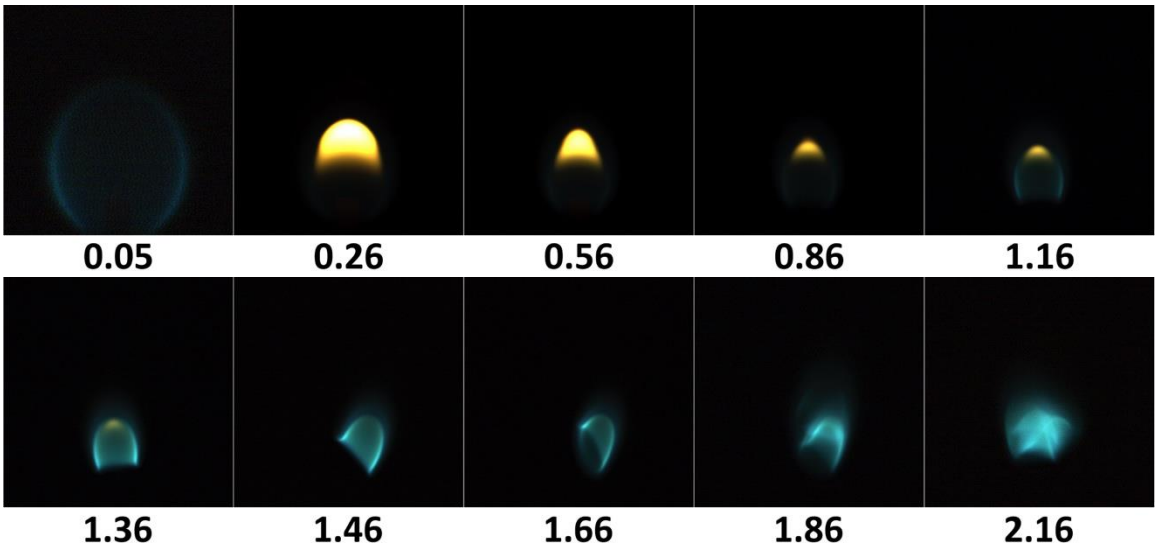


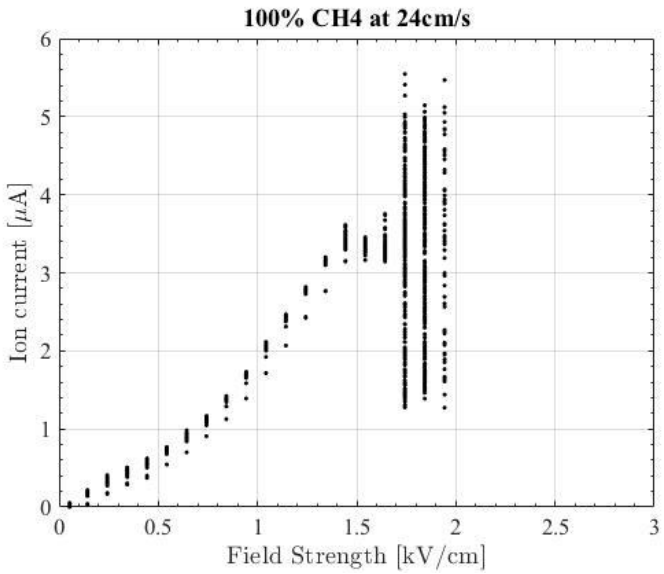
Figure 1. Flame images observed from the data camera at 19 cm/s methane jet with positive field applied. (Unit: kV/cm)

The very first image of each condition, e.g., at 0.05 kV/cm, demonstrates the initial voltage stage before the voltage sweep test commences. The jet flames are relatively more stable initially and less stable under high voltage than the previously reported flames using a coflow burner with no coflow [Chien et al., Ion Current and Flame Changes with Electric Fields in Microgravity, WSSCI Pasadena 2019]. As the positive field is applied to the jet, the flame becomes both narrower and shorter. An immediate change is the appearance of bright soot at the flame tip. With an increase of electric field strength, the soot region diminishes as the flame shrinks.

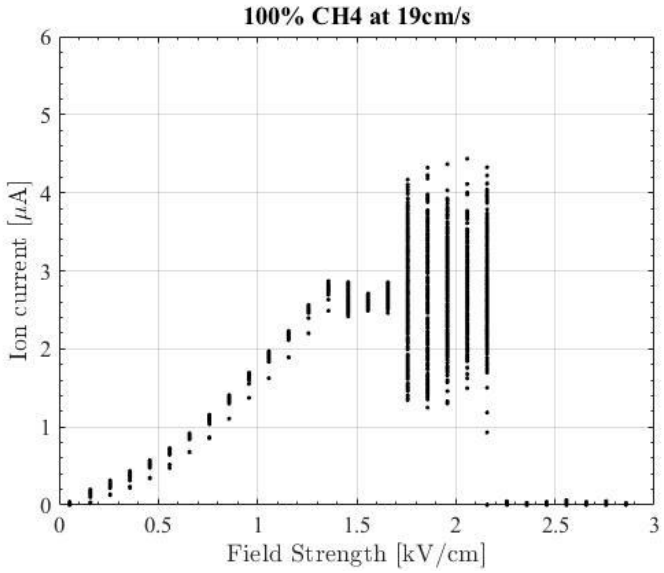
The voltage was increased 350 volts per step and held at each voltage for two seconds. The ion current data rate was 125 Hz and the monitored current fluctuation at the same voltage step is shown in the graph. The ion current to electric field strength (i.e., i-v) graph is plotted for 24 and 19 cm/s flow velocity in Figure 2, where the current initially increases parabolically with the applied voltage. The ion current corresponds to the flame appearance in that the current increases and reaches a peak at the most compact flame, e.g., at a field strength of 1.36 kV/cm for a 19 cm/s burner flow (as seen in Figure 2b). At

that maximum, the soot is almost gone (as seen in Figure 1). The flame base starts rising slightly around the current peak and lifts higher, with one side of the reaction wing attached to the tube, as the electric field increases further. The flame is not always lifted at the same edge of the burner tube; therefore, the attached reaction wing sometimes rotates along the tube edge with approximately the same lift angle under the same electric field strength. Otherwise, the lifted edge flame remains relatively stable. With further increases in the electric field strength, the flame starts fluctuating and a distinct dancing phenomenon can be seen from the i-v curve where the ion current oscillates strongly at the same field strength. The fluctuation crosses more than 3.5 μA (between ~1.2 to 5 μA) and reaches 5.0 μA for the 24 cm/s case. Similarly, this oscillation at the same voltage is observed for 19 cm/s with the ion current spanning around 2.5 μA (~1.4 to 4 μA) and reaching 4.0 μA; corresponding flame images are shown as 1.86 and 2.16 kV/cm in Figure 1. These very high peak values may represent an incipient corona or discharge event. The combustion extinguishes with the oscillations at 1.95 kV/cm for 24 cm/s and 2.16 kV/cm for 19 cm/s.

The flame geometries and soot distribution under the electric field influence clearly differ when comparing the gas-jet burner results with those from the earlier coflow burner tests without a coflow (i.e., with the same jet flow for each), where the annulus of the coflow burner acted as a planar electrode. But the maximum amount of ions produced within the flame (as evidenced by the ion current reaching the plateau) is similar for the case where the fuel flow velocity matches at 19 cm/s. The current also peaks around the same field strength per unit distance. Moreover, the oscillation behavior only exists in the jet burner experiment and furthermore influences the ion current monitoring due to the unstable flame behavior responding to the high field strength.



(a)



(b)

Figure 2. Ion current as a function of the field strength and fuel velocity. (a) 24 cm/s methane. (b) 19 cm/s methane.

<u>Negative electric field for 100% methane fuel with jet burner</u>

Unlike the positive field, the methane jet encounters the downward ion wind around the flame and the flame is not sustained after 1.86 kV/cm for 24 cm/s, being blown out by the electric field. The 19 cm/s flame lasts until 2.04 kV/cm. When the flame experiences a negative field, its height decreases and the full flame is moved onto the tube, similar to the downward pointing jet flame in 1g [Papac, M. et al., Ann. N. Y. Acad. Sci, 1077, 2006]. The change of the flame width is not as obvious as with a positive field but it still changes, as revealed in Figure 3 for 0.05, 0.14, and 0.24 kV/cm.

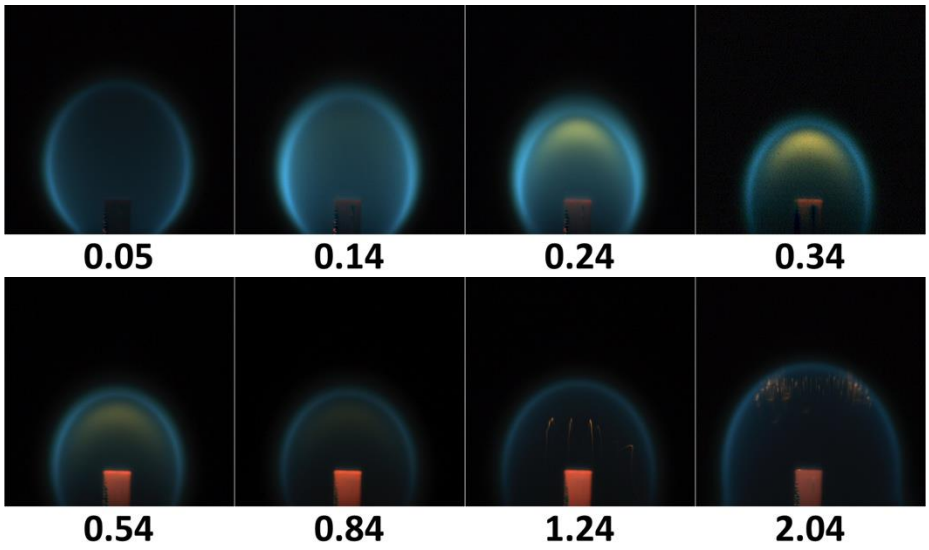
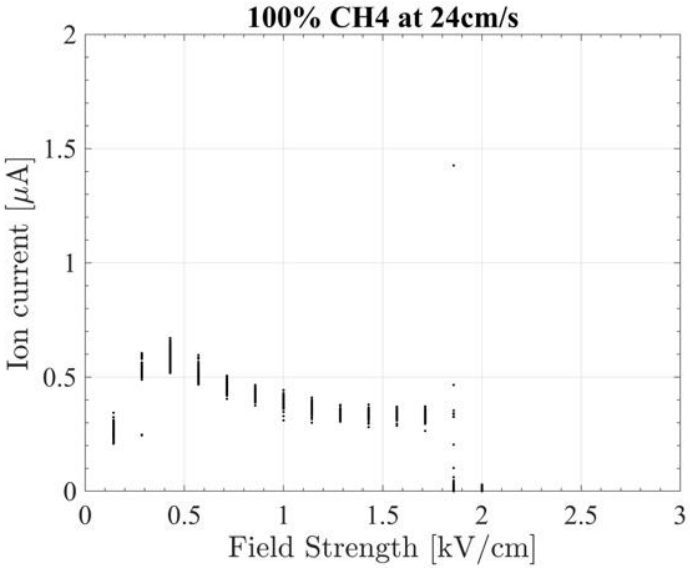
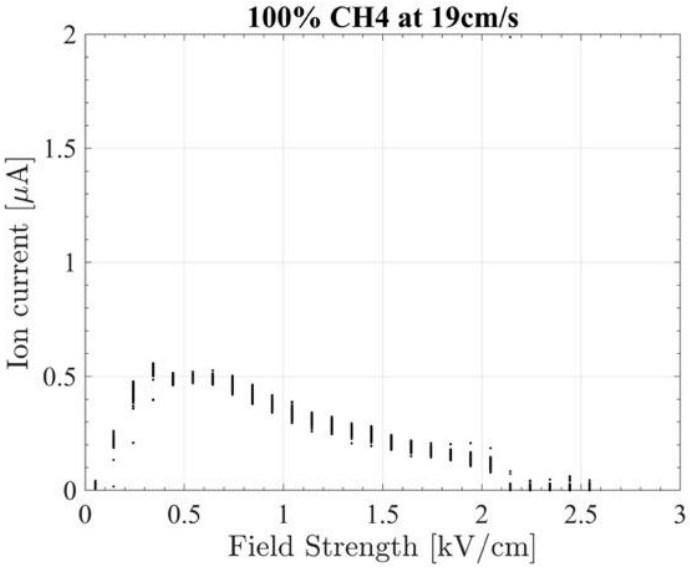


Figure 3. Flame appearance as a function of field strength observed from the data camera at 19 cm/s methane jet with negative field applied. (Unit: kV/cm)

The ion current (as seen in Figure 4's i-v curve) also increases with the field strength but differently than with the positive field at is has no clear curved slope. Instead, it decreases parabolically after reaching the current peak. The current peak corresponds to the most compact flame at 0.34 kV/cm for 19 cm/s and 0.43 kV/cm for 24 cm/s. The peak ion current is 0.53 and 0.55 μA, as shown in Figures 6b and 6a, respectively. Soot also appears at the flame tip but with a wider distribution along the reaction sheet. The soot reduces with increasing negative field strength, and it disappears at 0.84 kV/cm (as seen in Figure 3) along the parabolic decay line after passing the ion current peak. The flame width and height also expand after the most compact flame (at the peak current). This trend is demonstrated in the second-row of images in Figure 4. Notice that there are several sparkler-like particles, presumably of soot, at 1.24 and 2.04 kV/cm. Further evaluation of this sparking is underway.



(a)



(b)

Figure 4. Ion current as a function of the field strength and fuel velocity. (a) 24 cm/s methane. (b) 19 cm/s methane.

Positive electric field for 70% methane fuel with jet burner

The flame behavior with a lower hydrocarbon content of 70% methane diluted with nitrogen, shown in Figure 5, has a distinct reduction of soot at the flame tip in comparison to the cases with 100% methane (seen in Figure 1). The flame reaction wing starts lifting away from the burner edge when 1.06 kV/cm is applied, which is earlier than occurs in the 100% case, and the flame fluctuation also continues with increasing positive field strength.

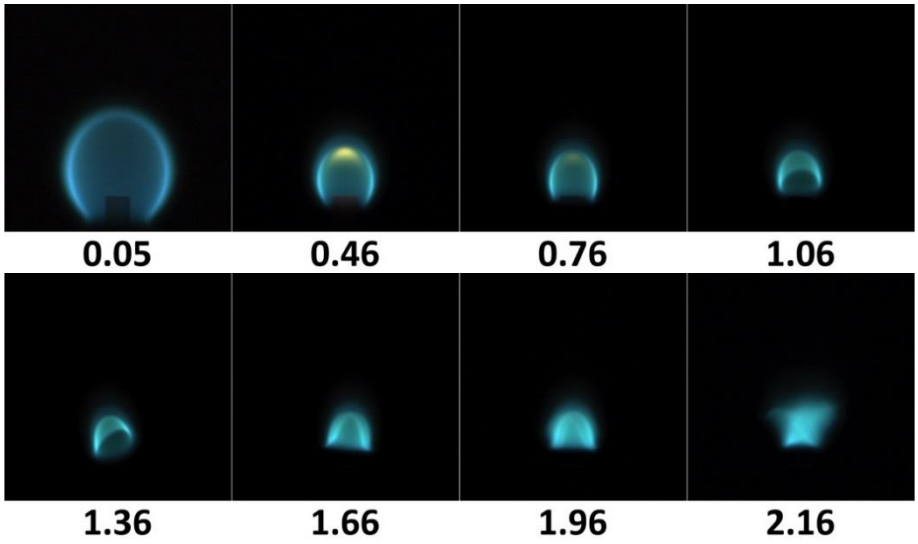


Figure 5. Flame appearance as a function of field strength observed from the data camera at 19 cm/s methane jet with negative field applied. (Unit: kV/cm)

The ion current to the electric field strength graph of 19 cm/s, shown in Figure 6, reveals a similar trend to Figure 2 for the 100% methane jet, and further details appeared in this condition. The ions are extracted from the flame gradually by the electric field, and with the increasing electric field the current rises. After reaching the ion current peak around 1.1 kV/cm, the flame has not yet oscillated as had occurred in the pure methane case. The ion current reduces and then rises back up in a parabolic behavior. The flame starts fluctuating around 1.75 kV/cm with the positive field strength increase. Further analysis is needed to understand the field strength to current behavior as a function of hydrocarbon content, and also with the influence from burner geometries in the microgravity environment.

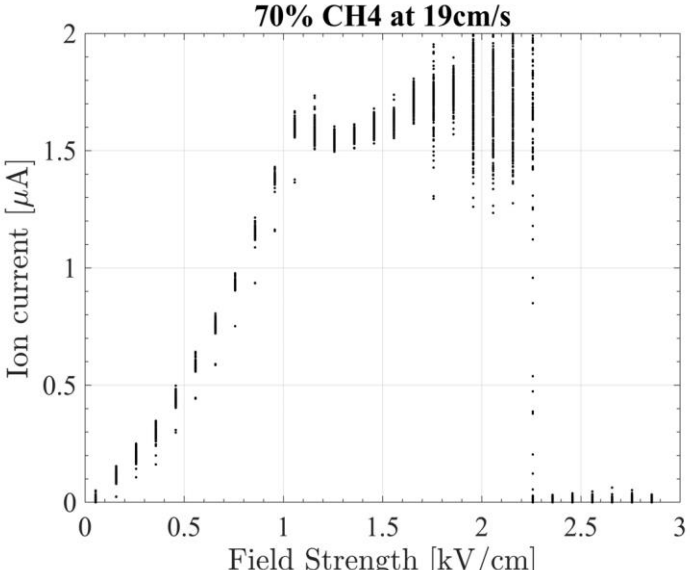


Figure 6. Ion current as a function of the field strength and fuel velocity at 19 cm/s methane.

In all of the i-v curves, the ion current data points at each field strength stretch across ~0.1 μA for each step, which are the jagged features from the spikes following a voltage step change that are mostly associated with the transient capacitance of the system when a rapid voltage step is requested. These overshoots notwithstanding, the i-v curves follow smooth and reproducible trajectories. The wide range of the ion current after the saturation plateau is due instead to the flame oscillation.

This report shows the initial results of the E-FIELD Flames experiment using the gas-jet burner in the International Space Station (ISS). The positive field allows the soot to concentrate and compresses the flame into a high current density thermal source. The flames demonstrate an unpredicted oscillation before extinguishing with increasing field strength. This oscillating phenomenon was not observed during the tests with the coflow burner from the first round, with the exception of a case that introduced the airflow at 30 cm/s which increased the surface reaction while the flame tended to balance between electric field effects, ions, and charged soot. The negative field allows the flame to maintain its spherical characteristic in microgravity. The yellow soot appears for both of the electric field directions within the flame around the tip and disappears with increasing field strength after reaching the current peak. The role of soot in the i-v behavior of these flames requires further investigation.

Early E-FIELD Flame results from the ISS testing were presented in following oral and poster presentations:

1. Derek Dunn-Rankin and Yu-Chien (Alice) Chien, “Electrical Field Effects on Laminar Diffusion Flames,” 11th Multi-Agency Coordinating Committee for Combustion Research (MACCCR) Annual Fuel and Combustion Research Review, Livermore, CA, 9-11 April 2018. (oral presentation)
2. Chien, Y.-C., Tinajero, J., Stocker, D., Hegde, U. and Dunn-Rankin, D. “Electric Field Effects on Flames in Microgravity on the International Space Station,” Central States Section Combustion Institute Spring Technical Meeting, St. Paul, MN, 20-22 April 2018. (conference paper and oral presentation)
3. Chien, Y.-C., Tinajero, J., Stocker, D., Hegde, U. and Dunn-Rankin, D. “Small Diffusion Flames under the influences of Electric-Field: Microgravity and Gravity,” 37th International Symposium on Combustion, Dublin, Ireland, 29 July to 3 Aug. 2018. (Work-in-Progress poster)
4. Yu-Chien (Alice) Chien, Jesse Tinajero, Dennis Stocker, Uday Hegde, Derek Dunn-Rankin, “Microgravity Experiments of Electric Field Effects on Laminar Ethylene/Air Diffusion Flames,” 34th Annual Meeting of the American Society for Gravitational and Space Research (ASGSR) Meeting, Rockville, MD, 31 Oct. to 3 Nov. 2018. (oral presentation)
5. Chien, Y.-C., Tinajero, J., Stocker, D., Hegde, U. and Dunn-Rankin, D. “Ion Current and Flame Changes with Electric Fields in Microgravity,” 11th U.S. National Combustion Meeting, Pasadena, CA, 24-27 March 2019. (conference paper and oral presentation)

Upcoming presentations of ISS results include the following:

6. Chien, Y.-C., Stocker, D., Hegde, U. and Dunn-Rankin, D. "Microgravity Experiments with Methane Jet Flames under Electric Field Influence on-board the International Space Station (ISS)," 12th Asia-Pacific Conference on Combustion Meeting 2019, 1-5 July, Fukuoka, Japan. (final conference paper submitted)
7. Chien, Y.-C., Stocker, D., Hegde, U. and Dunn-Rankin, D. "Microgravity Experiments Examining Electric Field Effects on Laminar Methane Gas-Jet Diffusion Flames," 35th Annual Meeting of the American Society for Gravitational and Space Research (ASGSR) 20-23 Nov. 2019, Denver, Colorado. (abstract submitted)

F. Communication Imagery

1. Still Imagery

Photos are available online at www.flickr.com/photos/space-flames. Although the Facebook page at www.facebook.com/space.flames is meant to provide updates on ACME and related combustion research, it is good source for recent photos, e.g., of flames, that may not be available on the Flickr site. Additional photographs may be requested from the Project Scientist, Dennis Stocker (dennis.p.stocker@nasa.gov).

2. Video

'The Power of Studying Combustion on the ISS,' which was developed with input from the ACME Project Scientist, is available at https://science.nasa.gov/science-news/news-articles/the-power-of-studying-combustion-on-the-iss.

Videos describing the ACME project, its hardware, and the value of combustion research can be found at:
https://spaceflightsystems.grc.nasa.gov/sopo/ihho/psrp/fcf/fcf-investigations/acme/.

Meanwhile, E-FIELD Flames videos can be found at:
https://spaceflightsystems.grc.nasa.gov/sopo/ihho/psrp/fcf/fcf-investigations/acme/e-field-flames/

Videos related to E-FIELD Flames and other microgravity combustion research can also be seen on Facebook at www.facebook.com/space.flames.

Contact Dennis Stocker (dennis.p.stocker@nasa.gov) for access to those videos or for additional videos.

IV. Additional Comments from science team

None.

Appendix D. Operational Timeline

All dates are in 2018, when the E-FIELD Flames operations were conducted. The following text was taken from the 'ACME Activity Log (selected)' section of the informal "ACME Operations (Ops) Log" prepared by Dennis Stocker. The only addition is the identification of the investigation's two rounds and intervening second round of the Coflow Laminar Diffusion Flame (CLD Flame) experiment. The red text indicates the key days when flames were ignited.

Date	GMT	Activity
E-FIELD Flames round 1 with the coflow burner		
March 5	064	HiBMs reconfigured by Norishige Kanai for start of E-FIELD Flames
March 8	067	ACME chamber insert reconfigured by Scott Tingle for E-FIELD Flames
March 9	068	Engineering check-out tests for E-FIELD Flames
March 14	**073**	**E-FIELD Flames tests using coflow burner, negative mesh, 100% CH4**
March 19	**078**	**E-FIELD Flames tests using coflow burner, negative mesh, 100% CH4**
March 20	079	Bottle swaps by Shkaplerov (with Tingle's assistance)
March 21	**080**	**E-FIELD Flames tests using coflow burner, negative mesh, 40% CH4, launch of Feustel, Arnold, and Artemyev**
March 26	085	Bottle swap by Shkaplerov
March 28	**087**	**E-FIELD Flames tests using coflow burner, negative mesh, 100% C2H4**
March 30	089	Bottle swap by Shkaplerov
April 2	**092**	**E-FIELD Flames tests using coflow burner, negative mesh, 30% C2H4; MWA prep by Arnold, hardware gather by Kanai; launch of additional igniters & spare fiber arrays on SpaceX CRS-14**
April 3	093	Installation of positive power supply and igniter tip replacement by Tingle
April 5	**095**	**E-FIELD Flames tests using coflow burner, positive mesh, 30% C2H4**
April 10	100	Bottle swap by Shkaplerov
April 12	**102**	**E-FIELD Flames tests using coflow burner, positive mesh, 40% CH4**
April 16	**106**	**E-FIELD Flames tests using coflow burner, positive mesh, 40% CH4**
April 18	108	Bottle swap by Feustel
April 23	**113**	**E-FIELD Flames tests using coflow burner, positive mesh, 100% CH4**
April 27	117	E-FIELD Flames science deferred because of the CIR's IPSU-G failure to boot
May 7	127	Recovery of the CIR's IPSU-G by Tingle
May 8	128	IPSU-G checked out
May 10	**130**	**E-FIELD Flames tests using coflow burner, positive mesh, 100% CH4**
May 14	134	CIR chamber front end cap seals replaced and bottles swapped by Arnold
May 17	**137**	**E-FIELD Flames tests using coflow burner, positive mesh, 100% C2H4**
May 21	141	ACME hardware for BRE, Flame Design, and s-Flame launched on OA-9
May 22	142	ACME chamber insert reconfigured for CLD Flame by Arnold
May 23	143	MFC testing and tuning for CLD Flame
May 25	**145**	**CLD Flame preparatory tests using 100% C2H4**
CLD Flame round 2 featuring ethylene		
E-FIELD Flames round 2 featuring the 2.1-mm gas-jet burner		
Sept. 18	261	Bottle swaps by Artemyev
Sept. 19	262	ACME insert reconfigured for E-FIELD Flames by Arnold assisted by Fuestel
Sept. 24	267	No-flame characterization and testing for E-FIELD Flames
Sept. 26	**269**	**E-FIELD Flames tests using 40% CH4**
Sept. 27	270	Bottle swap by Prokopev (with Arnold's assistance)

Oct. 1	**274**	**E-FIELD Flames tests using 30% C2H4**
Oct. 3	276	Bottle swap by Prokopev
Oct. 9	**282**	**E-FIELD Flames tests using 100% CH4**
Oct. 10	283	Bottle swap by Prokopev
Oct. 12	**285**	**E-FIELD Flames tests using 100% C2H4**
Oct. 17	**290**	**E-FIELD Flames tests using 100% C2H4**
Oct. 18	291	Bottle swap by Prokopev
Oct. 22	295	Installation of positive power supply, igniter tip replacement, etc. by Gerst
Oct. 25	**298**	**E-FIELD Flames tests using 40% CH4**
Oct. 26	299	Bottle swap by Prokopev
Oct. 29	**302**	**E-FIELD Flames tests using 30% C2H4**
Oct. 31	304	Bottle swap by Prokopev
Nov. 1	305	Gas chromatograph (GC) preparation by Gerst
Nov. 5-6	309-310	Gas chromatograph (GC) bakeout
Nov. 8	**312**	**E-FIELD Flames tests using 100% CH4**
Nov. 9	313	Bottle swap by Prokopev
Nov. 15	**319**	**E-FIELD Flames tests using 100% C2H4**

Appendix E. Astronauts and Cosmonauts who Conducted ISS Operations for E-FIELD Flames

This information is derived from the 'ACME Names and Numbers Log' of the informal "ACME Operations (Ops) Log" prepared by Dennis Stocker.

Eight astronauts and cosmonauts participated in ISS operations for this investigation, where they are listed with the dates of their activity (at least as planned). They are listed by the number of days in which they conducted supporting on-orbit activities and alphabetically. The crew activity on each of those dates is very briefly summarized in the Appendix B. Additional information about the crew activity can be found in a number of the subsequent appendices.

Sergey Prokopev (RSA)	GMT 270, 276, 283, 291, 299, 304, 313	7 days
Ricky Arnold (NASA)	GMT 092, 134, 142, 262, 270 (with Prokopev)	5 days
Anton Shkaplerov (RSA)	GMT 079, 085, 089, 100	4 days
Scott Tingle (NASA)	GMT 067, 079 (with Shkaplerov), 093, 127	4 days
Drew Feustel (NASA)	GMT 108, 262 (with Arnold)	2 days
Alex Gerst (ESA)	GMT 295, 305	2 days
Norishige Kanai (JAXA)	GMT 064, 092	2 days
Oleg Artemyev (RSA)	GMT 261	1 day

Together the eight men represented four space agencies and four nations, namely Germany, Japan, Russia, and the United States.

Appendix F. ISS Operations per the Payload Weekly Science Summary

This text is from the 'ACME Ops per the Payload Weekly Science Summary' section of the informal "ACME Operations (Ops) Log" prepared by Dennis Stocker. The only addition is the identification of the investigation's two rounds and intervening second round of the Coflow Laminar Diffusion Flame (CLD Flame) experiment. As a minor note, the sizes of some figures were adjusted in this report, but none were cropped.

Increment 55-56 (with round 1 featuring the coflow burner and the start of round 2 featuring the 2.1-mm gas-jet burner)

Week 02: March 2-8

Nemo: Excellent job replacing the imaging module in the CIR HiBMS Imaging Package on GMT 064! Thank you for all your hard work and diligence during the many intricate tasks. You replaced this imaging module in support of the next ACME experiment, E-FIELD Flames (Electric-Field Effects on Laminar Diffusion Flames).

Maker: Great job installing the ACME Mesh and replacing the ACME Bypass Controller on GMT 067! Thank you for all your hard work and attention to detail. You performed these tasks in support of the next ACME experiment, E-FIELD Flames (Electric-Field Effects on Laminar Diffusion Flames).

E-FIELD Flames studies how electric fields at different voltages influence flames. An electric field can strongly influence the motion of the ions produced by a combustion reaction. The resulting ion-driven wind impacts flame stability and soot production and reduces pollutant emissions. The results will lead to a better understanding of more efficient and less polluting combustion technology for use on Earth.

Week 03: March 9-15

On GMT 073, we performed the first ACME E-FIELD Flames (Electric-Field Effects on Laminar Diffusion Flames) test points. This first day was very successful with ten flames ignited for studying the effects of an electric field on laminar diffusion flames. Preliminary results show that the high-voltage electric field created forces similar to buoyancy, which dramatically affected the flame's shape, intensity, and soot production. The ability to control flames with an electric field may have practical applications on Earth. For example, an electric field could be used to stabilize a flame in fuel-lean conditions, which would lead to reduced pollutant emissions compared to a typical combustor.

CAPTION TO VIDEO, *WHICH IS NOT INCLUDED IN THIS LOG*

While the light is still on at the beginning of this video, you can see the copper mesh, which is used to create the electric field. Before the electric field is activated, the flame is dim and spherical. However, as the negatively-charged copper mesh is energized at a steadily increasing potential, the flame becomes brighter and narrower—transforming into a shape similar to that of flames in normal gravity. The bright yellow tip is caused by soot. Near the end of the test, the flame suddenly becomes dim and spherical again when the electric field is deactivated. (This video is at 2x normal speed.)

Week 04: March 16-22

Anton & Maker: Great job replacing the CIR Manifold Bottles this week! The new Manifold #2 bottle allows us to continue performing test points in a combustion chamber environment of air, and the new Manifold #4 bottle allows us to perform test points with diluted methane instead of pure methane.

On GMT 078 and 080, we continued performing ACME E-FIELD Flames (Electric-Field Effects on Laminar Diffusion Flames) test points to study the effects of an electric field on laminar diffusion flames. All tests so far have used pure and nitrogen-diluted methane, the dominant component of natural gas.

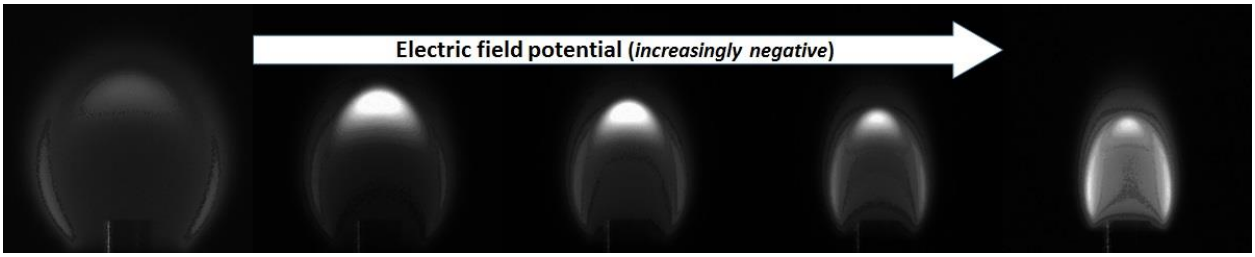


Figure 1. Composite Image of E-FIELD Flames at Increasingly Negative Electric-field Potential (100% Methane)

Following ignition, the copper electrode mesh (downstream of the burner) is energized at a relatively low voltage and is then gradually stepped up to several kilovolts. In the current testing, the mesh is negatively charged relative to the burner. As a result, the electric field enhances the upward motion of the transient positive ions within the flame, inducing an "ion wind," which affects the overall gas motion.

Figure 1 shows how an increasingly negative electric-field potential affects a flame during a single test point. The resultant "ion wind" dramatically affects the flame's shape, intensity, and soot production (the bright flame tip is caused by soot). This flame used 100% methane fuel; whereas, the tests conducted with 40% methane were soot-free under all conditions tested.

(Note that this is a preliminary image, where the apparent flame luminosity is not consistent from flame to flame.)

Week 05: March 23-29

Anton: Great job replacing the CIR Manifold Bottle this week! The new bottle allows us to perform test points with ethylene fuel instead of methane.

On GMT 087, we performed ACME E-FIELD Flames (Electric-Field Effects on Laminar Diffusion Flames) test points using pure ethylene fuel.

Week 06: March 30 - April 5

Anton: Thank you for replacing the CIR Manifold Bottle on GMT 089. The new bottle allows us to perform test points with diluted ethylene fuel instead of pure ethylene.

Maker: On GMT 093, great job installing the ACME Mesh Pos Power Supply and replacing the ACME igniter tip! You removed the negative power supply and installed the positive one to allow us to perform test points using a positively charged electric field to influence the flame. You replaced the igniter tip for preventative maintenance. We returned to performing test points later in the week.

On GMT 092, we performed ACME E-FIELD Flames (Electric-Field Effects on Laminar Diffusion Flames) test points using diluted ethylene fuel. Our first test points using ethylene fuel instead of methane were performed on GMT 087. These used pure ethylene. Ethylene produces a lot of soot when it burns, and we are investigating how an electric field influences this soot production.

CAPTION TO VIDEO, *WHICH IS NOT INCLUDED IN THIS LOG*

This video shows two test points side-by-side to compare the burning characteristics of 100% and 30% ethylene. The 100% ethylene flame is much bigger and brighter than the 30% flame (the copper mesh is illuminated by the brightness). The yellow color is caused by soot with the 30% flame having only a yellow tip. As the negatively charged copper mesh is energized at a steadily increasing potential, both flames become narrower in shape and bluer in color. The changes in the 100% flame are visually more dramatic.

When the electric field is deactivated near the end of the tests, the flames suddenly become broader and brighter again. The yellow tip of the 30% flame returns as well as the soot "horns" of the 100% flame. These soot "horns" are caused by heavy soot production just within the flame. Looking closely at the video, you can see where soot escaping from the 100% flame forms a discolored ring on the copper mesh. This soot escaped because of the sudden end to the intensified combustion created by the electric field.

Week 07: April 6-12

Anton: Great job replacing the CIR Manifold Bottles this week! The new Manifold #2 bottle allows us to continue performing test points in a combustion chamber environment of air. The new Manifold #4 bottle allows us to again perform test points with diluted methane but this time with a positively charged electric field.

On GMT 095, we performed our first ACME E-FIELD Flames (Electric-Field Effects on Laminar Diffusion Flames) test points with a positively charged electrode mesh. These tests demonstrated the ability of the electric field to control the soot characteristics of a flame. We observed that the flames (30% ethylene) became soot-free at high electric-field potentials. While soot can be advantageous for some combustion processes, e.g., to enhance radiant heating, it is a health concern and often unwanted. Today's tests demonstrated the potential for the practical application of electric fields to control soot within flames. One of the goals of E-FIELD Flames is to gain a fundamental understanding of this capability, which could lead to more efficient and less polluting combustion technology for use on Earth.

CAPTION TO VIDEO, *WHICH IS NOT INCLUDED IN THIS LOG*

At the start of this test, before the positively charged electric field is activated, the top half of the flame is yellow because of the glowing soot. As the voltage is increased—ultimately to 10 kilovolts—the soot fades away (so the flame is all blue), the flame shifts toward the burner, and the base dramatically broadens. The effect of the electric field on the flame can truly be seen when the electric field is deactivated near the end of the test, and the flame quickly returns to its natural state.

Week 08: April 13-19

Drew: Thank you for replacing the CIR Manifold Bottle on GMT 108. The new bottle will allow us to again perform test points with pure methane but this time with a positively charged electric field.

On GMT 102 and 106, we performed ACME E-FIELD Flames (Electric-Field Effects on Laminar Diffusion Flames) test points using diluted methane fuel. The first set of E-FIELD Flames tests uses a burner that can simultaneously flow air and a gaseous fuel mixture.

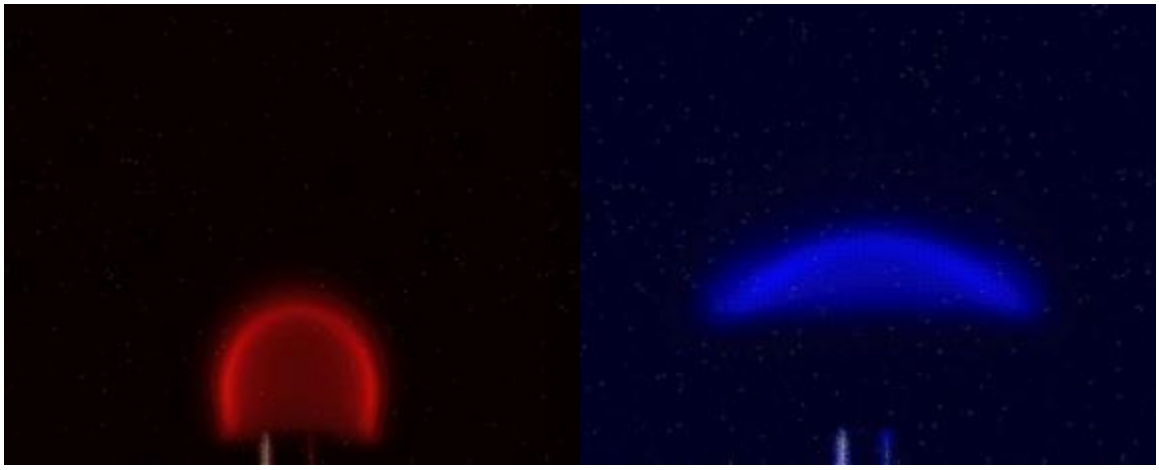


Figure 1. E-FIELD Flames with (left) and without (right) a Simultaneous Flow of Air
(Flames actually dim blue; colored & brightened here for clarity)

The “red” flame in Figure 1 benefited from the oxygen supplied by the simultaneous flow of air from the burner; in contrast, the “blue” flame relied on entrainment and diffusion for its oxygen supply. Although the fuel velocity is the same in both tests, the small “red” flame is attached to the burner, while the “blue” oxygen-starved flame is much wider and is lifted from the burner. These characteristics help the “blue” flame get the oxygen required to consume the fuel. Meanwhile, the “red” flame gets that oxygen in a much smaller zone thanks to the presence of the simultaneous flow of air.

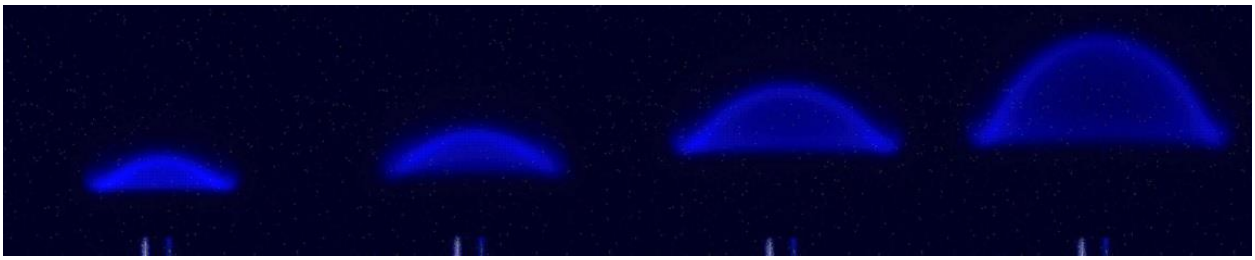


Figure 2. E-FIELD Flames at Increasing Fuel Velocity (No Simultaneous Flow of Air)
(Flames actually dim blue; colored & brightened here for clarity)

Figure 2 shows flames at increasing fuel velocities. Although the positive electric field tends to force the flames downward toward the burner, the flames are lifted from the burner because they are oxygen starved without the simultaneous flow of air. More oxygen is required to burn higher fuel flow rates, and the diameter at the base of the flame correspondingly increases. The distance that the flame is lifted from the burner also increases with the fuel velocity, where the separation allows more oxygen to reach the flame. The flame length is another measure of the oxygen transport, where the flame extends until it gets enough oxygen to consume all of the fuel.

Week 09: April 20-26

On GMT 113, we again performed ACME E-FIELD Flames (Electric-Field Effects on Laminar Diffusion Flames) test points using pure methane, but this time with a positively charged electrode mesh. Like we observed with 30% ethylene tests (GMT 095), these tests also showed the strong effect that an electric field can have on soot within the flame.

CAPTION TO VIDEO, *WHICH IS NOT INCLUDED IN THIS LOG*

In this test, just after ignition, the electric field is energized at a very low voltage. The small flame, with its yellow/orange top, stabilizes for several seconds. Then, the voltage is quickly increased to 1 kilovolt. Under the influence of this electric field, you can see the flame broaden, turn all blue in color (i.e., become soot free), and become quite dim at the tip. When the electric field is deactivated just before the end of the test, the flame quickly returns to its initial condition—with its sooty top.

Week 10: April 27 – May 3

On GMT 117, the boot drive in the FCF Image Processor failed. As this hard drive stores all our science image data, we were unable to perform the planned ACME E-FIELD Flames (Electric-Field Effects on Laminar Diffusion Flames) test points. Next week, the crew will remove this hard drive, so we can boot up the FCF Image Processor from its other hard drive. Once this hard drive is ready to go as the new boot drive, we plan to get back to performing test points.

ACME E-FIELD Flames Test Point Early Results

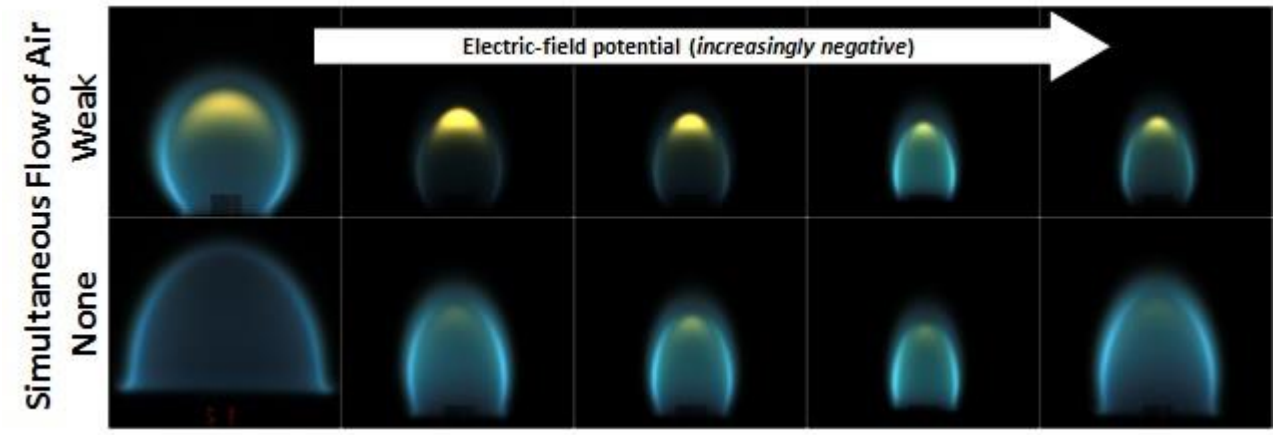


Figure 1. E-FIELD Flames with and without a Simultaneous Flow of Air at Increasingly Negative Electric-field Potential (100% Methane)

Figure 1 shows how the flame changes as the potential of the negatively-charged electric field is increased. The top row of flames benefitted from the oxygen supplied by air flowing from the burner at the same time as the fuel. The bottom row of flames had no extra flow of air—only the oxygen in the air-like environment of the Combustion Chamber. In both cases, the effect of the electric field on the flame shape and size is dramatic. The ion wind, an upward flow created by the negatively-charged electric field, enhances oxygen transport to the flame causing it to become narrower and shorter.

The negatively-charged electric field also affects the ion current, or the electric current that passes through the flame due to the motion created by the ion wind. As the electric field continues to increase in strength, the ion current peaks then sharply drops to a lower ion current and the flame size increases correspondingly. In comparable testing in normal gravity, the ion current peaked at a lower current then plateaued instead of sharply dropping like the microgravity tests. Figure 2 below compares the microgravity tests with and without the simultaneous flow of air. The flames without the simultaneous flow of air peak at a lower ion current and at a lower field strength compared to flames with a simultaneous flow of air.

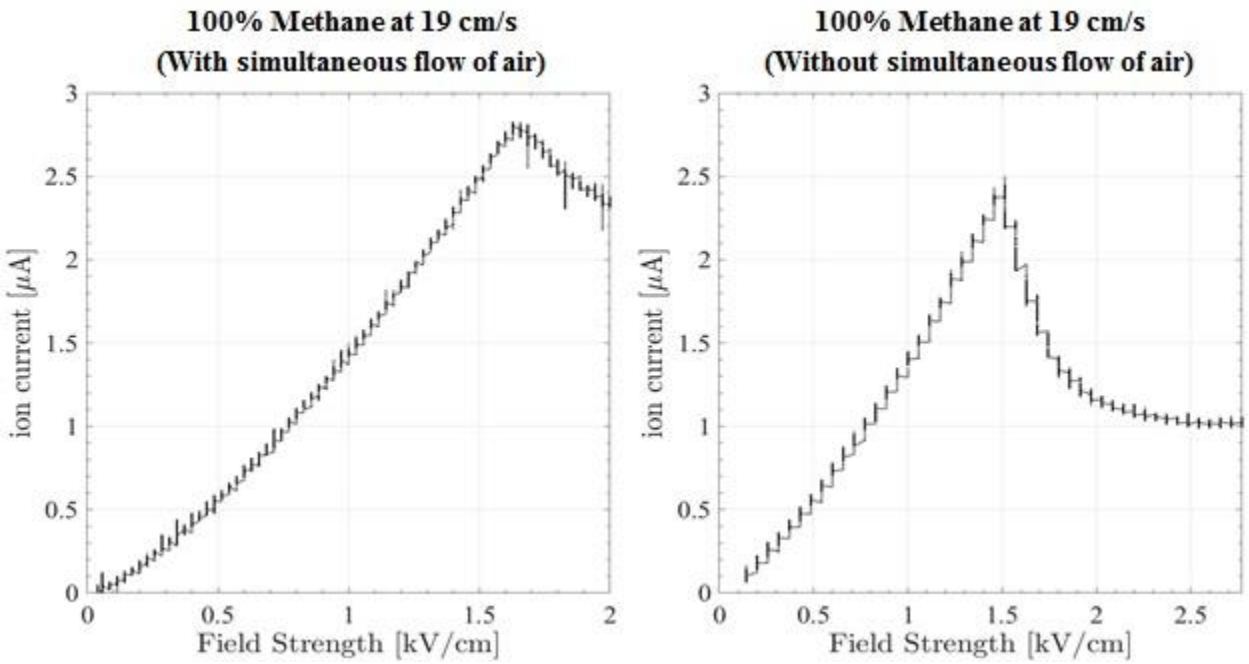


Figure 2. Ion Current as a Function of the Field Strength (for the two tests represented in Figure 1)

Week 12: May 11-17

Ricky: Great job replacing the CIR Combustion Chamber Seals and the Manifold Bottles on GMT 134! These seals are replaced every five years for preventative maintenance. The new Manifold #2 bottle allows us to continue performing test points in a combustion chamber environment of air. The new Manifold #4 bottle allows us to again perform test points with pure ethylene but this time with a positively charged electric field.

On GMT 130, we performed ACME E-FIELD Flames (Electric-Field Effects on Laminar Diffusion Flames) test points using 70% methane fuel. This data will be used to compare with the data we've already obtained with 40% and 100% methane fuel. The results from all these tests—performed in a positively charged electric field—will also be compared to results from tests performed in a negatively charged electric field using the same fuel concentrations.

Week 13: May 18-24

Ricky: On GMT 142, great job removing the ACME Mesh, replacing the igniter tip, and replacing two controllers. Thank you for all your hard work and attention to detail! You performed these tasks in support of the next ACME experiment, CLD Flame Part 2 (Coflow Laminar Diffusion Flame).

On GMT 137, we performed the final test points for the first round of testing for ACME E-FIELD Flames (Electric-Field Effects on Laminar Diffusion Flames). Since the start of E-FIELD Flames in early March 2018, we've ignited about 120 flames! These tests burned both weakly sooting methane and strongly sooting ethylene at various nitrogen dilutions and injected at different velocities. Some tests were performed with a simultaneous flow of air and others were performed with only the flow of gaseous fuel.

CAPTION TO VIDEO, *WHICH IS NOT INCLUDED IN THIS LOG*

At the start of this test, the 100% ethylene flame is broad and bright with soot "horns." These soot "horns" are caused by heavy soot production just within the flame. As the positively charged electric-field starts to strengthen, the flame starts to vigorously oscillate. You can see hot soot particles stream from the flame, where they would normally be consumed as they pass through the flame sheet.

At about 3 kilovolts, the flame steadies and flattens because the positively charged electric-field tends to push the flame toward the burner. It appears that the soot is trapped within the flame by thermophoresis, in which the soot particles are driven from hot to cold. At about 8 kilovolts, the flame begins to pulse strongly in an axisymmetric fashion; whereas, the earlier oscillations were asymmetric. Finally, when the electric field is deactivated, the soot streams from the top of the flame.

One of the goals of E-FIELD Flames is to gain a fundamental understanding of using electric fields to control soot production in flames. Results could ultimately lead to more efficient and less polluting combustion technology for use on Earth.

CLD Flame round 2 featuring ethylene

E-FIELD Flames round 2 featuring the 2.1-mm gas-jet burner

Week 31: September 21-27

Sergey & Ricky: Great job replacing the CIR Manifold Bottle this week! The new Manifold #4 bottle allows us to perform test points with ethylene fuel instead of methane fuel.

On GMT 269, we successfully began the second round of testing for ACME E-FIELD Flames (Electric-Field Effects on Laminar Diffusion Flames). The first round of testing was conducted with the coflow burner, which can simultaneously flow air and a gaseous fuel mixture. This round of testing uses a gas-jet burner, which can only flow a gaseous fuel mixture. Comparison of the results will allow an improved understanding of the importance of the coflow and the air entrainment on the flames.

Today, we successfully ignited eleven flames using 40% methane fuel, and the electrode mesh was used to create a negatively charged electric field. A gradual increase in the electric-field strength caused the dim blue flames to intensify, lift from the burner and tilt—where lifted flames are very sensitive to flow conditions. At higher field strengths, the flames strongly oscillated and sometimes blew out.

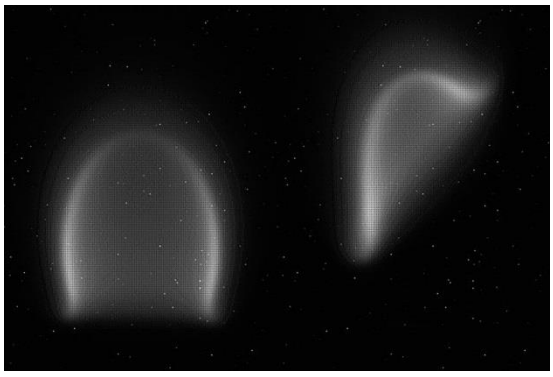


Figure 1. Two Preliminary Monochrome E-FIELD Flames Images from the Same Test (40% Methane)
Due to a higher negatively charged electric-field strength, the right-hand flame in Figure 1 is lifted from the burner and tilts to the side. An electric field can strongly influence the motion of the ions produced by a combustion reaction. This ion motion affects the motion of neutral molecules, creating an ion wind. The overall result is similar to gravity's buoyant effect on flames, where the hot gases are accelerated upward.

Week 32/01: September 28 – October 4

Sergey: Great job replacing the CIR Manifold Bottle this week! The new Manifold #4 bottle allows us to perform test points with pure methane fuel instead of diluted ethylene fuel.

On GMT 274, we performed ACME E-FIELD Flames (Electric-Field Effects on Laminar Diffusion Flames) test points using 30% ethylene fuel. In contrast to the 40% methane flames from last week's tests, these flames were more stable despite their greater dilution. These tests demonstrated that a moderate negatively charged electric field can be used with the gas-jet burner to intensify a flame without causing it to become unstable.

CAPTION TO VIDEO, *WHICH IS NOT INCLUDED IN THIS LOG*

While the light is still on at the beginning of this video, you can see the copper mesh, which is used to create the electric field. Before the electric field is activated, the flame is extremely dim. It brightens significantly and gains a sooty tip as the mesh is activated, where the electric field induces an ion wind enhancing the transport of oxygen to the flame. As the electric field ramps up in voltage, the flame shrinks dramatically in size and becomes much more intense because of the electric field's strong effect on the oxygen transport. Flames grow to reach the oxygen needed to burn the fuel, so better oxygen transport leads to smaller and more intense flames. At a sufficiently high voltage, the slightly lifted flame becomes unstable and partially detaches from the burner. As the voltage ramps down, the flame stabilizes and grows in size. We study the repeated ramping of the field strength because depending on the ramp (e.g., up vs. down) subtle differences can be seen in the flame behavior at the same voltage.

Increment 57-58 (with the rest of round 2, which features the 2.1-mm gas-jet burner)

Week 02: October 5-11

Sergey: Great job replacing the CIR Manifold Bottle this week! The new Manifold #4 bottle allows us to perform test points with ethylene fuel instead of methane fuel.

On GMT 282, we performed ACME E-FIELD Flames (Electric-Field Effects on Laminar Diffusion Flames) test points using 100% and 70% methane fuel. With these tests, we observed that the high-concentration methane flames on the gas-jet burner were extremely faint and soot-free until the electric field was activated. As we increased the field strength of the negatively charged copper mesh, the flames narrowed, brightened significantly, and gained sooty tips…until they became soot-free after lifting off the burner. Many of the flames extinguished at higher voltages, further demonstrating the strong effect that an electric field can have on a flame.

Methane, which is the primary component of natural gas, is an ideal gas to study for combustion research because its chemistry is well understood. However, our understanding of the role of ions—especially

under the influence of an electric field—is limited. The E-FIELD Flames experiment seeks to overcome this shortfall and explore how electrical fields may be used to manipulate combustion for practical benefit.

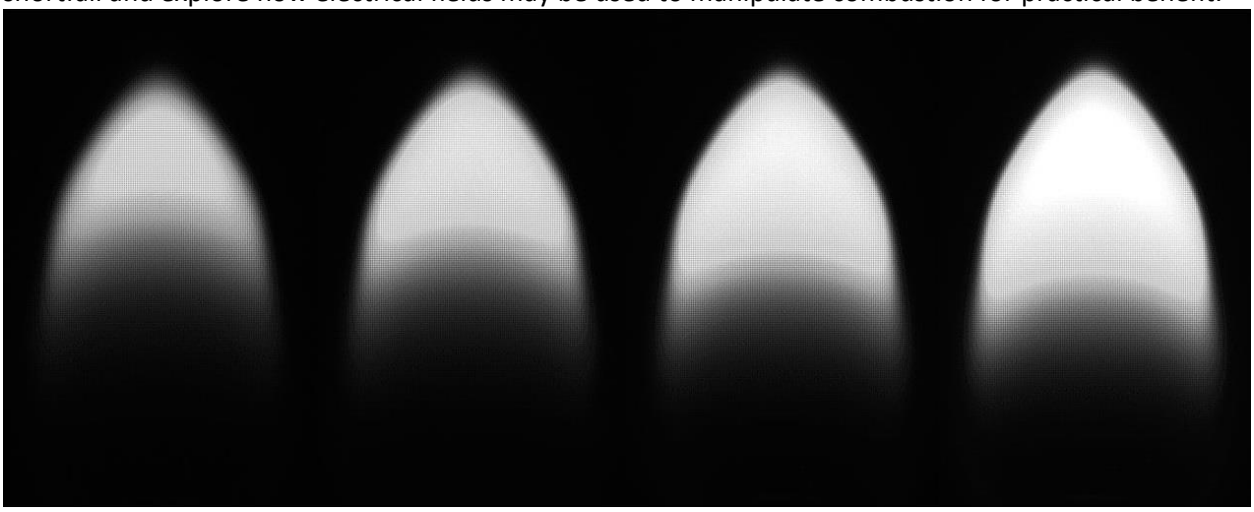


Figure 1 shows consecutive images of a 100% methane flame taken with increasingly longer exposure times. The high-voltage electric field transformed the flame into a shape similar to that of flames in normal gravity (i.e., narrower and with a pointed tip). Without the electric field, the flame tip is rounded, and the flame is much broader and soot-free.

Week 03: October 12-18

Sergey: Great job replacing the CIR Manifold Bottle this week! The new Manifold #4 bottle allows us to perform test points with diluted methane fuel instead of pure ethylene fuel.

On GMT 285 and 290, we performed ACME E-FIELD Flames (Electric-Field Effects on Laminar Diffusion Flames) test points with ethylene concentrations of 100% and 80%. With these tests, we observed how a negatively-charged electrode mesh affected sooty flames at high ethylene concentrations and at different fuel velocities.

E-FIELD Flames studies how electric fields influence flames. An electric field can strongly influence the motion of the ions produced by a combustion reaction. The resulting ion-driven wind can dramatically affect flame stability, structure, and soot production. The results show a strong potential for an innovative approach to reducing pollutant emissions in practical combustion processes on Earth.

Week 04: October 19-25

Alex: On GMT 095, great job installing the ACME Mesh Pos Power Supply and replacing the ACME igniter tip! You removed the negative power supply and installed the positive one to allow us to perform test points using a positively charged electric field to influence the flame. You replaced the igniter tip for preventative maintenance. We returned to performing test points later in the week.

Also, thank you for checking the CIR I/O Processor Custom Cable #1 connection to the Optics Bench connector saver. You checked this connection because we were having communication issues with one of our image processors on the back of the Optics Bench. Good news! On GMT 298, we successfully established communication with this package! We would like a couple more successful interactions with this package before we're certain the anomaly is solved, but we are hopeful!

Week 05: October 26 – November 1

Sergey: Great job replacing the CIR Manifold Bottles this week! The new bottles allowed us to continue performing test points.

Alex: Thank you for removing the CIR burst disks and opening all the gas chromatograph (GC) manifold bottles. These tasks have prepared the CIR GC system for a bakeout next week. The next ACME experiment—BRE (Burning Rate Emulator)—plans to use the GC to sample the chamber environment before and after test points.

On GMT 298 and 302, we performed ACME E-FIELD Flames (Electric-Field Effects on Laminar Diffusion Flames) test points with 40% methane and 30% ethylene. These tests were conducted with the positively-charged electrode mesh, which tends to push the flame toward and around the narrow burner. This technology could have practical terrestrial applications in stabilizing flames burning at fuel-lean conditions, which generate less nitrogen oxides (NOx) and sulfur oxides (SOx). NOx and SOx cause acid rain. On Earth, combustion is the principal human cause of both acid rain and greenhouse gas production.

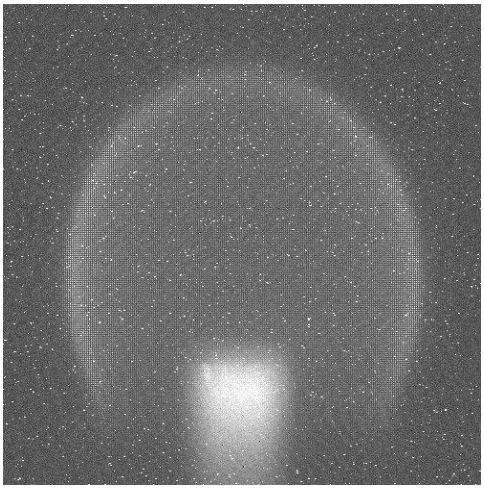


Figure 1. Image of an E-FIELD Flame Around the Burner Tip (40% Methane)

Figure 1 shows how the flame gets pushed "down" and around the burner tip due to the electric field resulting from the positively-charged electrode mesh.

Week 07: November 9-15

Sergey: Great job replacing the CIR Manifold Bottle this week! The new Manifold #4 bottle allows us to perform test points with pure ethylene fuel instead of pure methane fuel.

On GMT 312, we ignited 13 flames with 100% methane fuel for ACME E-FIELD Flames (Electric-Field Effects on Laminar Diffusion Flames). To date, 500 flames have been ignited for ACME! ACME, which is planned to run through the end of 2019, includes E-FIELD Flames and five other experiments that use gaseous fuels.

CAPTION TO VIDEO, *WHICH IS NOT INCLUDED IN THIS LOG*

While the light is still on at the beginning of this video, you can see the positively-charged electrode mesh, which is used to create an electric field. The flame starts off extremely dim and "standing" off the end of the burner tip. The electric field pushes the flame onto the burner tip.

You can see the burner tip glow when it gets heated by the surrounding flame. This test was particularly interesting because it resulted in the glowing burner as well as a small "fireworks" show, including both arcing and glowing particles (e.g., soot particles). The particles are charged and can be seen to move away from the electrode mesh and towards the methane gas flowing from the burner. Similar tests on this day didn't result in such "fireworks," so this phenomenon will be studied more closely.

Week 08: November 16-22

On GMT 319, we performed the final test points for the second round of testing for ACME E-FIELD Flames (Electric-Field Effects on Laminar Diffusion Flames). These test points used ethylene fuel concentrations ranging from 25% to 100%. Today's plan was to focus on high concentrations, namely 80% and 100%, where ethylene's carbon double bond can make highly sooty flames. The soot particles are naturally charged and cause the flame to be strongly affected by the electric field. However, we adjusted the plan because the high fuel concentrations at a high electric-field strength led to electrical breakdown, sparks and arcing, which interfered with our measurement objectives. To reduce the extent of the electrical breakdown, we performed the day's remaining test points at lower ethylene concentrations, which provided more useful data.

CAPTION TO VIDEO, *WHICH IS NOT INCLUDED IN THIS LOG*

This test point video shows the strong effect of an electric field on an 80% ethylene flame. When the electric field is activated, the flame intensifies, and the soot, which causes the yellow flame color, gets brighter. As the field strength is increased, the flame becomes shorter and narrower. The flame then engulfs the burner outlet as the positively-charged electric field pushes the flame away. You can see sparks at the flame tip and later throughout the flame. The flame tip—in both position and soot intensity—oscillates at varying speeds, sometimes becoming steady for a moment. The flame continues to shrink in size until it basically encircles just the tip of the burner. You can see bright spots intermittently at the flame tip. At the highest voltages, soot periodically bursts toward the electrode mesh and arcing becomes visible at the mesh. When the electric field is deactivated, the flame gets bigger and dimmer until it goes out when the fuel is shut off.

Appendix G. Crew Activity per the Weekly Recap from the Expedition Lead Scientist

*The subsequent text and images are from the 'ACME Crew Ops per the Weekly Recap from the Expedition Lead Scientist' section of the informal "ACME Operations (Ops) Log" prepared by Dennis Stocker. Neglecting minor reformatting, e.g., re-sizing of images, the only change was an updating of a web site address and the addition of text identifying the investigation's two rounds and intervening second round of the Coflow Laminar Diffusion Flame (CLD Flame) experiment. However, this appendix only includes a selected portion of the activity reported in the "ACME Operations (Ops) Log" because the listings are often limited to the E-FIELD Flames, ACME, and/or CIR being listed after the statement, "*Other work was done on these investigations.*"*

The weekly recap is available from e-mail or www.nasa.gov/mission_pages/station/research/index.html.

Note that hyperlinks were generally not included in the excerpts below, but they typically go to web pages accessible via https://www.nasa.gov/mission_pages/station/research/experiments/explorer/. Those pages are secondary sources of information for the ACME project, for which the primary web site is https://www1.grc.nasa.gov/space/iss-research/iss-fcf/cir/acme/.

E-FIELD Flames round 1 featuring the coflow burner

Week of March 12, 2018

Crew begins first week of new ACME operations

The Advanced Combustion Microgravity Experiment (**ACME**) investigation is a set of studies of gaseous flames to be conducted in the Combustion Integration Rack (**CIR**), one of which being Electric-Field Effects on Laminar Diffusion Flames (**E-FIELD Flames**).

In E-FIELD Flames, an electric field with voltages as high as 10,000 volts is established between the burner and a mesh electrode. The motion of the charged ions, which are naturally produced within the flame, are strongly affected by a high-voltage electric field. The resulting ion-driven wind can dramatically influence the stability and sooting behavior of the flame. Measurements are made of electric-field strength, the ion current passing through the flame, and flame characteristics such as the size, structure, temperature, soot, and stability. Conducting the tests in microgravity enables new understanding and the development of less polluting and more efficient combustion technology for use on Earth.

This week marks the first week of E-FIELD flames operations with a very successful first set of ten flames ignited for studying the effects of an electric field on laminar diffusion flames. Preliminary results show that the high-voltage electric field created forces similar to buoyancy. This effect dramatically affected the flame's shape, intensity, and soot production.

Week of March 26, 2018

Other work was done on these investigations: … ACME E-FIELD Flames ….



Roscosmos cosmonaut and current space station commander Anton Shkaplerov replaces a bottle in the Combustion Integration Rack (CIR) as a part of the ACME E-FIELD Flames investigation. ***Credits: NASA***

Week of April 2, 2018

ACME chamber prepared for initiation of experiment

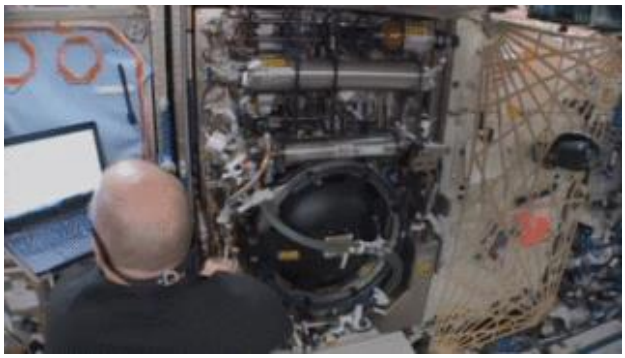
The Advanced Combustion Microgravity Experiment (**ACME**) investigation is a set of five independent studies of gaseous flames to be conducted in the Combustion Integration Rack (**CIR**), one of which being **E-Fields Flame**. ACME's goals are to improve fuel efficiency and reduce pollutant production in practical combustion on Earth and to improve spacecraft fire prevention through innovative research focused on materials flammability.

This week, crew members removed the ACME Mesh negative Power Supply and installed the ACME Mesh Positive Power Supply, replaced an ignitor tip and replaced other hardware within the chamber, all in preparation for the next series of data collection with a positive field.

Week of May 7, 2018

Other work was done on these investigations: … CIR ….

> **CAPTION TO VIDEO, *WHICH IS NOT INCLUDED IN THIS LOG***
> NASA astronaut Scott Tingle works within the CIR as a part of the ACME E-FIELD Flames investigation. ***Credits: NASA***



NASA astronaut Scott Tingle works within the CIR as a part of the ACME E-FIELD Flames investigation. ***Credits: NASA***

CLD Flame round 2 featuring ethylene

E-FIELD Flames round 2 featuring the 2.1-mm gas-jet burner

Week of September 17, 2018

Crew prepares for ACME operations

The Advanced Combustion Microgravity Experiment (**ACME**) investigation is a set of five independent studies of gaseous flames to be conducted in the Combustion Integration Rack (**CIR**), one of which being Electric-Field Effects on Laminar Diffusion Flames (**E-FIELD Flames**).

In E-FIELD Flames, an electric field with voltages as high as 10,000 volts is established between the burner and a mesh electrode. The motion of the charged ions, which are naturally produced within the flame, are strongly affected by a high-voltage electric field. The resulting ion-driven wind can dramatically influence the stability and sooting behavior of the flame. Measurements are made of electric-field strength, the ion current passing through the flame, and flame characteristics such as the size, structure, temperature, soot, and stability. Conducting the tests in microgravity allows for simplifications in the analysis, enabling new understanding and the potential development of less polluting and more efficient combustion technology for use on Earth.

CAPTION TO VIDEO, *WHICH IS NOT INCLUDED IN THIS LOG*

Oleg Artemyev of Roscosmos works within the Combustion Integration Rack (CIR) as a part of the ACME investigation. The crew conducted maintenance on the rack in order to prepare for E-FIELD Flames to begin. ***Credits: NASA***

Appendix H. Crew Activity per the ISS Daily Summary Report

*The subsequent text is from the 'ACME Crew Ops per the ISS Daily Summary Report' section of the informal "ACME Operations (Ops) Log" prepared by Dennis Stocker, where the source information is available from https://blogs.nasa.gov/stationreport. Text was added identifying the investigation's two rounds and intervening second round of the Coflow Laminar Diffusion Flame (CLD Flame) experiment. In contrast, E-FIELD Flames operations listed under the heading "*All activities are complete unless otherwise noted*" were not included in this report but can be seen in the ACME Ops Log. Many days of crew activity were only listed in that way, so this should not be seen as comprehensive.*

E-FIELD Flames round 1 featuring the coflow burner

2018 March 5: GMT 064

Combustion Integrated Rack (CIR) Advanced Combustion via Microgravity Experiments (ACME) Electric-Field Effects on Laminar Diffusion (**E-Field) Flames:** The crew replaced the compensator module in the CIR Hi Bit-depth Multi-Spectral (HiBMS) Imaging package today. This is in preparation for the upcoming ACME E-Field Flames investigation. The ACME investigation is a set of five independent studies of gaseous flames to be conducted in the CIR. ACME's primary goal is to improve fuel efficiency and reduced pollutant production in practical combustion on Earth. For E-Field Flames an electric field with voltages as high as 10,000 volts is established between the burner and a mesh electrode. The motion of the charged ions, which are naturally produced within the flame, are strongly affected by a high-voltage electric field. The resulting ion-driven wind can dramatically influence the stability and sooting behavior of the flame. Conducting the tests in microgravity allows for great simplifications in the analysis, enabling new understanding and the development of less polluting and more efficient combustion technology for use on Earth.

2018 March 8: GMT 067

Advanced Combustion via Microgravity Experiments (ACME) Electric-Field Effects on Laminar Diffusion (**E-Field) Flames:** In preparation for the upcoming ACME E-Field Flames investigation the crew installed the ACME Mesh and replace an ACME controller today. The ACME investigation is a set of five independent studies of gaseous flames to be conducted in the Combustion Integrated Rack (CIR). ACME's primary goal is to improve fuel efficiency and reduce pollutant production in practical combustion on Earth. For E-Field Flames an electric field with voltages as high as 10,000 volts is established between the burner and a mesh electrode. The motion of the charged ions, which are naturally produced within the flame, are strongly affected by a high-voltage electric field. The resulting ion-driven wind can dramatically influence the stability and sooting behavior of the flame. Conducting the tests in microgravity allows for great simplifications in the analysis, enabling new understanding and the development of less polluting and more efficient combustion technology for use on Earth.

2018 March 20: GMT 079

Combustion Integrated Rack (CIR): The crew removed and replaced manifold bottles today to support the ongoing microgravity combustion research operations in the CIR. Testing for the ACME investigation is currently being performed in the CIR. ACME includes a set of five independent studies of gaseous flames. The primary goals of ACME are to improve fuel efficiency and reduce pollutant production in routine fuel combustion activities on Earth. Its secondary goal is to improve spacecraft fire prevention through innovative research focused on materials flammability.

2018 March 30: GMT 089

Combustion Integrated Rack (CIR) Manifold #4 Bottle Replacement: As part of ISS joint research, a Russian crewmember removed and replaced a CIR manifold #4 bottle on the front of the optics bench. CIR provides sustained, systematic microgravity combustion research and it houses hardware capable of performing combustion experiments to further research of combustion in microgravity.

2018 April 3: GMT 093

Advanced Combustion via Microgravity Experiments (ACME) Electric-Field Effects on Laminar Diffusion (**E-Field) Flames:** In preparation for the upcoming ACME E-Field Flames investigation the crew installed the ACME Mesh Positive Power Supply. The ACME investigation is a set of five independent studies of gaseous flames to be conducted in the Combustion Integrated Rack (CIR). ACME's primary goal is to improve fuel efficiency and reduce pollutant production in practical combustion on Earth. For E-Field Flames an electric field with voltages as high as 10,000 volts is established between the burner and a mesh electrode. The motion of the charged ions, which are naturally produced within the flame, are strongly affected by a high-voltage electric field. The resulting ion-driven wind can dramatically influence the stability and sooting behavior of the flame. Conducting the tests in microgravity allows for great simplifications in the analysis, enabling new understanding and the development of less polluting and more efficient combustion technology for use on Earth.

2018 April 10: GMT 100

Combustion Integrated Rack (CIR): The crew removed and replaced manifold bottles today to support the ongoing microgravity combustion research operations in the CIR. The ACME experiments being performed in the CIR include five independent studies of gaseous flames. The primary goals of ACME are to improve fuel efficiency and reduce pollutant production in routine fuel combustion activities on Earth. Its secondary goal is to improve spacecraft fire prevention through innovative research focused on materials flammability.

2018 May 7: GMT 127

Combustion Integrated Rack (CIR): During activation of the CIR on April 27th, a hard drive for the Image Processing and Storage Unit (IPSU) failed. Today the crew removed the failed drive, which will allow the IPSU to boot up from a second hard drive. The final E-Field Flames run will be executed after verification of proper operation of the CIR. The ACME experiments being performed in the CIR include five independent studies of gaseous flames. The primary goals of ACME are to improve fuel efficiency and reduce pollutant production in routine fuel combustion activities on Earth. Its secondary goal is to improve spacecraft fire prevention through innovative research focused on materials flammability.

2018 May 14: GMT 134

Combustion Integrated Rack (CIR): The crew replaced CIR manifold bottles today to support the ongoing microgravity combustion research operations in the CIR. They also replaced a seal as part of routine CIR maintenance. The Advanced Combustion via Microgravity Experiments (ACME) experiments being performed in the CIR include five independent studies of gaseous flames. The primary goals of ACME are to improve fuel efficiency and reduce pollutant production in routine fuel combustion activities on Earth. Its secondary goal is to improve spacecraft fire prevention through innovative research focused on materials flammability.

CLD Flame round 2 featuring ethylene

E-FIELD Flames round 2 featuring the 2.1-mm gas-jet burner

2018 September 27: GMT 270

Combustion Integrated Rack (CIR): The crew removed and replaced the CIR manifold bottle #4. The new manifold bottle contains a mixture of ethylene fuel and nitrogen diluent to support the Electric-Field Effects on Laminar Diffusion Flames (E-Field flames) investigation, part 2. The Combustion Integrated Rack (CIR) includes an optics bench, combustion chamber, fuel and oxidizer control, and five different cameras for performing combustion experiments in microgravity.

2018 October 10: GMT 283

Advanced Combustion via Microgravity Experiments (ACME) / Electric-Field Effects on Laminar Diffusion (E-FIELD) Flames: The crew replaced the Manifold 4 bottle and the ground will be downlinking files from ACME test points and monitoring the bottle pressure. For E-Field Flames an electric field with voltages as high as 10,000 volts is established between the burner and a mesh electrode. The motion of the charged ions, which are naturally produced within the flame, are strongly affected by a high-voltage electric field. The resulting ion-driven wind can dramatically influence the stability and sooting behavior of the flame. Conducting the tests in microgravity allows for great simplifications in the analysis, enabling new understanding and the development of less polluting and more efficient combustion technology for use on Earth.

2018 October 18: GMT 291

Combustion Integrated Rack (CIR): The ISS crew swapped out CIR Manifold #4 (1-letre) Bottles. One bottle contains 100% C2H4 and the other contains 40% CH4. The CIR includes an optics bench, combustion chamber, fuel and oxidizer control, and five different cameras for performing combustion experiments in microgravity.

2018 October 22: GMT 295

Combustion Integration Rack (CIR) / Advanced Combustion via Microgravity Experiments (ACME) / Electric-Field Effects on Laminar Diffusion (E-FIELD) Flames Hardware Replacement: Today the crew removed the negative power supply and installed the positive power supply in support of the continuing E-Field Flames part 2 investigation. For E-Field Flames an electric field with voltages as high as 10,000 volts is established between the burner and a mesh electrode. The motion of the charged ions, which are naturally produced within the flame, are strongly affected by a high-voltage electric field. The resulting ion-driven wind can dramatically influence the stability and sooting behavior of the flame. Conducting the tests in microgravity allows for great simplifications in the analysis, enabling new understanding and the development of less polluting and more efficient combustion technology for use on Earth.

2018 October 31: GMT 304

Combustion Integration Rack (CIR) Manifold Bottle Replacement: In support of part 2 of the CIR / Advanced Combustion via Microgravity Experiments (ACME) / Electric-Field Effects on Laminar Diffusion (E-FIELD) Flames operations, the crew removed the 30% ethylene, 70% nitrogen fuel bottle and replaced it with a 100% methane fuel bottle today. For E-Field Flames, an electric field with voltages as high as 10,000 volts is established between the burner and a mesh electrode. The motion of the charged ions,

naturally produced within the flame, are strongly affected by a high-voltage electric field. The resulting ion-driven wind can dramatically influence the stability and sooting behavior of the flame. Conducting the tests in microgravity allows for simplifications in the analysis, enabling new understanding and the development of less pollution and more efficient combustion technology for use on Earth.

2018 November 1: GMT 305

Combustion Integration Rack (CIR) Burst Disc Removal: The crew removed the burst discs from the argon and checked gas manifolds in the CIR Gas Chromatography (GC) system. These burst discs are considered redundant with the pressure relief valves in the system and issues with the helium burst disc have occurred in the past. Following this activity, the valves for all three gas bottles were opened. The GC will be used with the future Advanced Combustion via Microgravity Experiment (ACME) investigation entitled Burning Rate Emulator (BRE), which focuses on spacecraft fire prevention.

Appendix I. Crew Activity per the NASA Space Station Blog

The subsequent text is from the 'ACME Crew Activity per the NASA Space Station Blog' section of the informal "ACME Operations (Ops) Log" prepared by Dennis Stocker, where the source materials is available from https://blogs.nasa.gov/spacestation/. Neglecting minor reformatting, the only change was the addition of text identifying the investigation's two rounds and intervening second round of the Coflow Laminar Diffusion Flame (CLD Flame) experiment.

Note that hyperlinks were generally not included in the excerpts below.

E-FIELD Flames round 1 featuring the coflow burner

March 8, 2018 - Plant and Flame Studies Alongside Plumbing, Life Support Work

After lunch, Tingle opened up the Combustion Integrated Rack and installed new gear to get ready for the Advanced Combustion Microgravity Experiment (ACME). ACME is a set of five independent studies researching gaseous flames in space that may enable more fuel efficient and less polluting technologies.

April 3, 2018 - Resupply Ship Midway to Station Amid Maintenance and Science Work

Tingle swapped out Combustion Integrated Rack hardware in the Destiny lab module.

April 10, 2018 - Crew Researches Biology and Physics, Practices for Emergency

Commander Anton Shkaplerov swapped manifold bottles inside the Combustion Integrated Rack, a device that enables the safe observation of flames and soot on the orbital laboratory. Shkaplerov's work today is in support of the Advanced Combustion Microgravity Experiment (ACME). ACME is a set of five independent studies researching gaseous flames in space that may enable more fuel efficient and less polluting technologies on Earth.

April 18, 2018 - Lab Tests and Life Science as Station Orbits Higher Today

Feustel installed new firewall gear in the Harmony module before replacing manifold bottles in the Combustion Integrated Rack.

CLD Flame round 2 featuring ethylene

E-FIELD Flames round 2 featuring the 2.1-mm gas-jet burner

September 19, 2018 - Physics, Human Research on Lab as Japan Announces Launch Date

NASA astronauts Ricky Arnold and Serena Auñón-Chancellor worked on separate science gear today that enables research into flames, fuels and high temperatures in space. Arnold spent most of Wednesday replacing experiment hardware inside the Combustion Integrated Rack.

October 18, 2018 - Liquid and Flame Science Work amid Japanese, Russian Maintenance

Fluid physics and combustion research on the International Space Station helps scientists understand how well-known phenomena on Earth behaves in microgravity. ... The way flames burn and create soot in space can also create safety issues for crews. … Sergey Prokopyev of Roscosmos opened up the Combustion Integrated Rack in the afternoon and replaced manifold bottles that contain gases for flame experiments. The flight engineer also packed items for disposal on a Japanese cargo ship and checked on Russian ventilation and air conditioning systems.

October 22, 2018 - Genetics, Vision and Earth Studies Aboard Station Today

Gerst started his day in the U.S. Destiny lab module replacing hardware inside the Combustion Integrated Rack that enables gas and flame studies.

October 26, 2018 - Spacesuits and High-Temp, Fire Science Focus of Crew Today

Fellow flight engineers Serena Auñón-Chancellor of NASA and Sergey Prokopyev of Roscosmos worked on advanced science hardware. The two devices, the Electrostatic Levitation Furnace (ELF) and the Combustion Integrated Rack (CIR), enable the safe research of high temperatures, flames and gases.

October 31, 2018 - Physics, Combustion and Biology Science Ahead of Station's 20th Anniversary

Cosmonaut Sergey Prokopyev worked inside the U.S. Destiny lab module replacing the Combustion Integrated Rack's (CIR) fuel bottles. The CIR has been enabling research and observations into how fuels and flames burn in space on the orbital lab for over ten years. Results may guide the development of rocket engines and fire safety aboard spacecraft.

November 9, 2018 - Crew Ends Week Researching Space Physics, Biology and Time

Flight Engineer Sergey Prokopyev from Roscosmos … swapped fuel bottles inside the Combustion Integrated Rack to maintain ongoing flame and gas research aboard the station.

Appendix J. Summary of the ISS data collection

The E-FIELD Flames data is all digital and includes both image and non-image data, where this is true for the other current ACME experiments. To be clear, there is no collection of soot or other physical samples that are subsequently transported to the ground.

The E-FIELD Flames image data includes images from the ACME DataCam, the CFI intensified camera (used with the CH* filter), and the High-Bitdepth Multispectral (HiBMs) camera (equipped with a red filter corresponding to the opposing CIR illumination package). The DataCam images are in a Halcon Iconic Object (hobj) format, while the images from the CFI and HiBMs cameras are in a Tagged Image File (tif) format. In contrast, the analog video was recorded on the ground in a Transport Stream (ts) format.

The key non-imaging data from a test is a comma-delimited file that is described in detail in the Software Interface Design Description (ACME-DOC-010 Rev. D).

The data is currently archived at NASA and work is underway to make it available publicly through NASA's Physical Science Informatics System (PSI).

Appendix K. Publications, presentations, awards, and outreach

K.1 Publications and Presentations with ISS Results

Invited Technical Lectures/Presentations

108. Lund University, Center for Combustion Science and Technology (CECOST) - Boutique Combustion - electric flames; burning ice, seeing through the diesel mist, hot air balloons, December 3, 2019.

107. University of Maryland, Fire Protection Engineering Department Seminar - Boutique Flames – in microgravity with electric fields; burning ice; and hot-air balloons, May 10, 2019.

106. Princeton University, Mechanical and Aerospace Engineering Seminar - Electric Field Effects on Laminar Diffusion Flames, February 8, 2019.

105. ME Graduate Student Association Research Symposium (MEGSA), University of California, Riverside, Invited Keynote - Electrical Aspects of Flames: Experiments on the International Space Station, May 25, 2018.

104. Multi Agency Coordinating Committee for Combustion Research (MACCCR), Sandia National Laboratories, Livermore, CA - Electric Field Effects on Flames in Microgravity on the International Space Station, April 11, 2018.

36th Annual Meeting of the American Society for Gravitational and Space Research (ASGSR) - E-FIELD Flames - Electric Field Effects on Laminar Diffusion Flames (presented in the Physical Sciences Plenary), Nov. 5-6, 2020.

36th Annual Meeting of the American Society for Gravitational and Space Research (ASGSR) - ACME E-FIELD Flames experiment — new data in PSI, Nov. 5-6, 2020.

36th Annual Meeting of the American Society for Gravitational and Space Research (ASGSR) - PSI Update and Demonstration," Nov. 5-6, 2020. (*where E-FIELD Flames was used as the PSI demonstration experiment*)

North America Taiwanese Professors' Association (NATPA) – Southern California (SoCal) Spring Meeting, CSU San Marcos — Microgravity Electric-Field Flames on International Space Station (ISS), April 28, 2018.

Conference Papers (full papers, acceptance based on abstract)

D.179 Chien, Y.-C., Tinajero, J., Stocker, D., Hegde, U., and Dunn-Rankin, D. (2018) "Electric-Field Effects on Flames in Microgravity on the International Space Station," Spring Technical Meeting, Central States Section/The Combustion Institute, Minneapolis, MN, May 20- 22.

D.185 Chien, Y.-C., Tinajero, J., Stocker, D., Hegde, U., and Dunn-Rankin, D. (2019) "Current and Flame Changes with Electric Fields in Microgravity," Paper 2F18, 11th US National Combustion Meeting, Pasadena, CA, March 24-27.

D.188 Chien, Y.-C., Stocker, D., Hegde, U. and Dunn-Rankin, D. (2019) "Microgravity Experiments with Methane Jet Flames under Electric Field Influence on-board the International Space Station (ISS)" 12th Asia-Pacific Conference on Combustion, Fukuoka, Japan, July 1-5.

Pichugin, S., Artemiev, O., Prokopiev, S., Kononenko, O., Novitsky, O., and Frolov, S. (2019), "Investigations at ISS on Combustion of Gas Mixtures. Sample Results and Technical Experience," XIII International Scientific and Practical Conference 'Manned Flights in Space,' Star City, Russia, Nov. 13-15

Conference Papers (only abstract required)

E.110 Chien, Y.-C., Tinajero, J., Stocker, D., Hegde, U., and Dunn-Rankin, D. (2018) "Small Diffusion Flames under the Influences of Electric Fields: Microgravity and Gravity," Work-in-Progress Poster 1P065, 37th International Symposium on Combustion, Dublin, Ireland, July 29-Aug. 3.

E.111 Chien, Y.-C., Stocker, D., Hegde, U. and Dunn-Rankin, D. (2019) “Microgravity Experiments Examining Electric Field Effects on Laminar Methane Gas-Jet Diffusion Flames," Annual Meeting of the American Society for Gravitational and Space Research (ASGSR), Denver, CO, November 20-23.

Chien, Y.-C., Tinajero, J., Stocker, D., Hegde, U. and Dunn-Rankin, D. (2018) “Microgravity Experiments of Electric Field Effects on Laminar Ethylene/Air Diffusion Flames.” 34th Annual Meeting of the American Society for Gravitational and Space Research (ASGSR), Rockville, MD, Oct. 31 to Nov. 3.

K.2 Publications and Presentations (comprehensive list)

Graduate Students

Ph.D. Dissertations

A.19 Karnani, S. (2011) “Electrical Control of Combustion in Microgravity." https://search.proquest.com/docview/919052642

A.24 Chien, Yu-Chien (Alice) (2015) “Electrical Aspects of Impinging Flames." https://escholarship.org/uc/item/0ms7b11c

A.29 Tinajero, J. (2017) “Flame Dynamics and Chemi-Ion Flows Driven by Applied Electric Fields." https://escholarship.org/uc/item/3273d8pg

Escofet-Martin, D. (2017) “Laser Diagnostics for High Pressure Combustion,” Ph. D. dissertation, University of California, Irvine.

Sauer, V. (2017) “Analysis of Laminar Nonpremixed Stagnation Flow and Swirl-Type Tubular Flames,” Ph. D. dissertation, University of California, Irvine. https://escholarship.org/uc/item/48h0j39b

Lopez-Camara, C.-F. (2020) “Numerical Study of Non-Premixed Methane/Air Flames Behavior under Different Body Forces: Buoyancy and Electric Field,” Ph. D. dissertation, University of California, Irvine (in Civil and Environmental Engineering).

M.S. Theses and Projects

B.45 Tsai, H.-J. (2011) “Attempts to Model Electrical Field Effects on Flames." (project)

B.53 Martin, David Escofet (2014) “OH PLIF Measurements in an Impinging Non-Premixed Flame." (project)

B.58 Tinajero, Jesse (2014) “Chemi-Ion Driven Flows in an Electric Field." https://escholarship.org/uc/item/8qv144fb

B.62 Ricchuiti, Valentina (2016) “OpenFOAM Simulations of Impinging Coflow Flames, including Chemi-Ionization and Electric Fields." https://escholarship.org/uc/item/9bc396vc

Senior Scholars/Researchers

Kiyotaka Yamashita, Post Doctoral Scientist, University of Tokyo, summer 2008 - Numerical Simulation of Electric Effects in Diffusion Flames

Valentina Ricchiuti - Researcher 2014 - CFD of Combustion using OpenFOAM; Researcher 2016 - CFD of Electrical Properties of Flames using LBNL code

Yu-Chien (Alice) Chien, Post Doctoral Project Scientist, 2016-present - Lasers, Flames, and Aerosols Laboratory manager and NASA Electric Field Effects on Flames project

Visiting Researchers (Thesis Study)

C.4 Francesco Borgatelli, Polytechnic Milano - Engineering Degree student, 2006 - Feedback control of flames with electric fields, “Behavior of a Small Diffusion Flame Affected by an Electric Field," degree conferred 2008/2009.

C.8 Julian Glorian, Universite D'Orleans, France - Engineering Degree student 2011 - Computational study of ions and excited state species in a methane/air laminar diffusion ame

C.9 Claudio Giani, Polytecnico Milano - Engineering Degree student 2011/2012 - Swirl designs for miniature film combustor

C.10 David Escofet Martin, Universitat Politcnica de Catalunya, Terrassa, Engineering Degree student 2011/2012 - Laser Spectroscopy Techniques: 2D OH Planar Laser Induced Fluorescence

C.11 Laia Torregrosa Sauret, Universitat Politcnica de Catalunya, Terrassa, Engineering Degree student 2012/2013 - Automated Measurements in a Counterow Flame

C.12 Joan Santacana Vall, Universitat Politcnica de Catalunya, Terrassa, Engineering Degree student 2012/2013 - Automated Measurements in a Hydrate Flame

C.13 Marco Minniti, Polytecnico Milano - Engineering Degree student 2012/2013 - Thin Filament Pyrometry for Combustion System Temperatures

C.14 Valentina Ricchiuti, Polytecnico Milano - Engineering Degree student 2012/2013 - Chemical Kinetics in a Water-Laden Non-Premixed Counterow Flame

C.17 Claudia Lopez Camara, Escola Tècnica Superior d’Enginyeria Química, Universitat Rovira i Virgili, Tarragona, Spain visiting M.Sc. student 2014/2015 - Numerical Simulation of a Co-Flow Methane/Air Flame Including Ions and Excited Species

C.18 Albert Aguilera Roman, Universitat Politcnica de Catalunya, Terrassa, Engineering Degree student 2014/2015 - Relating CH* Chemiluminescence to Charged Species in a Nonpremixed Methane Flame

C.21 Filippo Colagrande, Politecnico Milano - Engineering Degree student 2015/2016 - Study on Plasma Treatment of NOx in Engine Exhaust Gas; thesis date: September 2016

Visiting Scholar Internship Projects (with reports)

D.2 Benjamin Debareix, ISAE, ENSMA, France - Intern student 2011 (no formal report) – OpenFOAM Computation of Jet Diffusion Flame Impinging on a Surface

D.3 Mishal Francis, University of Glasgow - Intern student 2011/2012 - IR detection of electrical effects on small diffusion flames

D.4 Joshua Jacobs, University of Glasgow - Intern student 2011/2012 - Soot imaging in small diffusion flames

D.8 Fabien Plongeron, ENSMA, 2012: Image processing for chemiluminescense in sooting flames

D.34 Guillaume Eplenier, ENSMA, 2015; Chemical kinetics calculations of flames with ions and excited species

D.35 Sarah Benhaddou, ENSMA, 2016; Treatment of exhaust gases using non-thermal plasma

D.36 Louise Autef, ENSMA, 2016; Experiments and modeling of small diffusion flames in electric fields

D.43 Guillaume Bernard, ENSMA, 2016; Electrically controlled flames

Undergraduate Students

Francisco Villegas - CAMP Summer Scholar (2017) - ethylene flames under electric field influence

Igor Alves - Brazil Science without Borders Scholar (2016) - schlieren imaging of flames under the influence of an electric field

Miguel Plascencia - CAMP Summer Scholar (2013) - construction of microgravity electric-field burner system

Daniel Jaimes - CAMP Summer Scholar (2012) - sodium line reversal for temperature measurements in flames

Adrian Garcia - UC LEADS Scholar (2011) - Prosilica camera image analysis for zero gravity flame experiment

Tony Eguavoen - CAMP Summer Scholars Program (2010) - optical pyrometry for non-invasive temperature measurements in combustion;

Maribel Jaquez - UC-LEADS Scholar, pre B.S. (2010) - electrical effects on diffusion flames (E-FIELD Flames)

Martin Tajiboy - CAMP Summer Scholars Program (2008) - enclosure for electrical study of diffusion flames

Raymond Valdes - CAMP Summer Scholars Program (2007) - ow panel for magnetic effects on flames study

Invited Technical Lectures/Presentations

108. Lund University, Center for Combustion Science and Technology (CECOST) - Boutique Combustion - electric flames; burning ice, seeing through the diesel mist, hot air balloons, December 3, 2019.

107. University of Maryland, Fire Protection Engineering Department Seminar - Boutique Flames – in microgravity with electric fields; burning ice; and hot-air balloons, May 10, 2019.

106. Princeton University, Mechanical and Aerospace Engineering Seminar - Electric Field Effects on Laminar Diffusion Flames, February 8, 2019.

105. ME Graduate Student Association Research Symposium (MEGSA), University of California, Riverside, Invited Keynote - Electrical Aspects of Flames: Experiments on the International Space Station, May 25, 2018.

104. Multi Agency Coordinating Committee for Combustion Research (MACCCR), Sandia National Laboratories, Livermore, CA - Electric Field Effects on Flames in Microgravity on the International Space Station, April 11, 2018.

102. University of California, San Diego, Combustion and Engineering Physics Seminar - Electrical Field Effects on Laminar Diffusion Flames, November 6, 2017.

101. KAUST Clean Combustion Research Center - New Combustion Concepts, Saudi Arabia, Invited Keynote - Electrical Aspects of Flames, Methane Hydrate Combustion, and Miniature Film Combustors, March 7, 2017.

100. University of California, Irvine, MAE Department Seminar - Lasers, Flames, and Aerosols: A Few Fringe Topics in Combustion, March 3, 2017

97. University of Southern California, Mechanical Engineering Seminar - Fringe Topics in Combustion, April 20, 2016

94. Seoul National University, Seoul, Korea, Aerospace Engineering Seminar - Electrical Aspects of Combustion, December 8, 2015

86. Louisiana State University, Baton Rouge, Department of Mechanical and Industrial Engineering Seminar - Electrical Actuation of Small Diffusion Flames, April 4, 2014

78. China Lake Naval Warfare Center Lecture Series - Recent Research in Combustion - August 10, 2011

72. UCLA Mechanical Engineering Departmental Seminar - Using Old Fuels in New Ways and New Fuels in Old Ways, December 11, 2009

71. UCI MAE Departmental Seminar - Recent Research in the Lasers, Flames, and Aerosols Laboratory, October 14, 2009

70. NASA Glenn Research Center, Cleveland, Ohio - Electric Field Effects in a Small Co-Flow Diffusion Flame, (with S. Karnani), May 21, 2009

69. Hiroshima University, Hiroshima, Japan - Secondary Air Injection in a Miniature Film Combustor, and other topics, December 10, 2008
68. Osaka University, Osaka, Japan - Graduate Research in the LFA Group, December 8, 2008
67. Japanese Combustion Symposium, Kyoto, Japan - Electrical Manipulation of Flames, December 4, 2008
66. Aoyama Gakuin University, Tokyo, Japan - Using Old Fuels in New Ways, November 28, 2008
65. Keio University, Yokohama, Japan - Lasers, Flames and Aerosols: recent results in combustion research, November 27, 2008
64. Aoyama Gakuin University, Tokyo, Japan - Graduate Research at the University of California, Irvine, November 26, 2008
62. University of California, Riverside Departmental Seminar - Recent Combustion Research in Lasers, Flames, and Aerosols, October 10, 2008.
60. Universidad Nacional Estadio Sao Paulo (UNESP), Brazil - Lasers, Flames, and Aerosols: introduction to research in combustion at UCI, July 10, 2008.
58. Universidad Federal Rio de Janeiro, Brazil - Lasers, Flames, and Aerosols: current combustion research at UCI, July 8, 2008.
57. National Taiwan University, Taipei, Taiwan - Lasers, Flames, and Aerosols, June 5, 2008.
56. Industrial Technology Research Institute, Hsinchu, Taiwan - Lasers, Flames, and Aerosols: combustion research innovation, May 29, 2008
55. National Cheng Kung University, Tainan, Taiwan - Lasers, Flames, and Aerosols: current research topics at UCI, May 28, 2008.
52. University of Hawaii, Honolulu, HI, “Flame and Corona Ion Driven Winds," October 19, 2007.
48. University of California, San Diego Fluid Seminar, San Diego, CA - Convective Transport in Flame and Corona Ion-Driven Winds, January 22, 2007.
44. National Cheng-Kung University, Tainan, Taiwan - Control of Ion Winds from Flames and Corona Discharges, March 23, 2006.

36th Annual Meeting of the American Society for Gravitational and Space Research (ASGSR) - E-FIELD Flames - Electric Field Effects on Laminar Diffusion Flames (presented in the Physical Sciences Plenary), 5-6 Nov. 2020.

36th Annual Meeting of the American Society for Gravitational and Space Research (ASGSR) - ACME E-FIELD Flames experiment — new data in PSI, Nov. 5-6, 2020.

36th Annual Meeting of the American Society for Gravitational and Space Research (ASGSR) - PSI Update and Demonstration,” Nov. 5-6, 2020. (*where E-FIELD Flames was used as the PSI demonstration experiment*)

Escola Puigcerver, Reus, Spain - Highlights of Numerical Simulation from the ACME-NASA Project, May 2, 2019.

University of Rovira i Virgili, Tarragona, Spain - Numerical simulation of non-premixed laminar flame behavior with electric fields and buoyancy, May 11, 2018.

North America Taiwanese Professors’ Association (NATPA) – Southern California (SoCal) Spring Meeting, CSU San Marcos — Microgravity Electric-Field Flames on International Space Station (ISS), April 28, 2018.

KAUST Research Conference in New Combustion Concepts (KRCNCC) - Fringe Combustion Topics: electrical aspects of flames, burning hydrates, and miniature combustors (Invited Presentation), March 6-8, 2017.

Institut Supérieur de l'Aéronautique et de lÉspace, ENSMA, France - Characterizing Small Flames With Chemiluminescence, November 14, 2016.

Mechanical Engineering, National Cheng Kung University, Tainan, Taiwan - Electric Aspects of Impinging Flames/ Hydrate and Keck Laboratory, March 28, 2016.

Seminar at Department of Aeronautics and Astronautics, National Cheng Kung University, Tainan, Taiwan - Electric Aspects of Impinging Flames/ Hydrate and Keck Laboratory, March 29, 2016.

Publications

Refereed Publications

A.49 Weinberg, F., Carleton. F., Kara, D., Xavier, A., Rickard, M., and Dunn-Rankin, D. (2006) "Inducing Gas Swirl in Tubes using Ion Wind from Corona Discharges," *Experiments in Fluids*, 40, 231-237.
https://doi.org/10.1007/s00348-005-0062-0

A.51 Rickard, M.A., Dunn-Rankin, D., Weinberg, F.J., and Carleton, F. (2006) "Maximizing Ion Driven Gas Flows," *Journal of Electrostatics*, 64, 368-376.
https://doi.org/10.1016/j.elstat.2005.09.005

A.54 Papac, M.J. and Dunn-Rankin, D. (2008) "Modelling Electric Field Driven Convection in Small Combustion Plasmas and Surrounding Gases," *Combustion Theory and Modelling*, 12, 23-44.
https://doi.org/10.1080/13647830701383814

A.55 Rickard, M.A. and Dunn-Rankin, D. (2007) "Numerical Simulation of a Tubular Ion-Driven Wind Generator," *Journal of Electrostatics*, 65, 646-654.
https://doi.org/10.1016/j.elstat.2007.04.003

A.56 Weinberg, F.J., Carleton, F., and Dunn-Rankin, D. (2008) "Electric Field-Controlled Mesoscale Burners," *Combustion and Flame*, 152, 186-193.
https://doi.org/10.1016/j.combustflame.2007.07.007

A.64 Yamashita, K., Karnani, S., and Dunn-Rankin, D. (2009) "Numerical prediction of ion current from a small methane jet flame," *Combustion and Flame*, 156, 6, 1227-1233.
https://doi.org/10.1016/j.combustflame.2009.02.002

A.73 Borgatelli, F. and Dunn-Rankin, D. (2012) "Behavior of a Small Diffusion Flame as an Electrically Active Component in a High-Voltage Circuit," *Combustion and Flame*, 159, 210-220.
https://doi.org/10.1016/j.combustflame.2011.06.002

A.83 Karnani, S. and Dunn-Rankin, D. (2015) "Detailed Characterization of DC Electric Field Effects on Small Non-Premixed Flames," *Combustion and Flame*, 162, 2865-2872.
https://doi.org/10.1016/j.combustflame.2015.03.019

A.91 Chien, Y.-C., Escofet-Martin, D., and Dunn-Rankin, D. (2016) "CO Emission from an Impinging Non-Premixed Flame," *Combustion and Flame*, 174, 16-24.
https://doi.org/10.1016/j.combustflame.2016.09.004

A.94 Tinajero, J., Bernard, G., Autef, L., and Dunn-Rankin, D. (2017) "Characterizing I-V Curves For Non- Premixed Methane Flames Stabilized On Different Burner Configurations," *Combustion Science and Technology*, 189:10, 1739-1750.
https://doi.org/10.1080/00102202.2017.1331218

A.100 Chien, Y.-C. and Dunn-Rankin, D. (2018) "Electric Field Changes of a Diffusion Flame near an impinging surface: Heat and Thermal Perspective," *Energies*, Special Issue.
https://doi.org/10.3390/en11051235

A.105 Tinajero, J. and Dunn-Rankin, D. (2018) "Non-Premixed Axisymmetric Flames Driven by Ion Currents," *Combustion and Flame*, 199, 365-376.
https://doi.org/10.1016/j.combustflame.2018.10.036

A.108 Chien, Y.-C., Escofet-Martin, D., and Dunn-Rankin, D. (2019) "Ion Current and Carbon Monoxide Release from an Impinging Methane/Air Coflow Flame in an Electric Field," *Combustion and Flame*, 204, 250-259.
https://doi.org/10.1016/j.combustflame.2019.03.022

Refereed Conference Proceedings (full peer-review)

B.14 Dunn-Rankin, D. and Weinberg, F.J. (2006) "Using Large Electric Fields to Control Transport in Microgravity," Annals of the New York Academy of Sciences, 1077, 570-584. https://doi.org/10.1196/annals.1362.037

B.15 Papac, M.J. and Dunn-Rankin, D. (2006) "Canceling Buoyancy of Gaseous Fuel Flames in a Gravitational Environment using an Ion Driven Wind," Annals of the New York Academy of Sciences, 1077, 585-601. https://doi.org/10.1196/annals.1362.038

B.19 F.J. Weinberg, F.J., Dunn-Rankin, D., Carleton, F.B., Karnani, S., Markides, C., and Zhai, M. (2013) "Electrical aspects of flame quenching," Proceedings of the Combustion Institute, 34, 3295-3301. https://doi.org/10.1016/j.proci.2012.07.007

Sauer, V.M. and Dunn-Rankin, D. (2016) "Impinging Nonpremixed Coflow Methane-Air Flames with Unity Lewis Number," Proceedings of the Combustion Institute, vol. 36. https://doi.org/10.1016/j.proci.2016.06.193

Conference Papers (full papers, acceptance based on abstract)

D.83 Borgatelli, F. and Dunn-Rankin, D. (2006) "Feedback Control of Ion Current from a Small Diffusion Flame," Paper 06S-44, Western States Section/The Combustion Institute Spring Meeting, University of Idaho, Boise, March 27-28.

D.101 Karnani, S., Bennett, M., and Dunn-Rankin, D. (2008) "Electrical Properties of Small Diffusion Flames," Western States Section/The Combustion Institute Spring Meeting paper 08S-4, University of Southern California, Los Angeles, March 17-18.

D.106 Dunn-Rankin, D. (2008) "Electrical Manipulation of Flames," invited lecture, Japanese Combustion Symposium, Kyoto, Japan, December 4.

D.107 Karnani, S. and Dunn-Rankin, D. (2009) "Electric Field Effects on a Small Co-Flow Diffusion Flame," Paper 13A1, U.S. Combustion Meeting, Ann Arbor, Michigan, May 18-20.

D.115 Karnani, S., Coffin, P., Schoen, M., Dunn-Rankin, D., Takahashi, F., Yuan, Z.-Y., and Stocker, D. (2009) "Exploring the effects of gravity on a coflow diffusion flame in an electric field," Paper 09F-79 at the Western States Section/The Combustion Institute Fall Meeting, University of California, Irvine, October 26-27.

D.126 Swenson, K., Karnani, S., Dunn-Rankin, D., Takahashi, F., Stocker, D., and Guang, Z.-G. (2011) "Electric Field Induced Convection in Microgravity Combustion," 7th US National Combustion Meeting, Georgia Institute of Technology, Atlanta, GA, March 20-23.

D.129 Karnani, S., Dunn-Rankin, D., Takahashi, F., Yuan, Z.G., and Stocker, D. (2011) "Simulated Gravity Using Electric Fields in Microgravity Combustion," 23rd International Colloquium on the Dynamics of Explosions and Reactive Systems, Irvine, CA, July 24-29.

D.141 Chien, Y.-C., Yamashita, K., Karnani, S., and Dunn-Rankin, D. (2013) "The Influence of Electric Field Power Systems on Flame Behavior," Paper 070MI-0368, US Combustion Meeting, Park City, Utah, May 20-23.

D.142 Karnani, S. and Dunn-Rankin, D. (2013) "Visualizing CH* Chemiluminescence in Sooting Flames via Image Post-Processing," Paper 070DI-0393, US Combustion Meeting, Park City, Utah, May 20-23.

D.144 Chien, Y.-C., Yamashita, K., and Dunn-Rankin, D. (2013) "Electrical Aspects of Impinging Flames," 24th International Colloquium on the Dynamics of Explosions and Reactive Systems, Paper 126, Taipei, Taiwan, July 28-August 2.

D.150 Takahashi, F., Nagapetyan, H., Stocker, D.P., and Dunn-Rankin, D. (2014) “Transient Responses of Jet Diffusion Flames to a DC Electric Field in Microgravity," Spring Technical Meeting of the Central States Section/The Combustion Institute, Tulsa, OK, March 16-18.

D.151 Escofet-Martin, D., Chien, Y.C., and Dunn-Rankin, D. (2014) “OH PLIF of an Impinging Flame," Western States Section/The Combustion Institute Spring Meeting paper 087DI-0032, California Institute of Technology, Pasadena, CA, March 24-25.

D.152 Tinajero, J., Dunn-Rankin, D., Jie, B., and Plascencia, M. (2014) “Schlieren Imaging of Chemi-Ion Driven Flows in a Time Varying Electric Field," Western States Section/The Combustion Institute Spring Meeting paper 087LF-0048, California Institute of Technology, Pasadena, CA, March 24-25.

D.153 Chien, Y.C. and Dunn-Rankin, D. (2014) “Temperature measurement over a quenching plate with electric field flames," Spring Meeting, Western States Section/The Combustion Institute, Paper 087LF-0060, California Institute of Technology, Pasadena, CA, March 24-25.

D.156 Escofet-Martin, D., Chien, Y.-C., and Dunn-Rankin, D. (2015) “Two-Photon CO PLIF of Flames near a Quenching Plate," 9th U.S. National Combustion Meeting, Cincinnati, OH, May 17-20.

D.157 Aguilar, A., Tinajero, J., and Dunn-Rankin, D. (2015) “Relating CH* Chemiluminescence with Charged Species in a Non-Premixed Methane Flame," 9th U.S. National Combustion Meeting, Cincinnati, OH, May 17-20.

D.159 Chien, Y.-C., Escofet-Martin, D., and Dunn-Rankin, D. (2015) “CO Emission from an Impinging Non-Premixed Flame under the Influence of an Electric Field," 9th U.S. National Combustion Meeting, Cincinnati, OH, May 17-20.

D.161 Escofet-Martin, D., Chien, Y.-C., and Dunn-Rankin, D. (2015) “Two-Line OH PLIF Temperature Measurements of Flames Near a Quenching Plate," 25th International Colloquium on the Dynamics of Explosions and Reactive Systems, Leeds University, England, August 2-7.

D.162 Chien, Y.-C., Escofet-Martin, D., and Dunn-Rankin, D. (2015) “Non-Premixed Impinging Flames and CO Release under the Influence of an Electric Field," 25th International Colloquium on the Dynamics of Explosions and Reactive Systems, Leeds University, England, August 2-7. https://www.semanticscholar.org/paper/Non-premixed-Impinging-Flames-and-CO-Release-under-Chien-Escofet-Martin/f9e576785326e5a7fab6ea28c68f3acff8b8bbef

D.163 Lopez-Camara, C-F., Eplenier, G., Tinajero, J., and Dunn-Rankin, D. (2015) “Numerical Simulation of Methane/Air Flames Including Ions and Excited Species," Western States Section/The Combustion Institute Fall Meeting Paper 134CK-0055, Brigham Young University, Provo, UT, October 5-6.

D.168 Tinajero, J., Bernard, G., Autef, L., and Dunn-Rankin, D. (2017) “Enhanced Flame Ion Production Through External Electric Fields," 10th Paper 1G13 - U.S. National Combustion Meeting, College Park, Maryland, April 23-26.

D.177 Lopez-Camara, C.-F. and Dunn-Rankin, D. (2017) “Numerical Simulations of a Co-Flow Methane/Air Flame including Ions and Excited Species under Different Gravity Conditions," Paper 29LF-0053, Western States Section/The Combustion Institute Fall Meeting, University of Wyoming, Laramie, WY, October 1-3.

D.179 Chien, Y.-C., Tinajero, J., Stocker, D., Hegde, U., and Dunn-Rankin, D. (2018) “Electric-Field Effects on Flames in Microgravity on the International Space Station," Spring Technical Meeting, Central States Section/The Combustion Institute, Minneapolis, MN, May 20- 22.

D.182 Lopez-Camara, C.-F., Belhi, M., Im, H.G., and Dunn-Rankin, D. (2019) “Numerical Simulations of Laminar Nonpremixed CH4-Air Jet Flames Influenced by Varying Electric Fields," Paper 2F16, 11th US National Combustion Meeting, Pasadena, CA, March 24-27.

D.185 Chien, Y.-C., Tinajero, J., Stocker, D., Hegde, U., and Dunn-Rankin, D. (2019) “Current and Flame Changes with Electric Fields in Microgravity," Paper 2F18, 11th US National Combustion Meeting, Pasadena, CA, March 24-27.

D.188 Chien, Y.-C., Stocker, D., Hegde, U. and Dunn-Rankin, D. (2019) “Microgravity Experiments with Methane Jet Flames under Electric Field Influence on-board the International Space Station (ISS)" 12th Asia-Pacific Conference on Combustion, Fukuoka, Japan, July 1-5.

Sauer, V. and Dunn-Rankin, D. (2015) “Impinging Coflow Nonpremixed Methane-Air Flames with Unity Lewis Number,” 9th U.S. National Combustion Meeting, Cincinnati, OH, May 17-20.

Ricchiuti, V. and Marc Day, M. (2017) “A Spectral Deferred Correction Strategy for Low Mach Number Reacting Flows Subject to Electric Fields,” 16th Sixteenth International Conference on Numerical Combustion (NC17), Orlando, Florida, April 3-5.

Pichugin, S., Artemiev, O., Prokopiev, S., Kononenko, O., Novitsky, O., and Frolov, S. (2019), “Investigations at ISS on Combustion of Gas Mixtures. Sample Results and Technical Experience,” XIII International Scientific and Practical Conference ‘Manned Flights in Space,’ Star City, Russia, Nov. 13-15.

Conference Papers (only abstract required)

E.55 Weinberg, F.J., Carleton, F.A., and Dunn-Rankin, D. (2006) “Electric Field-Controlled Mesoscale Burners," Poster 1A14, 31st International Combustion Symposium, Heidelberg, Germany, August 7- 13.

E.72 Proctor, C., Karnani, S., Dunn-Rankin, D., Takahashi, F., Yuan, Z.-G., and Stocker, D. (2010) “Exploring Electric Field Effects on Laminar Diffusion Flames in Microgravity," Poster Session of the 33rd International Combustion Symposium, Beijing, China, August 2-6.

E.76 Chien, Y.C., Jaquez-Nunez, M., Francis, M., Jacobs, J., Karnani, S., and Dunn-Rankin, D. (2012) “Recent Progress in Studying Emissions from Electric Field Effects on Flames," W5P040- Work-in-Progress Poster, International Combustion Symposium, Warsaw, Poland, July 30-August 3.

E.80 Tinajero, J.A., Karnani, S.V., and Dunn-Rankin, D. (2013) “E-Field: Electric Field Effects on Laminar Diffusion Flames," American Space and Gravity Science Research 29th Annual Meeting, Orlando, FL, November 3-8.

E.82 Chien, Y.-C., Escofet-Martin, D., and Dunn-Rankin, D. (2014) “Electric Field Effects on CO Emission from Non-Premixed Quenching Flames," Work-in-Progress Poster W1P011, 35th International Combustion Symposium, San Francisco, CA, August 3-8.

E.88 Tinajero, J. and Dunn-Rankin, D. (2014) “Carbon Particle Behavior in Flames Exposed to an Electric Field," Work-in-Progress Poster W4P140, 35th International Combustion Symposium, San Francisco, CA, August 3-8.

E.89 Tinajero, J., Dunn-Rankin, D., Jie, B., and Plascencia, M. (2014) “Ground-Based Schlieren Experiments of Chemi-Ion Driven Flows in a Time Varying Electric Field," 30th Annual Meeting of the American Society for Gravitational and Space Research (ASGSR), Pasadena, CA, October 22-26.

E.92 Chien, Y.C., Escofet-Martin, D., and Dunn-Rankin, D. (2015) “Electric field effects on CO emission from nonpremixed quenching flames," Poster at the Gordon Research Conference on Fundamental of Laser Diagnostics in Combustion, Waterville Valley, NH, August 914.

E.95 Lopez-Camara, C.-F., Tinajero, J., and Dunn-Rankin, D. (2015) “Numerical Simulation of a Co-Flow Methane/Air Flame with an Applied Electrical Field," 31st Annual Meeting of the American Society for Gravitational and Space Research (ASGSR), Alexandria, VA, November 11-14.

E.98 Escofet-Martin, D., Chien, Y.C., and Dunn-Rankin, D. (2016) “Two-Photon CO LIF Thermometry and Concentration," Work-in-Progress Poster, 36th International Combustion Symposium, Seoul, South Korea, July 31-August 5.

E.99 Tinajero, J. and Dunn-Rankin, D. (2016) “Dynamics of Chemi-Ion Driven Flows," Work-in-Progress Poster, 36th International Combustion Symposium, Seoul, South Korea, July 31-August 5.

E.102 Tinajero, J. and Dunn-Rankin, D. (2016) “Using Voltage Current Curves to Characterize Electric Field Effects on Flames For Different Burner/Mesh Electrode Configurations," 32nd Annual Meeting of the American Society for Gravitational and Space Research (ASGSR), Cleveland, OH, October 25-29.

E.103 Lopez-Camara, C.-F. and Dunn-Rankin, D. (2016) “Numerical Simulation of Flames under the Influence of Electric Field," 32nd Annual Meeting of the American Society for Gravitational and Space Research (ASGSR), Cleveland, OH, October 25-29.

E.108 Lopez-Camara, C., Belhi, M., Im, H., and Dunn-Rankin, D. (2018) “Numerical Simulations of CH4-Air Flames Under the Influence of Buoyancy and Electric Fields," Work-in-Progress Poster 1P063, 37th International Symposium on Combustion, Dublin, Ireland, July 29-Aug. 3.

E.110 Chien, Y.-C., Tinajero, J., Stocker, D., Hegde, U., and Dunn-Rankin, D. (2018) “Small Diffusion Flames under the Influences of Electric Fields: Microgravity and Gravity," Work-in-Progress Poster 1P065, 37th International Symposium on Combustion, Dublin, Ireland, July 29-Aug. 3.

E.111 Chien, Y.-C., Stocker, D., Hegde, U. and Dunn-Rankin, D. (2019) “Microgravity Experiments Examining Electric Field Effects on Laminar Methane Gas-Jet Diffusion Flames," 35th Annual Meeting of the American Society for Gravitational and Space Research (ASGSR), Denver, CO, November 20-23.

E.112 Lopez-Camara, C.-F., Pitz, W.J., and Dunn-Rankin, D. (2019) Comprehensive chemical kinetic models for simple hydrocarbon fuels including important excited and charged species, 35th Annual Meeting of the American Society for Gravitational and Space Research (ASGSR), Denver, CO, November 20-23.

Roman, A.A. Dunn-Rankin, D., and Tianjero, J. (2015) “Relating Ch* chemilumiescence with charged species in a non-premixed methane flame,” poster, Fifth Brazilian School of Combustion and 2015 International Combustion Institute Summer School, Sao Paulo, Brazil, June 8-12.

Sauer, V. and Dunn-Rankin, D. (2016) “Parametric Uncertainty In The Analysis Of The Steady-State And Extinction Characteristics Of Nonpremixed Laminar Flames With Simplified Chemistry,” Work in Progress Poster, 36th International Symposium on Combustion, Seoul, Korea, 31 July to 5 August 2016.

Lopez-Camara, C.-F. (2017) "Simulation of Combustion Processes with Applied Electric Field in Microgravity," poster, KAUST Research Conference in New Combustion Concepts (KRCNCC), King Abdullah University of Science and Technology, Saudi Arabia, March 6-8.

Lopez-Camara, C.-F., Tinajero, J., and Dunn-Rankin, D. (2017) "Numerical Simulations of a Co-Flow Methane/Air Flame under Different Gravity Conditions,” 33rd Annual Meeting of the American Society for Gravitational and Space Research (ASGSR), Renton, WA, Oct. 25-28.

Chien, Y.-C., Tinajero, J., Stocker, D., Hegde, U. and Dunn-Rankin, D. (2018) “Microgravity Experiments of Electric Field Effects on Laminar Ethylene/Air Diffusion Flames.” 34th Annual Meeting of the American Society for Gravitational and Space Research (ASGSR), Rockville, MD, Oct. 31 to Nov. 3.

Lopez-Camara, C.-F., Belhi, M., Im, H.G., and Dunn-Rankin, D. (2018) “Numerical Simulations of Laminar Nonpremixed CH4-Air Flames Varying Buoyancy and Applied E-Field,” 34th Annual Meeting of the American Society for Gravitational and Space Research (ASGSR), Rockville, MD, 31 Oct. to 3 Nov. 2018.

K.3 Awards

In part based on his preparatory E-FIELD Flames research, Derek Dunn-Rankin was presented the A.K. Oppenheim Prize at the 2013 International Colloquium on the Dynamics of Explosions and Reactive Systems (ICDERS). The ICDERS prize is given for “for brilliant contributions to the theoretical or

interpretative aspects of the Dynamics of explosions and Reactive Systems. This is intended for an active scientist who is distinguished by showing how to analyze or what it means."

K.4 Outreach

NASA Project Scientists Dennis Stocker created the Space Flames Facebook page (www.facebook.com/space.flames) and Flickr site (https://www.flickr.com/photos/space-flames), where some entries are specific to the E-FIELD Flames. He continues to administer these sites, where most content has been about the ACME project of which E-FIELD Flames is a part. Meanwhile, Stocker has made educational and general information is available via https://tinyurl.com/space-flames. As an educational example, a simple board game related to the ACME project, can be downloaded. As another example, an ACME information sheet providing an overview and recent status can also be downloaded through that site. Such flyers have been distributed at meetings of The Combustion Institute and the American Society for Gravitational and Space Research (ASGSR).

UC Irvine promoted awareness of the E-FIELD Flames investigation through an social media, an online article, and in print.

- Samueli School of Engineering News article, UC Irvine "UCI Conducts Combustion Research aboard the International Space Station," by UCI campus writer Anna Lynn Spitzer, posted on June 18, 2018 and promoted on:
    - June 18 at https://www.facebook.com/space.flames/posts/2107196372825690, "E-FIELD Flames - A UC Irvine article (see the link below) was posted earlier today about the Electric-Field Effects on Laminar Diffusion Flames (E-FIELD Flames) experiment. The article explains, for example, "Our project is to see if we can get a better feel for where these ions are coming from and how we can pull on them to make flames more efficient and less polluting," says [Prof. Derek] Dunn-Rankin [the experiment's Principal Investigator]. The first half of the testing - using the coflow burner - has been completed, where the second half - using the gas-jet burner - will be conducted in the coming months."
    - June 21 at https://twitter.com/UCIEngineering/status/1009864603684048896
    - July 2 at www.facebook.com/ucirvineengineering/posts/10156412390887350, "Among the nearly 6,000 pounds of scientific research, supplies and hardware hurtling toward space aboard the Falcon 9 rocket was equipment integral to a UCI-designed-and-run combustion experiment nearly 20 years in the making."
- Dean's Report, 2018, UCI Samueli School of Engineering "Extraterrestrial", produced by Communication Department in Engineering, writer Anna Lynn Spitzer, published on September 27th, 2018. The online copy of the report and the 4-page article about the ISS research was promoted as follows and 16,000 printed reports were mailed out.
    - Oct. 5 at www.facebook.com/ucirvineengineering/posts/10156625247567350, "The wait is over! Check out the @UCIEngineering 2017-18 Dean's Report here: https://issuu.com/ucisamuelischool/docs/dean_s_report
    - Oct. 5 at https://twitter.com/UCIEngineering/status/1048268163387777024, "The wait is over! Check out the @UCIEngineering 2017-18 Dean's Report here: https://issuu.com/ucisamuelischool/docs/dean_s_report …"
    - Oct. 16 at https://www.facebook.com/space.flames/posts/2210385345840125, "E-FIELD Flames - A great opportunity to learn about combustion research in space - and more specifically the Electric-Field Effects on Laminar Diffusion Flames (E-FIELD Flames) project - is found in the new annual report from the Dean of Engineering at

> UC Irvine (https://uci.edu). The experiment which is currently operating in the space station's Combustion Integrated Rack (CIR) is led by Prof. Derek Dunn-Rankin (right) and Dr. Yu-Chien (Alice) Chien (left). The fifth test day in the current round of testing is scheduled for tomorrow, Oct. 17. Next week, the high-voltage power supply will be swapped by the space station commander, Alex Gerst. The subsequent four days of tests will be carried out with a positively-charged electrode mesh, where the recent tests have been carried out with a negatively-charged mesh. The ion wind resulting from the electric field tends to lift the flame when the mesh is negatively charged and push it back toward the burner when it is positively charged."

Educational outreach related to the experiment was carried out by doctoral student Claudia Lopez-Camara, where these presentations are just two examples:

McBride, S., Munoz Abujder, R., Putman, E., Ramya Bhaskar, R., Lopez-Camara, C.-F., and Helmig, J. (2018) "A Space Science Curriculum to Initiate Student Interest in STEM," 2018 ISS Research and Development Conference, San Francisco, July 23-26.

McBride, S., Munoz Abujder, R., Putman, E., Ramya Bhaskar, R., Lopez-Camara, C.-F., and Helmig, J. (2018) "Space Science Outreach to Initiate Early Interest in STEM," Annual Meeting of the American Society for Gravitational and Space Research Annual Meeting, Rockville, MD, Oct. 31 - Nov. 3.

Meanwhile, NASA promoted awareness of ISS combustion research with the video 'The Power of Studying Combustion on the ISS' which was posted on March 13, 2018 – during the first round of E-FIELD Flames testing.

Awareness of ISS combustion research, including the ACME project and specifically the E-FIELD Flames experiment, was promoted on Oct. 15, 2018 by The Combustion Institute in their online news article, "International Space Station (ISS) Combustion Experiments."

Stocker is involved in a broad range of NASA educational outreach to students and teachers, Scouts of both genders, and the general public. His primary outreach focus is the fundamentals of microgravity (for which resources are available via http://tinyurl.com/SEECmicrogravity) and human spaceflight. But he periodically addresses other topics, e.g., where he sometimes speaks on 'NASA and the Science of Superman' – for example at the Superman Celebration in Metropolis, IL. In this regard, his NASA outreach is carried out locally, regionally, nationally, and online. For example, he regularly presents on microgravity at Space Center Houston's Space Exploration Educators Conference (SEEC). He has exhibited the last five National Scout Jamborees, four national Boy Scout conferences, the last three national Girl Scout conventions, and the 2019 World Scout Jamboree - which was hosted in the U.S. Stocker has been a key lead for the agency in NASA's participation in these Scouting events, where as an example the latter outreach is visually described at www.nasa.gov/feature/stem/world-scout-jamboree.html.